Supporting Materials for "Automatic Differentiation for Inference on Partially Observed Markov Process Models"

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S1 Proof of Theorem 1 (DMOP-α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1:n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i.

Lemma S1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:N,j}^{A,F,\theta};\theta\big), \tag{S1}$$

with the same undiscounted path-space score target as the bootstrap filter estimators of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider MOP-α with $\alpha = 1$. Hold ϕ, the simulator draws, and the reference-filter resampling indices fixed when differentiating, then evaluate at $\theta = \phi$. Write $k_{n,j}$ for k_j at time n. Algorithm 1 gives

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta}, \qquad w_{n,j}^{F,\theta} = w_{n,k_{n,j}}^{P,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = w_{n-1,k_{n,j}}^{F,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}}. \tag{S2}$$

The after-resampling likelihood estimate therefore telescopes:

$$\begin{aligned}\hat{\mathcal{L}}^1(\theta) &= \prod_{n=1}^N L_n^{A,\theta,1} = \prod_{n=1}^N \left(L_n^{\phi}\frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}\right)\\ &= \left(\prod_{n=1}^N L_n^{\phi}\right)\prod_{n=1}^N \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} = \left(\frac{1}{J}\sum_{j=1}^J w_{N,j}^{F,\theta}\right)\prod_{n=1}^N L_n^{\phi}.\end{aligned} \tag{S3}$$

Here each intermediate weight sum cancels, and the initial sum is J because $w_{0,j}^{F,\theta} = 1$. The log-derivative identity gives

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{J^{-1}\sum_{j=1}^J \nabla_\theta w_{N,j}^{F,\theta}\prod_{n=1}^N L_n^{\phi}}{J^{-1}\sum_{j=1}^J w_{N,j}^{F,\theta}\prod_{n=1}^N L_n^{\phi}} = \frac{\sum_{j=1}^J \nabla_\theta w_{N,j}^{F,\theta}}{\sum_{j=1}^J w_{N,j}^{F,\theta}}. \tag{S4}$$

We next expand the filtered weight along its ancestry. Using the existing A convention, let $g_{i,j}^{A,F,\theta}$ denote the measurement density at time i along the filtered trajectory $x_{1:n,j}^{A,F,\theta}$; the endpoint n is implicit. At time n, this density is $g_{n,k_{n,j}}^{\theta}$. At earlier times, the trajectory agrees with that of the selected parent $x_{n-1,k_{n,j}}^{F,\theta}$. Starting from $w_{0,j}^{F,\theta} = 1$ and repeatedly substituting the filtered-weight recursion therefore gives

$$w_{n,j}^{F,\theta} = w_{n-1,k_{n,j}}^{F,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = \left(\prod_{i=1}^{n-1}\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)\frac{g_{n,j}^{A,F,\theta}}{g_{n,j}^{A,F,\phi}} = \prod_{i=1}^{n}\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}. \tag{S5}$$

The product through $n-1$ is empty when $n=1$; for $n>1$, it is the product already accumulated by the selected parent. The reference factors and ancestor indices are fixed under differentiation, so

$$\begin{aligned}\frac{\nabla_\theta w_{n,j}^{F,\theta}}{w_{n,j}^{F,\theta}} &= \nabla_\theta \log w_{n,j}^{F,\theta} = \nabla_\theta \log\left(\prod_{i=1}^{n} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right) \\ &= \nabla_\theta \sum_{i=1}^{n}\left(\log g_{i,j}^{A,F,\theta} - \log g_{i,j}^{A,F,\phi}\right) = \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,F,\theta}. \end{aligned} \tag{S6}$$

For the terminal filtered trajectory at time N, conditional independence of the observations gives

$$\begin{aligned}\sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} &= \nabla_\theta \log\left(\prod_{n=1}^{N} g_{n,j}^{A,F,\theta}\right) \\ &= \nabla_\theta \log\left(\prod_{n=1}^{N} f_{Y_n|X_n}\left(y_n^* \mid x_{n,j}^{A,F,\theta};\theta\right)\right) \\ &= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right). \end{aligned} \tag{S7}$$

The derivative includes the dependence of the simulated particle locations on θ through the chain rule. Multiplying the log-weight derivative by $w_{N,j}^{F,\theta}$ yields

$$\nabla_\theta w_{N,j}^{F,\theta} = w_{N,j}^{F,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} = w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right). \tag{S8}$$

Finally, at $\theta=\phi$, every ratio in Equation (S5) equals one, so $w_{N,j}^{F,\theta}=1$ and $\sum_j w_{N,j}^{F,\theta}=J$. Substituting into Equation (S4) after differentiating gives

$$\begin{aligned}\nabla_\theta \hat{\ell}^1(\theta)\Big|_{\theta=\phi} &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta}\Big|_{\theta=\phi} \\ &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right)\Big|_{\theta=\phi}, \end{aligned} \tag{S9}$$

as claimed. This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. □

For $\theta \neq \phi$, the same calculation instead gives the normalized weighted path score

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right)}{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta}}. \tag{S10}$$

Growth of the unnormalized weight products alone does not establish a variance rate for this normalized expression. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite-J upper bound.

Lemma S2. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)\right), \tag{S11}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{g^\theta_{n,k_{n,j}}}{g^\phi_{n,k_{n,j}}} \tag{S12}$$

as shorthand for the selected measurement density ratios, with the reference parameter and resampling indices held fixed. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} &= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) && \text{(S13)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) && \text{(S14)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\theta}; \theta\right)}{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\phi}; \phi\right)} \right). && \text{(S15)}
\end{aligned}$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\begin{aligned}
\nabla_\theta \hat{\ell}^0(\theta) &:= \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) && \text{(S16)} \\
&= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} && \text{(S17)} \\
&= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} && \text{(S18)} \\
&= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} && \text{(S19)}
\end{aligned}$$

$$= \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right), \tag{S20}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. □

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n, the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^\theta(Z_n) = f_{Y_n|X_n}(y_n^* \mid X_n^\theta;\theta), \qquad W_\theta(Z_N) = \prod_{n=1}^{N} G_n^\theta(Z_n), \tag{S21}$$

$$\psi_\theta(Z_N) = \nabla_\theta \log W_\theta(Z_N), \qquad \Xi_\theta(Z_N) = \nabla_\theta^2 \log W_\theta(Z_N). \tag{S22}$$

Differentiate through X_n^θ in $G_n^\theta(Z_n)$, holding the simulator draws and resampling indices fixed.

Write $\rho_N^J = J^{-1}\sum_{j=1}^{J}\delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_\theta(Z_N)F(Z_N)\}}{\mathbb{E}\{W_\theta(Z_N)\}}. \tag{S23}$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_\theta| \le N\bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_\theta \ell(\theta) = \rho_N(\psi_\theta), \qquad \nabla_\theta \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_\theta). \tag{S24}$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1,\dots,N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \le q \le N$, write $\rho_q^J = J^{-1}\sum_{j=1}^{J}\delta_{\mathsf{Z}_{q,j}}$, where the $\mathsf{Z}_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $\mathsf{Z}_{N,j} = Z_{N,j}$. Write $V_q^\theta(\mathsf{Z}_q) = G_{q+1}^\theta(\mathsf{Z}_q)$ for $q = 0,\dots,N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^\theta$ append the next independent simulator innovation and its gradients for $0 \le q \le N-2$, and put $\mathcal{M}_N^\theta = \mathrm{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^\theta(\mu)(dz') = \frac{\int \mu(dz)V_q^\theta(z)\mathcal{M}_{q+1}^\theta(z,dz')}{\mu(V_q^\theta)}, \qquad 0 \le q < N. \tag{S25}$$

Conditionally on the current cloud, the J children are independent with common law $\Phi_{q+1}^\theta(\rho_q^J)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V_{N-1}^\theta = G_N^\theta$ followed by the identity mutation. Thus ρ_N^J is exactly the terminal empirical

measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \le q \le N$, let

$$Q^{\theta}_{q,N}F(z_q) = \mathbb{E}\left[F(\mathsf{Z}_N)\prod_{r=q}^{N-1} V^{\theta}_r(\mathsf{Z}_r)\,\middle|\,\mathsf{Z}_q = z_q\right], \tag{S26}$$

$$P^{\theta}_{q,N}F = \frac{Q^{\theta}_{q,N}F}{Q^{\theta}_{q,N}1}, \qquad g_{q,N} = \sup_{z,z'}\frac{Q^{\theta}_{q,N}1(z)}{Q^{\theta}_{q,N}1(z')}, \qquad b_{q,N} = \sup_{\mathrm{osc}(F)\le 1}\mathrm{osc}(P^{\theta}_{q,N}F), \tag{S27}$$

where $Q^{\theta}_{N,N}F = F$ and $\mathrm{osc}(F) = \sup_{z,z'}|F(z) - F(z')|$. Thus $0 \le b_{q,N} \le 1$. With

$$R_G = \frac{\bar{G}}{\underline{G}}, \qquad S_N = \sum_{q=0}^{N} g^3_{q,N}b_{q,N}, \qquad K_N = 32S_N, \qquad K^{\mathrm{pf}}_N = 32\sum_{r=0}^{N} R^{3r}_G, \tag{S28}$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \le Q^{\theta}_{q,N}1(z_q) \le \bar{G}^{N-q}, \qquad g_{q,N} \le R^{N-q}_G, \qquad K_N \le K^{\mathrm{pf}}_N. \tag{S29}$$

In particular, the constant used in the theorem is the closed-form quantity

$$K^{\mathrm{pf}}_N = \begin{cases} 32(N+1), & R_G = 1, \\ 32\dfrac{R^{3(N+1)}_G - 1}{R^3_G - 1}, & R_G > 1. \end{cases} \tag{S30}$$

Here is the constant calculation behind the tail bound used below. Let r^{DM}_N and β^{DM}_N denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r^{\mathrm{DM}}_N \le 4S_N, \qquad (\beta^{\mathrm{DM}}_N)^2 \le 4\sum_{q=0}^{N} g^2_{q,N}b^2_{q,N} \le 4S_N, \tag{S31}$$

where the last inequality uses $g_{q,N} \ge 1$ and $0 \le b_{q,N} \le 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \le \log 6$, then $6e^{-x} \ge 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \le 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \le 8x$ to obtain

$$\frac{r^{\mathrm{DM}}_N}{J}\{1 + h^{-1}(x)\} \le \frac{u^2}{8x}\{1 + 2x + 2\sqrt{x}\} \le u^2 \le \frac{u}{2},$$
$$\beta^{\mathrm{DM}}_N\sqrt{\frac{2x}{J}} \le 2\sqrt{S_N}\sqrt{\frac{2x}{J}} = \frac{u}{2}.$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr\left(|(\rho^J_N - \rho_N)(F)| > \mathrm{osc}(F)u\right) \le 6\exp\left(-\frac{Ju^2}{K_N}\right) \le 6\exp\left(-\frac{Ju^2}{K^{\mathrm{pf}}_N}\right). \tag{S32}$$

for every F with $\mathrm{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho^J_N - \rho_N)(F) = 0$ identically. Equations S28– S32 replace all unnamed model-dependent concentration constants.

Lemma S3 (Explicit concentration of the DMOP-1 gradient)**.** *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1\le i\le p} \operatorname{osc}(\psi_{\theta,i}), \qquad u_g(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6p}{\delta}}. \tag{S33}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6p/\delta)$*, then, with probability at least* $1-\delta$*,*

$$\|\nabla_\theta \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta \ell(\theta)\|_2 \le A_N(\theta)\sqrt{p}\, u_g(J,\delta). \tag{S34}$$

Moreover, Assumption (A3) gives $A_N(\theta) \le 2NG'(\theta)$*. Consequently, the explicit condition*

$$J \ge K_N^{\mathrm{pf}} \log\frac{6p}{\delta}\max\left\{4, \frac{4pN^2G'(\theta)^2}{\epsilon^2}\right\} \tag{S35}$$

implies that the left side of equation S34 is at most ϵ *with probability at least* $1-\delta$*.*

Proof. The lower bound on J makes $u_g(J,\delta) \le 1/2$. Apply equation S32 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1\le i\le p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \le A_N(\theta) u_g(J,\delta)$$

with probability at least $1-\delta$. Now use $\|v\|_2 \le \sqrt{p}\|v\|_\infty$ and equation S24. Finally, $|\psi_{\theta,i}| \le NG'(\theta)$, so $A_N(\theta) \le 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}\, u_g(J,\delta) \le \epsilon$ gives equation S35. □

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian)**.** *Suppose in addition that Assumption (A3) gives the entrywise bound* $|\Xi_{\theta,ab}(Z_N)| \le NG''(\theta)$ *for* $1 \le a, b \le p$*. Define*

$$d_p = \frac{p(p+3)}{2}, \qquad H_N(\theta) = 2p\{NG''(\theta) + 3N^2G'(\theta)^2\}, \qquad u_H(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6d_p}{\delta}}. \tag{S36}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6d_p/\delta)$*, then, with probability at least* $1-\delta$*,*

$$\|\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta^2 \ell(\theta)\|_{\mathrm{op}} \le H_N(\theta) u_H(J,\delta). \tag{S37}$$

Fix $0 < \kappa < \gamma$*. If*

$$\gamma I_p \preceq -\nabla_\theta^2 \ell(\theta) \preceq \Gamma I_p \quad \textit{and} \quad J \ge K_N^{\mathrm{pf}} \log\frac{6d_p}{\delta}\max\left\{4, \frac{H_N(\theta)^2}{\kappa^2}\right\}, \tag{S38}$$

then, with probability at least $1-\delta$*,*

$$(\gamma-\kappa) I_p \preceq -\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma+\kappa) I_p. \tag{S39}$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log\left\{\frac{1}{J}\sum_{j=1}^{J}\exp(a_j(\theta))\right\}, \qquad a_j(\theta) = \log w_{N,j}^{F,\theta}, \qquad a_j(\phi) = 0,$$

where C_ϕ is constant in θ. Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla_\theta^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N^J(\psi_\theta)\rho_N^J(\psi_\theta)^T. \tag{S40}$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial^2_{\theta_a\theta_b} W_\theta| = W_\theta|\Xi_{\theta,ab} + \psi_{\theta,a}\psi_{\theta,b}| \le \bar{G}^N\{NG''(\theta) + N^2G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E}W_\theta$ gives the population identity

$$\nabla_\theta^2 \ell = \rho_N(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N(\psi_\theta)\rho_N(\psi_\theta)^T. \tag{S41}$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1, and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2. Apply equation S32 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned}\|(\rho_N^J - \rho_N)(\psi)\|_2 &\le A_1\sqrt{p}\, u_H(J,\delta),\\ \|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\mathrm{op}} &\le pA_2 u_H(J,\delta).\end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p}\, NG'(\theta)$. Therefore

$$\begin{aligned}&\|\rho_N^J(\psi)\rho_N^J(\psi)^T - \rho_N(\psi)\rho_N(\psi)^T\|_{\mathrm{op}}\\ &\qquad \le 2\sqrt{p}\, NG'(\theta)\,\|(\rho_N^J - \rho_N)(\psi)\|_2 \le 4pN^2G'(\theta)^2 u_H(J,\delta).\end{aligned}$$

Combining the last three displays with equation S40 and equation S41 proves equation S37. Condition equation S38 makes its right side at most κ; Weyl's eigenvalue inequality then gives equation S39. □

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\widetilde{B} = Q\,\mathrm{diag}(\lambda_1,\dots,\lambda_p)Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q\,\mathrm{diag}([\lambda_1]_c^C,\dots,[\lambda_p]_c^C)Q^T, \qquad [x]_c^C = \min\{C, \max\{c, x\}\}. \tag{S42}$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \le v^TBv \le C\|v\|_2^2, \qquad C^{-1}\|v\|_2^2 \le v^TB^{-1}v \le c^{-1}\|v\|_2^2. \tag{S43}$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S38; equation S42 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let $\mathcal{D}$ be the convex region in Theorem 2, and set

$$G'_{\mathcal{D}} = \sup_{\theta\in\mathcal{D}} G'(\theta), \qquad J_0 = \left\lceil K_N^{\mathrm{pf}} \log\!\left(\frac{6pM}{\delta}\right) \max\left\{4, \frac{4pN^2(G'_{\mathcal{D}})^2}{\epsilon^2}\right\}\right\rceil. \tag{S44}$$

Thus it is enough to take $J \geq J_0$. With $g_m = -\nabla_\theta \hat{\ell}_J^{A,1}(\theta_m)$, define

$$\tau = \min\left\{m \in \{0,\dots,M-1\} : \|g_m\|_2 \leq \sigma\epsilon\right\}, \qquad \min\varnothing = M, \tag{S45}$$

and

$$\mathcal{E}_M = \bigcap_{m=0}^{\min\{\tau,M-1\}} \left\{\|g_m - \nabla f(\theta_m)\|_2 \leq \epsilon\right\}. \tag{S46}$$

If $\mathcal{F}_m$ is the history before the mth particle-filter call, then θ_m and $\{m \leq \min(\tau, M-1)\}$ are $\mathcal{F}_m$-measurable. Lemma S3, with failure probability δ/M, gives

$$\begin{aligned}\Pr(\mathcal{E}_M^c \mid \theta_0) &\leq \sum_{m=0}^{M-1} \mathbb{E}\left[\mathbf{1}_{\{m\leq\min(\tau,M-1)\}} \Pr\left(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m\right) \middle| \theta_0\right] \\ &\leq \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta.\end{aligned} \tag{S47}$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1} g_m$. If $\|g_m\|_2 \leq \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon \leq (\sigma+1)\epsilon. \tag{S48}$$

Strong convexity implies $f(\theta) - f(\theta^*) \leq \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \leq \frac{(\sigma+1)^2\epsilon^2}{2\gamma}. \tag{S49}$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \leq \|g_m\|_2 \leq C\|p_m\|_2, \qquad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \tag{S50}$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned}f(\theta_m + \eta p_m) - f(\theta_m) &\leq \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2 \\ &\leq -\eta p_m^T B_m p_m + \eta\epsilon\|p_m\|_2 + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2.\end{aligned} \tag{S51}$$

The choices of σ and η imply

$$\frac{C}{\sigma}\|p_m\|_2^2 \leq \frac{c(1-\beta)}{2}\|p_m\|_2^2, \qquad \frac{\Gamma\eta}{2}\|p_m\|_2^2 \leq \frac{c(1-\beta)}{2}\|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c\|p_m\|_2^2$ in equation S51 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \tag{S52}$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma+1}{\sigma}\|g_m\|_2. \tag{S53}$$

Consequently, equation S43 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \tag{S54}$$

Combining equation S52 and equation S54 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q\{f(\theta_m) - f(\theta^*)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2. \tag{S55}$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1-\beta)\frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \qquad \text{so} \qquad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S49 gives its stopping conclusion after substituting $f = -\ell$. □

The result is deliberately restricted to DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$ be the prediction and filtering probability measures on the current state space $\mathcal{X}$. The posterior measure ρ_n is on simulator histories through time n, including the base variables, as defined in Equation (S123). Its state-path marginal is $f_{X_{0:n}|Y_{1:n}}$. These are the measures we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0,1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h \, d\eta = \int h(x)\eta(dx), \tag{S56}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{S57}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{S58}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \tag{S59}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(h) - \nu(h)| = \sup_{\mathrm{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \tag{S60}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y \in \mathcal{X}} \big\|M(x, \cdot) - M(y, \cdot)\big\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{S61}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y_n^*|x_{n,j}^P; \theta\big) = G_n(x_{n,j}^P)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*| \cdot)$ is the conditional density of the observed measurement at time n, evaluated at a state in $\mathcal{X}$. At the filtered particle, $G_n(x_{n,j}^F) = g_{n,k_{n,j}}$.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{S62}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{S63}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ ... \circ \Phi_{k+1}. \tag{S64}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^P}, \tag{S65}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^F}. \tag{S66}$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. For $\theta = \phi$ and multinomial resampling, the corresponding Markov transition kernel is

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \left(\Phi_n\left(\eta_{n-1}^J\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J} \delta_{x_{n-1,j}^P}\right)\right)^{\otimes J}, \tag{S67}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \tag{S68}$$

The kernel acts on the full prediction cloud. Conditional on the preceding prediction cloud, its law is

$$\text{Law}\left(x_{n,1:J}^P \mid x_{n-1,1:J}^P\right) = \mathbf{M}_n(x_{n-1,1:J}^P, \cdot). \tag{S69}$$

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP-α. That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J))\rfloor}, \tag{S70}$$

for some ϵ dependent on $\bar{G}, \underline{\text{G}}, \bar{M}, \underline{\text{M}}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP-α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general results, from which it will be clear that MOP-α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters)**.** *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model* $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$*, where we inductively define given some* $x_{n-1,j}^F$*,* $w_{n-1,j}^F$ *comprising the filtering measure approximation at time* $n-1$*:*

1. *Conditional on the filtered particles and weights at time* $n-1$*, draw independently* $x_{n,j}^P \sim M_n(x_{n-1,j}^F, \cdot)$ *using fresh simulator draws. Retain the corresponding parent weight:* $w_{n,j}^P = w_{n-1,j}^F$*.*
2. *The prediction measure at timestep* n *is given by*
$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \, \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \tag{S71}$$
3. *The particles are resampled at every timestep* n*, yielding indices* k_j *according to some arbitrary probabilities* $(p_{n,j})_{j=1}^J$*.*
4. *The filtering measure at timestep* n *is given by*
$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P \, g_{n,j}}{\sum_{j=1}^J w_{n,j}^P \, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \, \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \tag{S72}$$
5. *The prediction weights at the next timestep are given by* $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}$*.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting)**.** *A pair of random vectors* (X, W) *drawn from some measure* g ***targets*** *another measure* π *if, for any measurable and bounded functional* h*,*
$$E_g\big[h(X) \cdot W\big] = E_\pi\big[h(X)\big]. \tag{S73}$$
A particle approximation is a triangular array of pairs of random vectors $(X_j^J, W_j^J), j = 1, 2, \ldots, J$*. The particle approximation* ***targets*** π *if, for any measurable and bounded functional* h*,*
$$\frac{\sum_{j=1}^J h(X_j^J) \, W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{a.s.} E_\pi\big[h(X)\big] \tag{S74}$$
as $J \to \infty$*.*

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure)**.** *Suppose that* $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Now, let* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *be a multinomial sample with indices* k_j *drawn from* $\{(\tilde{X}_j^J, U_j^J)\}$ *where* $(\tilde{X}_j^J, U_j^J)$ *is represented, on average, proportional to* $\pi_j^J J$ *times, with* $\pi_j^J > 0$*. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{S75}$$

If the importance sampling weights U_j/π_j *are bounded uniformly in* j, J *after normalizing* $\sum_m U_m^J = 1$*, then* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *targets* f_X*.*

Proof. Normalize $\sum_j U_j^J = 1$, which leaves both weighted empirical measures unchanged. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{S76}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\big]. \tag{S77}$$

We want to show

$$\frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{S78}$$

For this, it is sufficient to show that

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}\pi_j^J \overset{a.s.}{\to} 0 \tag{S79}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence to 1 of J^{-1} times the denominator in (S78). Write $g_j^J = h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\frac{1}{J}\sum_j Z_j^J := \frac{1}{J}\sum_j \left(g_{k_j}^J - \sum_{m=1}^J g_m^J \pi_m^J \right) \overset{a.s.}{\to} 0. \tag{S80}$$

Because the functional h and importance sampling weights $U_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\sup_{J,j} \mathbb{E}\big[(Z_j^J)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent, identically distributed, and centered given the $\tilde{X}_j^J$, U_j^J, and π_j^J,

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_j Z_j^J\right)^4 \middle| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\right] &= J\,\mathbb{E}\left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\right] \\
&\quad + 3J(J-1)\left(\mathbb{E}\big[(Z_1^J)^2 \big| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\big]\right)^2 \quad \text{(S81)} \\
&\leq CJ^2, \quad \text{(S82)}
\end{aligned}$$

for some $C > 0$ independent of J. Taking expectations on both sides yields

$$\mathbb{E}\left[\left(\sum_j Z_j^J\right)^4\right] \leq CJ^2 \quad \text{(S83)}$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P}\left(\left|\frac{1}{J}\sum_{j=1}^J Z_j^J\right| > \epsilon\right) \leq \frac{1}{\epsilon^4 J^4}\mathbb{E}\left[\left(\sum_{j=1}^J Z_j^J\right)^4\right] \leq \frac{C}{\epsilon^4 J^2}, \quad \text{(S84)}$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\,\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\,\frac{U_j^J}{\pi_j^J}\,\pi_j^J \overset{a.s.}{\to} 0. \quad \text{(S85)}$$

Similarly, we also have that

$$\sum_j \frac{U_j^J}{\pi_j^J}\pi_j^J - \frac{1}{J}\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J} \overset{a.s.}{\to} 0, \quad \text{(S86)}$$

and the result follows by taking the ratio, since $\sum_j U_j^J = 1$.

□

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ targets f_X. Also suppose that conditional on $(\tilde{X}_m^J, U_m^J)_{m=1}^J$, the variables $\tilde{Z}_1^J, \ldots, \tilde{Z}_J^J$ are independent, with $\tilde{Z}_j^J \sim f_{Z|X}(\cdot \mid \tilde{X}_j^J)$, where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U_j^J are uniformly bounded above and away from zero, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$\frac{\sum_j h(\tilde{Z}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\big[\mathbb{E}_{f_{Z|X}}[h(Z) \mid X]\big]. \tag{S87}$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{S88}$$

Let $\bar{U}_j^J = JU_j^J / \sum_j U_j^J$. For constants $0 < c \le U_j^J \le C < \infty$, uniformly in j, J, we have $\bar{U}_j^J \le C/c$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} = \frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J}{J^{-1}\sum_j \bar{U}_j^J}. \tag{S89}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J. \tag{S90}$$

Set

$$D_J = \frac{1}{J}\sum_j \bar{U}_j^J \big\{h(\tilde{Z}_j^J) - \mathbb{E}[h(Z) \mid X = \tilde{X}_j^J]\big\}.$$

Conditioning on $(\tilde{X}_j^J, U_j^J)_{j=1}^J$, we apply the fourth-moment argument of Lemma S5 to the centered, uniformly bounded summands, obtaining

$$\mathbb{E}[D_J^4] \le \frac{C_h}{J^2}, \qquad \mathbb{P}(|D_J| > \epsilon) \le \frac{C_h}{\epsilon^4 J^2},$$

for some $C_h < \infty$ independent of J and every $\epsilon > 0$. Thus $D_J \overset{a.s.}{\to} 0$ by Borel–Cantelli. For the expectation of (S90), the tower property gives

$$\lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J\right] = \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\big[h(\tilde{Z}_j^J)\, \bar{U}_j^J \mid \tilde{X}_j^J, \bar{U}_j^J\big]\right] \tag{S91}$$

$$= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\big[h(Z) \mid X = \tilde{X}_j^J\big]\, \bar{U}_j^J\right]. \tag{S92}$$

Applying (S88) with $g(x) = \mathbb{E}[h(Z) \mid X = x]$, the average on the right hand side converges almost surely to $\mathbb{E}\{\mathbb{E}[h(Z) \mid X]\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem. The almost-sure convergence of this average, together with $D_J \overset{a.s.}{\to} 0$, gives

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{S93}$$

□

Remark. The conditional independence assumption in Lemma S6 holds for standard particle filters, and for our MOP filter it is guaranteed by step 1 of Definition S1.

Remark: The uniform boundedness condition in Lemma S6 can be guaranteed by construction in the context of Definition S1 for suitable choices of the resampling probabilities $\pi_{1:J}$ at an earlier iteration. In the context of DMOP-α, this is guaranteed if the measurement density is similarly bounded, which holds under Assumption (A2) for our theoretical analysis. It is trivially true for MOP-α at $\theta = \phi$, since in this case the weights at each propagation step are equal. The smoothness assumption (A5) therefore also guarantees this condition in a neighborhood of $\theta = \phi$.

Remark. For nonnegative weights with positive sum, the weight condition in Lemma S6 can be replaced by the weaker requirement

$$\max_{1\le j\le J} \frac{JU_j^J}{\sum_m U_m^J} \le B < \infty,$$

almost surely, for a constant B independent of J. The same proof applies.

Lemma S7 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$*. Then, if* $x \mapsto f_{Z|X}(z^*|x)$ *is bounded and* $0 < f_Z(z^*) < \infty$*,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$*.*

Proof. Again, we want to show that, for any measurable bounded h,

$$\frac{\sum_j h(X_j^J)\cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{S94}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\, U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{S95}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{S96}$$

The functions $x \mapsto f_{Z|X}(z^*|x)$ and $x \mapsto h(x) f_{Z|X}(z^*|x)$ are bounded. We will apply Equation (S95) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big], \tag{S97}$$

while the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{S98}$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} \;=\; \mathbb{E}_{f_X}\left[\frac{h(X)\, f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] \tag{S99}$$

$$= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\left[h(X)|z^*\right], \tag{S100}$$

to give the desired result.

□

Remark. In the MOP application, the boundedness and positivity conditions in Lemma S7 follow from Assumption (A3).

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter of Definition S1 targets the filtering distribution when the resampling probabilities $(p_{n,j})_{j=1}^J$ are chosen so that $g_{n,j}/(J\,p_{n,j})$ is uniformly bounded above and away from zero whenever $g_{n,j} > 0$.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X_{n-1,j}^F, w_{n-1,j}^F\right)\right\}$ targets π_{n-1}. Then, Lemma S6 tells us that $\left\{\left(X_{n,j}^P, w_{n,j}^P\right)\right\}$ targets η_n. Lemma S7 tells us that $\left\{\left(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta\right)\right\}$ therefore targets π_n. Lemma S5 guarantees that the resampling rule, given by

$$\left(X_{n,j}^F, w_{n,j}^F\right) = \left(X_{n,k_j}^P, w_{n,k_j}^P\, g_{n,k_j}/p_{n,k_j}\right), \tag{S101}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition S1 (MOP-1 Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, MOP-α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta}\right)\right\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*;\theta)$. Then, Lemma S6 tells us that $\left\{\left(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\right)\right\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. Lemma S7 tells us that $\left\{\left(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\, g_{n,j}^\theta\right)\right\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. Lemma S5 guarantees that the resampling rule, given by

$$\left(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}\right) = \left(X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta}\, g_{n,k_j}^\theta/g_{n,k_j}^\phi\right), \tag{S102}$$

with resampling weights proportional to $g_{n,j}^\phi$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting)**.** *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1_n-1}^*;\theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta\, w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \tag{S103}$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}}. \tag{S104}$$

where L_n^{ϕ} is as defined in the various algorithms.

Here, (S103) is a direct consequence of our earlier result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. To see (S104), define the corrected weights before resampling and the reference resampling probabilities by

$$w_{n,j}^{FC,\theta} = w_{n,j}^{P,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}}, \qquad p_{n,j} = \frac{g_{n,j}^{\phi}}{JL_n^{\phi}}, \qquad \sum_{j=1}^{J} p_{n,j} = 1. \tag{S105}$$

The numerator of (S103) can then be written as

$$\sum_{j=1}^{J} w_{n,j}^{P,\theta} g_{n,j}^{\theta} = JL_n^{\phi} \sum_{j=1}^{J} p_{n,j} w_{n,j}^{FC,\theta}. \tag{S106}$$

Resampling selects $w_{n,j}^{F,\theta} = w_{n,k_j}^{FC,\theta}$. With the pre-resampling particles, weights, and reference densities fixed, the expectation over multinomial resampling satisfies

$$\begin{aligned} \mathbb{E}_{\text{res}}\left[L_n^{\phi} \sum_{j=1}^{J} w_{n,j}^{F,\theta} \right] &= JL_n^{\phi} \sum_{m=1}^{J} p_{n,m} w_{n,m}^{FC,\theta} \\ &= \sum_{m=1}^{J} w_{n,m}^{P,\theta} g_{n,m}^{\theta}. \end{aligned} \tag{S107}$$

The same conditional triangular-array argument as in Lemma S5 gives consistency of this resampling approximation under its boundedness conditions. Dividing by the unchanged prediction-weight sum gives (S104). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^{N} \left(L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} \right). \tag{S108}$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^{J} w_{0,j}^{F,\theta} = J$ on the denominator and $\sum_{j=1}^{J} w_{N,j}^{F,\theta}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now show that, under sufficient local regularity conditions, one can interchange differentiation and the particle limit. We retain the Arzelà–Ascoli argument, working on a compact neighborhood

of the evaluation parameter. Consistency is used for the estimates themselves: differentiating a weighted particle estimate also differentiates its particles, weights, and normalizer, so boundedness alone does not make its derivative an empirical average to which Theorem S1 applies.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let K be a closed ball of positive radius centered at ϕ and contained in the interior of $\Theta \subset \mathbb{R}^p$. Let $F_J(\theta)$ be scalar particle estimates, such as $\eta_n^J(h_\theta)$, defined on a common probability space. On one probability-one event, suppose every F_J has two continuous derivatives on a neighborhood of K, with*

$$\sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta F_J(\theta)\|_2 \leq G^*, \qquad \sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta^2 F_J(\theta)\|_{\mathrm{op}} \leq H^*$$

for finite constants G^*, H^*. Suppose $F_J(\theta) \to h_\theta^*$ almost surely for each fixed $\theta \in K$, where h^* is deterministic. Then h^* is continuously differentiable in the interior of K, and almost surely

$$\nabla_\theta F_J(\theta) \longrightarrow \nabla_\theta h_\theta^*$$

uniformly on every compact subset of that interior, in particular at $\theta = \phi$.

Proof. For each fixed pair $x, y \in K$, pointwise almost-sure convergence and the gradient bound give $|h_y^* - h_x^*| \leq G^*\|y - x\|_2$. Thus the deterministic function h^* is Lipschitz, and in particular continuous, on K.

Choose a countable dense subset D of K. Intersect the probability-one event of the derivative bounds with the events of convergence at each $\theta \in D$, and fix ω in this intersection. Suppress ω below. By the gradient bound, the functions F_J are uniformly Lipschitz on the convex set K. Convergence on D, compactness of K, and continuity of h^* therefore imply $F_J \to h^*$ uniformly on K: for any $\epsilon > 0$, choose a sufficiently fine finite net in D, use convergence at its finitely many points, and use the common Lipschitz bound and uniform continuity of h^* between those points.

The gradients are uniformly bounded and, by the Hessian bound, uniformly equicontinuous. Explicitly, for $\theta, \theta' \in K$,

$$\|\nabla_\theta F_J(\theta) - \nabla_\theta F_J(\theta')\|_2 \leq H^*\|\theta - \theta'\|_2. \tag{S109}$$

Thus, by Arzelà–Ascoli, every subsequence of gradients has a further subsequence converging uniformly on K to a continuous function g. We claim that every such limit is $\nabla_\theta h^*$ in the interior of K. For $x, y \in K$, the fundamental theorem of calculus gives

$$F_J(y) - F_J(x) = \int_0^1 \nabla_\theta F_J\big(x + t(y-x)\big)^{\mathsf{T}}(y-x)\,dt. \tag{S110}$$

Passing to the limit along the convergent subsequence yields

$$h_y^* - h_x^* = \int_0^1 g\big(x + t(y-x)\big)^{\mathsf{T}}(y-x)\,dt. \tag{S111}$$

Taking $y = x + te_i$ for each coordinate direction and letting $t \to 0$ shows that $\partial_{\theta_i} h_x^* = g_i(x)$ for every interior x. Since g is continuous, h^* is continuously differentiable there. Every subsequential gradient limit is therefore the same on the interior, and hence on K by continuity. Relative compactness and uniqueness of this limit imply convergence of the full gradient sequence. In particular, almost surely, for interior θ,

$$\lim_{J\to\infty} \nabla_\theta F_J(\theta) = \nabla_\theta \lim_{J\to\infty} F_J(\theta) = \nabla_\theta h_\theta^*, \tag{S112}$$

with convergence uniform on every compact subset of the interior, as required. □

For $F_J(\theta) = \eta_n^J(h_\theta)$, the conclusion is convergence to $\nabla_\theta \eta_n(h_\theta)$. No identity with $\eta_n(\nabla_\theta h_\theta)$ is asserted, since the target distribution can itself depend on θ.

Theorem 4 follows by taking $F_J(\theta) = \hat{\ell}_J^1(\theta)$ and $h_\theta^* = \ell(\theta)$. Keep the reference parameter ϕ fixed throughout K. For $\alpha = 1$, the off-parameter consistency result and Lemma S8 give $\hat{\mathcal{L}}_J^1(\theta) \to \mathcal{L}(\theta) > 0$ almost surely for each fixed $\theta \in K$, not only at $\theta = \phi$. The continuous mapping theorem gives the corresponding convergence of the log likelihoods. The neighborhood form of Assumption (A4) supplies their uniform gradient and Hessian bounds, and Assumption (A5) supplies their pathwise smoothness with the simulator draws and reference-filter resampling indices held fixed. Applying the theorem to these log likelihoods establishes score consistency at ϕ directly.

S6 The case $\alpha = 0$

[KT:At $\alpha = 0$, the weight gradient contains only the current measurement term. This gives a simple preliminary case for the general argument in Section S7. Let $h_{n,j}$ be the full pathwise derivative of $\log f_{Y_n|X_n}(y_n^* \mid X_{n,j}^{P,\theta}; \theta)$ at $\theta = \phi$, including the derivative of the simulated state. Write $W_{n,j} = g_{n,j}^\phi / \sum_l g_{n,l}^\phi$, and let $K_{n,j}$ be independent draws from these weights conditional on the history $\mathcal{F}_n^-$ just before resampling.

Lemma S9 (Current measurement gradients)**.** *At $\theta = \phi$, the before- and after-resampling scores satisfy*

$$\nabla_\theta \hat{\ell}_J^{B,0} = \sum_{n=1}^N \sum_{j=1}^J W_{n,j} h_{n,j}, \qquad \nabla_\theta \hat{\ell}_J^{A,0} = \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J h_{n,K_{n,j}}. \tag{S113}$$

Under (A3), their difference has mean zero and

$$\mathbb{E} \left\| \nabla_\theta \hat{\ell}_J^{A,0} - \nabla_\theta \hat{\ell}_J^{B,0} \right\|_2^2 \le NpG'(\theta)^2 / J. \tag{S114}$$

Proof. The prediction weights are one when $\alpha = 0$. Differentiating the weighted measurement average, or its resampled average, gives Equation (S113). Put $\bar{h}_n = \sum_j W_{n,j} h_{n,j}$ and $\varepsilon_n = J^{-1} \sum_j h_{n,K_{n,j}} - \bar{h}_n$. Conditional multinomial sampling gives

$$\begin{aligned} \mathbb{E}(\varepsilon_n \mid \mathcal{F}_n^-) &= 0, \\ \mathbb{E}(\|\varepsilon_n\|_2^2 \mid \mathcal{F}_n^-) &= \frac{1}{J} \sum_j W_{n,j} \|h_{n,j} - \bar{h}_n\|_2^2 \le \frac{pG'(\theta)^2}{J}. \end{aligned} \tag{S115}$$

For $m < n$, ε_m is $\mathcal{F}_n^-$-measurable, so the cross terms in $\mathbb{E}\| \sum_n \varepsilon_n \|_2^2$ vanish. Summing the conditional bounds proves the result. □

This calculation controls the extra resampling error. Controlling error relative to the population score also requires control of the particle histories. Under (A6), we decompose each measurement gradient according to the age of its parameter dependence; under (A6-J), it is a function of the current state and its parameter derivative. Lemma S13 gives, for either estimator at $\alpha = 0$,

$$\mathbb{E}\|\nabla_\theta \hat{\ell}_J^0 - \nabla_\theta \ell^0\|_2^2 \lesssim pG'(\theta)^2 \left(\frac{N}{J} + \frac{N^2}{J^2} \right).$$

The difference between the population score $\nabla_\theta \ell^0$ and the full likelihood score is treated separately in the next section.]

S7 Proof of Theorem 5 (Bias-Variance Analysis)

[KT:Except where we explicitly average over observations, the data are fixed and expectations are over particle-filter randomness. At $\theta = \phi$, add and subtract the population discounted score:

$$\begin{aligned}&\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha - \nabla_\theta \ell\|_2^2 \\ &\quad \le 2\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha - \nabla_\theta \ell^\alpha\|_2^2 + 2\|\nabla_\theta \ell^\alpha - \nabla_\theta \ell\|_2^2. \end{aligned} \tag{S116}$$

The first term is particle error and the second is population discounting error. We first bound the population term for fixed observations and under the data-generating law, then bound the particle term. Write $\nabla_\theta \ell_n^\alpha(\theta)$ for the nth conditional score of the population recursion. At $\theta = \phi$ and $\alpha = 1$, it equals $\nabla_\theta \ell_n(\theta)$. Along each selected lineage,

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \alpha\, \nabla_\theta \log w_{n-1,k_j}^{F,\theta}\Big|_{\theta=\phi} + \nabla_\theta \log g_{n,k_j}^{\theta}\Big|_{\theta=\phi}. \tag{S117}$$

The last term is the measurement gradient h_n defined in Equation (S119). This follows by differentiating

$$\begin{aligned}\log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta}, \\ \log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^{\theta} - \log g_{n,k_j}^{\phi}. \end{aligned} \tag{S118}$$

at $\theta = \phi$.

Score identities. Differentiation holds the simulator draws and reference resampling indices fixed and includes the dependence of each state on θ. The measurement gradient on a simulator history satisfies

$$h_k = \nabla_\theta \log g_k^{A,F,\theta}\Big|_{\theta=\phi}, \tag{S119}$$

$$\|h_k\|_\infty \le G'(\theta). \tag{S120}$$

For fixed t, the product rule bounds every coordinate of the differentiated likelihood by

$$\left\|\nabla_\theta \prod_{s=1}^{t} G_s(X_s)\right\|_\infty \le \prod_{s=1}^{t} G_s(X_s) \sum_{s=1}^{t} \|\nabla_\theta \log G_s(X_s)\|_\infty \le t\bar{G}^t G'(\theta). \tag{S121}$$

The same local bound controls difference quotients by the mean-value theorem, so dominated convergence gives

$$\nabla_\theta \mathbb{E}_U\left[\prod_{s=1}^{t} G_s(X_s)\right] = \mathbb{E}_U\left[\nabla_\theta \prod_{s=1}^{t} G_s(X_s)\right]. \tag{S122}$$

Here $\mathbb{E}_U$ integrates the independent base simulator draws. At $\theta = \phi$, their posterior law through time t is

$$\rho_t(h) = \frac{\mathbb{E}_U[h \prod_{s=1}^{t} G_s(X_s)]}{\mathbb{E}_U[\prod_{s=1}^{t} G_s(X_s)]}. \tag{S123}$$

It includes the derivative history needed to evaluate h_k, and its current-state marginal is π_t. Particle-error expectations instead integrate the particle-filter law, including resampling. Let

$$m(\xi_t) = \frac{1}{J}\sum_{j=1}^{J} \delta_{\xi_{t,j}}, \qquad m(\xi_t)(h_k) = \frac{1}{J}\sum_{j=1}^{J} h_k(\xi_{t,j}), \quad 1 \le k \le t, \tag{S124}$$

where $\xi_{t,j}$ is the simulator history along particle j's ancestry. Iterating the filtered-weight recursion gives

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r h_{n-r}(\xi_{n,j}). \tag{S125}$$

The prediction step preserves j, whereas selection enters only in the filtered weight:

$$w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha, \qquad w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta}\frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \tag{S126}$$

All these weights equal one at $\theta = \phi$. Differentiating the two sample averages defining the after-resampling conditional likelihood therefore yields

$$\nabla_\theta \hat{\ell}_n^\alpha(\theta) = \frac{1}{J}\sum_j \nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} - \frac{\alpha}{J}\sum_j \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi} \tag{S127}$$

$$= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}. \tag{S128}$$

Define the particle and population lag coefficients by

$$\begin{aligned} \Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \le r < n. \end{aligned} \tag{S129}$$

Differentiation under the simulator expectation gives

$$\nabla_\theta \log \mathbb{E}_U\left[\prod_{s=1}^{n} G_s(X_s)\right] = \sum_{s=1}^{n} \rho_n(h_s). \tag{S130}$$

Subtracting its time-$(n-1)$ version, and differentiating the two population weighted averages in the MOP-α recursion, respectively, give

$$\nabla_\theta \ell_n(\theta) = \rho_n(h_n) + \sum_{r=1}^{n-1}\{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} = \sum_{r=0}^{n-1} \Delta_{n,r}, \tag{S131}$$

$$\nabla_\theta \ell_n^\alpha(\theta) = \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r\{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} = \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}. \tag{S132}$$

]
[KT:

Lemma S10 (Full-score summation by parts)**.** *For $\theta = \phi$ and $\alpha \in [0,1]$, the after-resampling score and its population counterpart satisfy*

$$\nabla_\theta \hat{\ell}^\alpha = (1-\alpha) \sum_{t=1}^{N-1} m(\xi_t) \left(\sum_{k=1}^{t} \alpha^{t-k} h_k \right) + m(\xi_N) \left(\sum_{k=1}^{N} \alpha^{N-k} h_k \right), \tag{S133}$$

$$\nabla_\theta \ell^\alpha = (1-\alpha) \sum_{t=1}^{N-1} \rho_t \left(\sum_{k=1}^{t} \alpha^{t-k} h_k \right) + \rho_N \left(\sum_{k=1}^{N} \alpha^{N-k} h_k \right). \tag{S134}$$

For every coordinate i and $t \le N$,

$$\left\| \sum_{k=1}^{t} \alpha^{t-k} h_{k,i} \right\|_\infty \le G'(\theta) \sum_{k=1}^{t} \alpha^{t-k}. \tag{S135}$$

Proof. Summing Equation (S128) by creation time gives

$$\nabla_\theta \hat{\ell}^\alpha = \sum_{k=1}^{N} \left[m(\xi_k)(h_k) + \sum_{t=k+1}^{N} \alpha^{t-k} \{ m(\xi_t)(h_k) - m(\xi_{t-1})(h_k) \} \right]. \tag{S136}$$

For each k, collect the coefficient at every endpoint:

$$\begin{aligned} & m(\xi_k)(h_k) + \sum_{t=k+1}^{N} \alpha^{t-k} \{ m(\xi_t)(h_k) - m(\xi_{t-1})(h_k) \} \\ &= (1-\alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \end{aligned} \tag{S137}$$

Interchanging the finite sums proves the particle identity. Replacing $m(\xi_t)$ by ρ_t proves the population identity. The coordinate bound follows from $\|h_{k,i}\|_\infty \le G'(\theta)$. □

]

[KT:

Lemma S11 (Lagwise first-derivative forgetting)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0,1)$ such that*

$$\|\Delta_{n,r}\|_2 \le C \sqrt{p}\, G'(\theta) \varrho^r, \qquad 0 \le r < n. \tag{S138}$$

Proof. Fix $1 \le r < n$, put $k = n - r$, and consider coordinate i. We first integrate out the history in $h_{k,i}$. Set

$$\bar{h}_{k,i}(X_k) = \mathbb{E}_{\rho_k}(h_{k,i} \mid X_k), \tag{S139}$$

$$L_{k,t}(X_k) = \mathbb{E}_U \left[\prod_{s=k+1}^{t} G_s(X_s) \,\middle|\, X_k \right], \qquad L_{k,k} = 1. \tag{S140}$$

Future simulator draws are independent of the history given X_k. Thus, for every bounded q and $t \ge k$,

$$\rho_t\{h_{k,i} q(X_k)\} = \frac{\rho_k\{h_{k,i} q(X_k) L_{k,t}(X_k)\}}{\rho_k(L_{k,t})} = \frac{\rho_k\{\bar{h}_{k,i} q(X_k) L_{k,t}(X_k)\}}{\rho_k(L_{k,t})}. \tag{S141}$$

Consequently $\mathbb{E}_{\rho_t}(h_{k,i} \mid X_k) = \bar{h}_{k,i}(X_k)$. This conditional independence also holds given a later state, so the physical backward kernels propagate the projected historical score. They are

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S142}$$

With $\epsilon = \underline{\mathrm{M}}/\bar{M}$, the transition bounds imply

$$B_s(x_s, \cdot) \geq \epsilon \pi_{s-1}(\cdot), \qquad \operatorname{osc}(B_s f) \leq (1-\epsilon)\operatorname{osc}(f). \tag{S143}$$

Indeed, subtracting the common mass $\epsilon\pi_{s-1}$ leaves $(1-\epsilon)$ times a probability kernel. It follows that

$$H(x) := \mathbb{E}_{\rho_{n-1}}(h_{k,i} \mid X_{n-1} = x) = (B_{n-1} \cdots B_{k+1} \bar{h}_{k,i})(x), \tag{S144}$$

$$\operatorname{osc}(H) \leq 2G'(\theta)(1-\epsilon)^{r-1}. \tag{S145}$$

An empty kernel product is the identity, and a zeroth power here is one, including when $\epsilon = 1$.

Integrating the last simulator draw gives the posterior update

$$\rho_n(h_{k,i}) = \frac{\rho_{n-1}\{h_{k,i} M_n(G_n)\}}{\rho_{n-1} M_n(G_n)}, \qquad M_n(G_n)(x) = \int M_n(x, du) G_n(u). \tag{S146}$$

Writing the numerator using H therefore gives

$$\Delta_{n,r,i} = \frac{\pi_{n-1}[(H - \pi_{n-1}H)\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1} M_n(G_n)}. \tag{S147}$$

The denominator is at least $\underline{\mathrm{G}}$. Also, $M_n(x, \cdot) \geq \epsilon M_n(x^\star, \cdot)$ for any fixed $x^\star$, so subtracting this common mass shows that

$$\operatorname{osc}\{M_n(G_n)\} \leq (1-\epsilon)(\bar{G} - \underline{\mathrm{G}}). \tag{S148}$$

Combining the last three displays with $|f - \pi_{n-1} f| \leq \operatorname{osc}(f)$ gives

$$|\Delta_{n,r,i}| \leq \frac{2(\bar{G} - \underline{\mathrm{G}})}{\underline{\mathrm{G}}} G'(\theta)(1-\epsilon)^r. \tag{S149}$$

Choose

$$\varrho = 1 - \frac{\underline{\mathrm{M}}}{2\bar{M}} \in (0,1). \tag{S150}$$

Since $1 - \epsilon \leq \varrho$, summing squared coordinate bounds gives the result for $r \geq 1$. For $r = 0$, $\|\rho_n(h_n)\|_2 \leq \sqrt{p}\, G'(\theta)$. If $\epsilon = 1$, $M_n(G_n)$ is constant and every positive-lag coefficient is zero. □

]

[KT:

Lemma S12 (Population discounting bias)**.** *Under the conditions of Lemma S11, suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then*

$$\|\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\|_2 \leq C\sqrt{p}\, G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r)\varrho^r$$

$$\leq C\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \tag{S151}$$

For the full score,

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad\leq C\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q \\
&\quad= C\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\frac{(N-1)-N\alpha\varrho+(\alpha\varrho)^N}{(1-\alpha\varrho)^2} \\
&\quad\leq C(N-1)\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}.
\end{aligned} \tag{S152}$$

For $N = 1$, the finite sum is empty, the displayed fraction is zero, and the full-score difference is zero. The dependence on the transition bounds can be sharpened further. With the existing $\varrho = 1 - \underline{M}/(2\bar{M})$ from Equation (S150), for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad\leq 2\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)\left\{\alpha\frac{2\varrho-1}{3-2\varrho}\right\}^q \\
&\quad\leq 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.
\end{aligned} \tag{S153}$$

If every transition is independent of its starting location, in particular if $\varrho = \frac{1}{2}$, the population discounting bias is zero for every N and α.

Proof. The score identities give the exact difference

$$\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) = \sum_{r=1}^{n-1}(1-\alpha^r)\Delta_{n,r}. \tag{S154}$$

In particular, $\nabla_\theta \ell_n^0(\theta) = \Delta_{n,0}$, whereas $\alpha = 1$ gives the exact score. Lemma S11 and the geometric-series identity

$$\sum_{r=1}^{\infty}(1-\alpha^r)\varrho^r = \frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho} = \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \tag{S155}$$

prove the conditional-score bound.

For the full score, summation by parts retains the cancellation across observation times. The contribution of h_s to the discounted score is

$$\rho_s(h_s) + \sum_{q=1}^{N-s}\alpha^q\{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} = (1-\alpha)\sum_{q=0}^{N-s-1}\alpha^q\rho_{s+q}(h_s) + \alpha^{N-s}\rho_N(h_s). \tag{S156}$$

Subtract this from $\rho_N(h_s)$ and sum over s to obtain

$$\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) = (1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}\alpha^q\{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S157}$$

We bound each comparison directly, using a likelihood-ratio inequality. For a probability measure μ and $0 < a \le L \le b$, put $m = \mu L$. When $a < b$, the chord bound for the convex function $|L - m|$ gives

$$\mu|L - m| \le \frac{2(b-m)(m-a)}{b-a}. \tag{S158}$$

Subtracting the midpoint of the range of f from $\mu[f(L-m)]$ therefore yields

$$\begin{aligned}\left|\frac{\mu(fL)}{\mu L} - \mu f\right| &\le \frac{\operatorname{osc}(f)}{2m}\mu|L-m| \\ &\le \operatorname{osc}(f)\frac{a+b-m-ab/m}{b-a} \le \operatorname{osc}(f)\frac{\sqrt{b}-\sqrt{a}}{\sqrt{b}+\sqrt{a}}.\end{aligned} \tag{S159}$$

The last inequality uses $m + ab/m \ge 2\sqrt{ab}$. For $a = b$, both sides are zero.

Write $\epsilon = \underline{M}/\bar{M} = 2(1-\varrho)$ and, for this proof,

$$\beta = \frac{1-\epsilon}{1+\epsilon} = \frac{2\varrho - 1}{3 - 2\varrho}, \qquad \gamma = \frac{1-\sqrt{\epsilon}}{1+\sqrt{\epsilon}} = \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}. \tag{S160}$$

For two backward kernels $B_t(x,\cdot)$ and $B_t(x',\cdot)$, the former is a likelihood tilt of the latter by

$$L(u) = \frac{f_{X_t|X_{t-1}}(x \mid u;\theta)}{f_{X_t|X_{t-1}}(x' \mid u;\theta)}, \qquad \epsilon \le L(u) \le \epsilon^{-1}. \tag{S161}$$

Equation (S159), with $a = \epsilon$ and $b = \epsilon^{-1}$, gives

$$\operatorname{osc}(B_t f) \le \beta \operatorname{osc}(f). \tag{S162}$$

For $s \le t = s + q < N$, future observations are conditionally independent of the extended history through t given X_t. The projection argument in Lemma S11 thus gives

$$H_{s,t,i}(x) := \mathbb{E}_{\rho_N}(h_{s,i} \mid X_t = x) = \mathbb{E}_{\rho_t}(h_{s,i} \mid X_t = x) = (B_t \cdots B_{s+1}\bar{h}_{s,i})(x). \tag{S163}$$

The kernel product is empty if $q = 0$. Since $\operatorname{osc}(\bar{h}_{s,i}) \le 2G'(\theta)$,

$$\operatorname{osc}(H_{s,t,i}) \le 2G'(\theta)\beta^q. \tag{S164}$$

The future likelihood in Equation (S140) satisfies

$$L_{t,N}(x) = \int M_{t+1}(x, du)G_{t+1}(u)L_{t+1,N}(u), \qquad \epsilon \le \frac{L_{t,N}(x)}{L_{t,N}(x')} \le \epsilon^{-1}. \tag{S165}$$

The ratio bound follows from the transition bounds on the first kernel. The two expectations being compared are consequently

$$\rho_N(h_{s,i}) = \frac{\pi_t(H_{s,t,i}L_{t,N})}{\pi_t(L_{t,N})}, \qquad \rho_t(h_{s,i}) = \pi_t(H_{s,t,i}). \tag{S166}$$

Here $\sup L_{t,N}/\inf L_{t,N} \le \epsilon^{-1}$, so the same likelihood-tilt inequality bounds their difference by $\gamma \operatorname{osc}(H_{s,t,i})$. Summing squared coordinate bounds gives

$$\|\rho_N(h_s) - \rho_{s+q}(h_s)\|_2 \le 2\sqrt{p}\, G'(\theta)\gamma\beta^q$$

$$= 2\sqrt{p}\, G'(\theta) \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \left(\frac{2\varrho-1}{3-2\varrho}\right)^q \leq 2\sqrt{p}\, G'(\theta)\varrho^{q+1}, \tag{S167}$$

since $\beta, \gamma \leq 1-\epsilon = 2\varrho - 1 \leq \varrho$. If transitions are independent of their starting location, $L_{t,N}$ is constant and these differences vanish. This includes $\epsilon = 1$.

Insert the sharper comparison into Equation (S157) and reverse the finite sums:

$$\begin{aligned}\|\nabla_\theta \ell - \nabla_\theta \ell^\alpha\|_2 &\leq 2\sqrt{p}\, G'(\theta)\gamma(1-\alpha) \sum_{q=0}^{N-2} (N-1-q)(\alpha\beta)^q \\ &\leq 2(N-1)\sqrt{p}\, G'(\theta) \frac{\gamma(1-\alpha)}{1-\alpha\beta}. \end{aligned} \tag{S168}$$

Substituting β and γ proves Equation (S153). Using $\gamma\beta^q \leq \varrho^{q+1}$ instead gives the full-score bound, with the finite sum evaluated by

$$\begin{aligned}(1-z)\sum_{q=0}^{N-2}(N-1-q)z^q &= (N-1) - \sum_{q=1}^{N-1} z^q, \\ \sum_{q=0}^{N-2}(N-1-q)z^q &= \frac{(N-1) - Nz + z^N}{(1-z)^2}, \qquad z = \alpha\varrho < 1. \end{aligned} \tag{S169}$$

For $N = 1$ all comparisons are empty. The factor $1-\alpha$ handles $\alpha = 1$, and the formulas at $\alpha = 0$ retain only $q = 0$. □

Remark 1 (Signed cancellation in the population bias)**.** *Retaining signs gives a first-order expansion of the population bias. The score is a finite polynomial in α, so*

$$\begin{aligned}\frac{\partial}{\partial\alpha}\nabla_\theta \ell^\alpha(\theta)\Big|_{\alpha=1} &= \sum_{n=1}^{N}\sum_{r=1}^{n-1} r\Delta_{n,r} \\ &= \sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1} \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \end{aligned} \tag{S170}$$

Differentiating the finite lag sum gives the first equality, and Equation (S157) gives the second. Subtracting this first-order term yields

$$\begin{aligned}&\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) - (1-\alpha)\frac{\partial}{\partial\alpha}\nabla_\theta \ell^\alpha(\theta)\Big|_{\alpha=1} \\ &\quad = -(1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}(1-\alpha^q)\{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \end{aligned} \tag{S171}$$

By Equation (S167), for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}&\left\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) - (1-\alpha)\frac{\partial}{\partial\alpha}\nabla_\theta \ell^\alpha(\theta)\Big|_{\alpha=1}\right\|_2 \\ &\leq 2\sqrt{p}\, G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(1-\alpha^q)\left(\frac{2\varrho-1}{3-2\varrho}\right)^q \end{aligned}$$

$$\le 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\left\{\frac{3-2\varrho}{4(1-\varrho)}-\frac{3-2\varrho}{3-2\varrho-\alpha(2\varrho-1)}\right\}$$
$$=(N-1)\sqrt{p}\,G'(\theta)\frac{(1-\sqrt{2(1-\varrho)})^2(3-2\varrho)(1-\alpha)^2}{2(1-\varrho)\{3-2\varrho-\alpha(2\varrho-1)\}}. \tag{S172}$$

The last equality uses $2\varrho-1=(1-\sqrt{2(1-\varrho)})(1+\sqrt{2(1-\varrho)})$. *For* $\varrho=\frac{1}{2}$, *the score difference and its signed derivative are both zero, so the same bound holds. In particular, the population bias is bounded by*

$$\|\nabla_\theta\ell(\theta)-\nabla_\theta\ell^\alpha(\theta)\|_2$$
$$\le(1-\alpha)\left\|\sum_{n=1}^{N}\sum_{r=1}^{n-1}r\Delta_{n,r}\right\|_2+(N-1)\sqrt{p}\,G'(\theta)\frac{(1-\sqrt{2(1-\varrho)})^2(3-2\varrho)(1-\alpha)^2}{2(1-\varrho)\{3-2\varrho-\alpha(2\varrho-1)\}}. \tag{S173}$$

For every fixed observation sequence, if the signed sum vanishes, this gives $O\{N(1-\alpha)^2\}$ *bias and* $O\{N^2(1-\alpha)^4\}$ *squared bias, with fixed model constants.*

]

S7.1 Population error averaged over observations

[KT:The preceding bounds hold for each fixed observation sequence. We now generate $Y_{1:N}$ from the POMP at θ_0 and evaluate both scores at $\theta=\phi=\theta_0$. The expectation $\mathbb{E}_{\theta_0}$ averages over these data. It does not average over particle-filter randomness. The posterior law in Equation (S123) then satisfies

$$\rho_n(h_s)=\mathbb{E}_{\theta_0}(h_s\mid\mathcal{F}_n^Y),\qquad \mathcal{F}_n^Y=\sigma(Y_1,\dots,Y_n),\qquad s\le n. \tag{S174}$$

Here h_s is the full pathwise measurement derivative in Equation (S119), evaluated with the observation held fixed. The causal simulator uses fresh, independent, parameter-free draws at each time, including initialization.]

Proposition S2 (Population error under the data-generating law)**.** *[KT:*Suppose (A2), (A3), and (A5) hold in a neighborhood of θ_0, with the same deterministic bounds over the support of the observations. In particular, let $G'(\theta_0)$ bound the full pathwise measurement derivatives throughout this neighborhood. Fix $\alpha\in[0,1]$ independently of the observations, and evaluate at $\theta=\phi=\theta_0$. For either population score,

$$\mathbb{E}_{\theta_0}\,\|\nabla_\theta\ell^\alpha(\theta_0)-\nabla_\theta\ell(\theta_0)\|_2^2$$
$$\le C(N-1)pG'(\theta_0)^2\left\{\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}\right\}^2\lesssim NpG'(\theta_0)^2(1-\alpha)^2, \tag{S175}$$

where ϱ is the forgetting constant of Lemma S11 and the constants are independent of N and α. Neither (A6) nor (A6-J) is required.]

Proof. [KT:By Equation (S132), the time-n score difference is

$$e_n^\alpha:=\nabla_\theta\ell_n(\theta_0)-\nabla_\theta\ell_n^\alpha(\theta_0)$$
$$=\sum_{s=1}^{n-1}(1-\alpha^{n-s})\left\{\mathbb{E}_{\theta_0}(h_s\mid\mathcal{F}_n^Y)-\mathbb{E}_{\theta_0}(h_s\mid\mathcal{F}_{n-1}^Y)\right\}. \tag{S176}$$

The zero-lag term cancels. Applying the tower property to each remaining term gives

$$\mathbb{E}_{\theta_0}(e_n^\alpha \mid \mathcal{F}_{n-1}^Y) = 0.$$

For $m < n$, e_m^α is $\mathcal{F}_{n-1}^Y$-measurable, so

$$\mathbb{E}_{\theta_0}\{(e_m^\alpha)^\mathsf{T} e_n^\alpha\} = \mathbb{E}_{\theta_0}\left[(e_m^\alpha)^\mathsf{T}\mathbb{E}_{\theta_0}(e_n^\alpha \mid \mathcal{F}_{n-1}^Y)\right] = 0,$$
$$\mathbb{E}_{\theta_0}\left\|\sum_{n=1}^N e_n^\alpha\right\|_2^2 = \sum_{n=1}^N \mathbb{E}_{\theta_0}\|e_n^\alpha\|_2^2. \tag{S177}$$

The observation-uniform version of Lemma S11 bounds each increment by

$$\begin{aligned}\|e_n^\alpha\|_2 &\le C\sqrt{p}\,G'(\theta_0)\sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r \\ &\le C\sqrt{p}\,G'(\theta_0)\left\{\frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho}\right\} = C\sqrt{p}\,G'(\theta_0)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.\end{aligned} \tag{S178}$$

Since $e_1^\alpha = 0$, substitution into Equation (S177) proves the result.] □

[KT:Thus averaging over observations replaces the squared sum of conditional errors by a sum of their squares. This uses the generating parameter: at a different evaluation parameter, the posterior expectations in Equation (S174) are not conditional expectations under the data law. The result also treats α as fixed before seeing the data.]

Remark 2 (Fixed observations and random observations)**.** *[KT:*A continuous-state example shows why the two population bounds have different powers of N. Let $X_n \in [0,1]$, write $s(x) = 2x - 1$, and take $|\theta| < 1/2$. Initialize X_0 uniformly on $[0,1]$, independently of θ and all later draws. At odd times reset X_n to an independent uniform draw. At even times use the transition density

$$q(z \mid x) = 1 + \tfrac{1}{2}s(x)s(z). \tag{S179}$$

The observations are Bernoulli, with success probabilities

$$\Pr_\theta(Y_n = 1 \mid X_n = x) = \begin{cases}\frac{1}{2}\{1+\theta s(x)\}, & n \text{ odd},\\ \frac{1}{2}\{1+\frac{1}{2}s(x)\}, & n \text{ even}.\end{cases} \tag{S180}$$

The parameter enters only the odd-time measurement model. Each two-observation block is independent of the others. For its observation signs $a = 2Y_{2i-1} - 1$ and $b = 2Y_{2i} - 1$, integration gives

$$\begin{aligned}\Pr_\theta(a,b) &= \int_0^1\int_0^1 \frac{1+a\theta s(x)}{2}\left\{1+\tfrac{1}{2}s(x)s(z)\right\}\frac{1+\frac{1}{2}bs(z)}{2}\,dz\,dx \\ &= \frac{1}{4}\left(1+\frac{ab\theta}{36}\right),\end{aligned} \tag{S181}$$

using $\int s(x)\,dx = 0$ and $\int s(x)^2\,dx = 1/3$. At $\theta = \phi = 0$, $h_{2i-1} = as(X_{2i-1})$ and $h_{2i} = 0$. The odd observation leaves the state uniform, while the even observation changes the posterior mean of

h_{2i-1} from zero to $ab/36$. All other positive-lag coefficients vanish because the next reset separates the blocks. Thus Equation (S132) gives full and discounted block scores $ab/36$ and $\alpha ab/36$. For $N = 2m$,

$$\nabla_\theta \ell(0) - \nabla_\theta \ell^\alpha(0) = \frac{1-\alpha}{36} \sum_{i=1}^{m} a_i b_i. \tag{S182}$$

For the fixed sequence $Y_n = 1$, the squared difference is exactly

$$|\nabla_\theta \ell(0) - \nabla_\theta \ell^\alpha(0)|^2 = \frac{N^2(1-\alpha)^2}{5184}. \tag{S183}$$

Under the generating parameter 0, the four outcomes of each block have probability 1/4. The products $a_i b_i$ are therefore independent with mean zero and variance one, giving

$$\mathbb{E}_0 \, |\nabla_\theta \ell(0) - \nabla_\theta \ell^\alpha(0)|^2 = \frac{N(1-\alpha)^2}{2592}. \tag{S184}$$

Both orders are consequently attained within the stated model class.

To check the assumptions, transition densities lie in $[1/2, 3/2]$, measurement probabilities in $[1/4, 3/4]$, and the first two log-measurement parameter derivatives are bounded by 2 and 4. The simulator can use the inverse conditional CDF

$$F(z \mid x) = z + \tfrac{1}{2} s(x)(z^2 - z), \qquad \left| \frac{\partial F^{-1}(u \mid x)}{\partial x} \right| = \frac{z(1-z)}{q(z \mid x)} \le \frac{1}{2}.$$

The odd-time Jacobian is zero and the spatial log-measurement derivative is at most 2, so (A6) holds with $C = 2$ and $\kappa = 1/2$. The positive CDF derivative gives a smooth extension at the interval endpoints. Since the simulator is independent of θ, its parameter tangent is zero. The joint process $(X_n, 0)$ has the same transition bounds relative to $dx \otimes \delta_0$, proving (A6-J). Smoothness gives (A1) and (A5). For each fixed N, the deterministic bounds $G^* = 2N^2$ and $H^* = 8 \sum_{n=1}^{N} n^2$ suffice for (A4), uniformly in J and α. The model and stability constants used in the rate bounds are independent of N.]

S7.2 Particle error

[KT:

(A6) Measurement-sensitivity decay. The simulator is C^1 in its incoming state. For a continuation $X_n^\theta(x_t, \omega_{t+1:n})$ from x_t, with $X_t^\theta(x_t) = x_t$,

$$\left\| D_{x_t} \log f_{Y_n|X_n}\big(y_n^* \mid X_n^\theta(x_t, \omega_{t+1:n}); \theta\big) \right\|_2 \le C\kappa^{n-t}, \qquad C < \infty, \quad 0 < \kappa < 1. \tag{S185}$$

The bound is uniform in $t \le n$, states, draws, and θ near ϕ. The parameter derivatives of the initial state, simulator, and log measurement density are bounded, holding incoming states fixed.

(A6-J) Joint-transition alternative. The log measurement density is C^1 in (x, θ). The joint process (X_n^θ, D_n^θ), $D_n^\theta = \partial_\theta X_n^\theta$, has transition densities satisfying

$$0 < \underline{M}_D \le f_{(X_n,D_n)|(X_{n-1},D_{n-1})}(x', v' \mid x, v; \theta) \le \bar{M}_D < \infty \tag{S186}$$

relative to a common finite measure on the invariant joint state space, uniformly in time, admissible pairs, and θ near ϕ.

The particle bounds assume either (A6) or (A6-J); the consistency results and population Lemmas S11 and S12 retain (A1)–(A5). Neither additional condition requires density evaluation by the algorithm. Their constants enter the particle bounds, while ϱ retains its physical-model definition.

Sufficient condition for (A6). Suppose the parameter derivatives in (A6) are bounded, the spatial log measurement derivatives are bounded by L_g, and the one-step simulator Jacobians are bounded by L. If, for fixed $b \ge 1$ and $q \in (0, 1)$, $\|\partial X_{t+b}^\theta / \partial X_t^\theta\|_{\mathrm{op}} \le q$ uniformly with the draws fixed, then writing $n - t = mb + v$, $0 \le v < b$, gives

$$\left\| D_{x_t} \log f_{Y_n|X_n}(y_n^* \mid X_n^\theta; \theta) \right\|_2 \le L_g L^v q^m \le C\kappa^{n-t}, \qquad \kappa = q^{1/b} \in (0, 1). \tag{S187}$$

Here $C = L_g \max_{0 \le v < b} (L/\kappa)^v$; a zero block Jacobian permits any $q \in (0, 1)$.

Example: physical mixing without (A6). Consider the annulus $\mathcal{X} = \{x \in \mathbb{R}^2 : 1 \le \|x\|_2 \le 2\}$. Let R_u denote planar rotation by u, and let independent simulator draws satisfy $U_n \sim \mathrm{Unif}[0, 2\pi]$ and $V_n \sim \mathrm{Unif}[1, 2]$. Set

$$X_0^\theta = R_\theta x_0, \qquad X_n^\theta = V_n R_{U_n} \frac{X_{n-1}^\theta}{\|X_{n-1}^\theta\|_2}, \qquad \Pr(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{x_1}{4\|x\|_2}. \tag{S188}$$

The physical transition forgets the incoming state in one step: its Lebesgue density on $\mathcal{X}$ is

$$f_{X_n|X_{n-1}}(x' \mid x; \theta) = \frac{1}{2\pi\|x'\|_2}, \qquad \frac{1}{4\pi} \le f_{X_n|X_{n-1}} \le \frac{1}{2\pi}. \tag{S189}$$

Both measurement probabilities lie in $[1/4, 3/4]$. For fixed draws, $X_n^\theta = R_\theta X_n^0$, so all parameter derivatives exist and are bounded; in particular the first two log measurement derivatives have magnitudes at most 1 and 2. The likelihood is smooth, and the MOP derivatives are bounded for each fixed horizon. Thus (A1)–(A5) hold.

Nevertheless, fix the observations $y_n^* = 1$. For $n > t$, the composition is

$$X_n^\theta(x, \omega_{t+1:n}) = V_n R_{U_{t+1} + \cdots + U_n} \frac{x}{\|x\|_2}. \tag{S190}$$

Its final angle is uniform, and a fixed-random-number change in the incoming angle is preserved. Differentiating the displayed measurement probability therefore gives

$$\mathbb{E} \left\| D_x \log f_{Y_n|X_n}\big(1 \mid X_n^\theta(x, \omega_{t+1:n}); \theta\big) \right\|_2^2 = \frac{1}{2\pi\|x\|_2^2} \int_0^{2\pi} \frac{\sin^2 u}{(2 + \cos u)^2}\, du > 0. \tag{S191}$$

The right side is independent of $n - t$, so (A6) fails despite (A1)–(A5). Here $D_n^\theta = R_{\pi/2} X_n^\theta$, and the joint transition has the physical density relative to $dx'\, \delta_{R_{\pi/2}x'}(dv')$. Thus (A6-J) holds and the particle bounds still apply. In particular, (A6-J) does not imply (A6).

Under (A6-J), the measurement gradient is

$$h_{t,i}(x,v) = \partial_{\theta_i} \log f_{Y_t|X_t}(y_t^* \mid x;\theta) + \sum_{a=1}^{d} \partial_{x_a} \log f_{Y_t|X_t}(y_t^* \mid x;\theta) v_{ai}, \qquad \theta = \phi, \tag{S192}$$

which is bounded by (A3).]

Lemma S13 (Particle error of the DMOP-α score)**.** *Suppose $J \geq 2$, $\theta = \phi$, $\alpha \in [0,1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J) holds. Then* [KT:for either the before- or after-resampling estimator,

$$\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2^2 \lesssim p G'(\theta)^2 \left[\frac{N}{J} \min\left\{ \sum_{r=0}^{N-1} \alpha^r, J \right\} + \frac{N^2}{J^2} \right], \tag{S193}$$

$$\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \lesssim N\sqrt{p}\, G'(\theta)/J. \tag{S194}$$

Both estimators have the same expectation. Under (A6), use the enlarged $G'(\theta)$ in Equation (S197). The MSE bound also bounds the variance.] *The constants suppressed by $\lesssim$ depend only on the fixed model and stability bounds and are independent of $N, J,$ and α.*

Proof. [KT:We first decompose the measurement gradient under (A6), then separate the particle error into its conditional mean and genealogical parts.]

[KT:*Gradient-age decomposition.* All derivatives are evaluated at $\theta = \phi$, with reference resampling indices fixed. Write $X_u^\theta = \texttt{process}_u(X_{u-1}^\theta;\theta,\omega_u)$, and let $\partial_{\theta_i}\texttt{process}_u$ denote its partial derivative holding X_{u-1}^θ fixed. The chain rule, iterated from time zero, gives

$$\partial_{\theta_i} X_s^\theta = \frac{\partial X_s^\theta}{\partial X_0^\theta} \partial_{\theta_i} X_0^\theta + \sum_{u=1}^{s} \frac{\partial X_s^\theta}{\partial X_u^\theta} \partial_{\theta_i} \texttt{process}_u(X_{u-1}^\theta;\theta,\omega_u). \tag{S195}$$

Under (A6), write the coordinate-i measurement gradient as a sum over the age r of its parameter derivative:

$$\begin{aligned}
h_{s,i}^{[0]} &= \partial_{\theta_i} \log f_{Y_s|X_s}(y_s^* \mid x;\theta)\big|_{x=X_s^\theta} + D_x \log f_{Y_s|X_s}(y_s^* \mid x;\theta)\big|_{x=X_s^\theta} \partial_{\theta_i}\texttt{process}_s(X_{s-1}^\theta;\theta,\omega_s), \\
h_{s,i}^{[r]} &= D_x \log f_{Y_s|X_s}\big(y_s^* \mid X_s^\theta(x,\omega_{s-r+1:s});\theta\big)\Big|_{x=X_{s-r}^\theta} \partial_{\theta_i}\texttt{process}_{s-r}(X_{s-r-1}^\theta;\theta,\omega_{s-r}), \quad 1 \leq r < s, \\
h_{s,i}^{[s]} &= D_x \log f_{Y_s|X_s}\big(y_s^* \mid X_s^\theta(x,\omega_{1:s});\theta\big)\Big|_{x=X_0^\theta} \partial_{\theta_i} X_0^\theta.
\end{aligned} \tag{S196}$$

The first term includes the direct measurement derivative and the current simulator derivative; the last includes the initial-state derivative. Under (A6), enlarge $G'(\theta)$ if necessary so that

$$\max\left\{ \|\partial_{\theta_i} X_0^\theta\|_2,\ \|\partial_{\theta_i}\texttt{process}_u(x;\theta,\omega_u)\|_2,\ \left|\partial_{\theta_i} \log f_{Y_u|X_u}(y_u^* \mid x;\theta)\right| \right\} \leq G'(\theta), \tag{S197}$$

uniformly in u, i, x and simulator draws, holding x fixed in the partial derivatives. Under (A6-J), the original bound on $h_{s,i}$ suffices. Multiplying Equation (S195) by the spatial log measurement derivative gives

$$h_{s,i} = \sum_{r=0}^{s} h_{s,i}^{[r]}, \qquad \|h_{s,i}^{[r]}\|_\infty \leq C G'(\theta) \kappa^r. \tag{S198}$$

For $r = 0$ use (A6) with $t = s$ and the bounded partial derivatives; for $1 \le r < s$ use it with $t = s - r$; for $r = s$ use $t = 0$ and the initial-gradient bound. For $r < s$, the displayed function depends only on X_{s-r-1} and the simulator variables from $s - r$ through s. For $r = s$ it also uses the initial simulator draw, whose particle replicates are independent. Set $h_{s,i}^{[r]} = 0$ for $r > s$.

Under (A6-J), take instead $h_{s,i}^{[0]} = h_{s,i}$, the bounded current function in Equation (S192), and $h_{s,i}^{[r]} = 0$ for $r \ge 1$. Use the joint pair as the state throughout the particle calculation. The sums over r below then contain only $r = 0$; κ is needed only under (A6).]

[KT:*After-resampling score.* We first treat the after-resampling score $\nabla_\theta \hat{\ell}_J^{A,\alpha}$, then transfer the bound to $\nabla_\theta \hat{\ell}_J^{B,\alpha}$. Put $H_N(\alpha) = \sum_{q=0}^{N-1} \alpha^q$. Lemma S10 gives

$$\nabla_\theta \hat{\ell}_J^{A,\alpha} = \sum_{s=1}^{N} \sum_{b=s}^{N} c_b \alpha^{b-s} m(\xi_b)(h_s), \qquad c_b = \begin{cases} 1-\alpha, & b < N, \\ 1, & b = N. \end{cases} \tag{S199}$$

Replacing $m(\xi_b)$ by ρ_b gives $\nabla_\theta \ell^\alpha$. For each creation time s, the endpoint weights satisfy

$$\begin{aligned} \sum_{b=s}^{N} c_b \alpha^{b-s} &= 1, \\ \sum_{b=s}^{N} c_b \alpha^{b-s}(b-s+1) = \sum_{q=0}^{N-s} \sum_{b=s+q}^{N} c_b \alpha^{b-s} &= \sum_{q=0}^{N-s} \alpha^q \le H_N(\alpha). \end{aligned} \tag{S200}$$

These identities include $\alpha = 0, 1$. Fix coordinate i and age r. Under (A6), Equation (S198) gives $\|h_{s,i}^{[r]}\|_\infty \le CG'(\theta)\kappa^r$ and a block starting at $a_s = \max(1, s-r-1)$, with the initial history included when needed. Under (A6-J), use the joint state, $a_s = s$, and only $r = 0$. Write Z_r for this component of equation S199 and z_r for its population value.

We separate the error in the prediction clouds from the remaining random genealogy. Under (A6), let

$$\mathcal{C} = \sigma(X_{u,j}^{P,\phi} : 1 \le u \le N+1,\ 1 \le j \le J).$$

Under (A6-J), use the joint states instead. The artificial transition at $N + 1$ is to a one-point space with no observation, and its parents record the actual time-N resampling. Parent indices and simulator draws are excluded from $\mathcal{C}$. We will bound the two terms in

$$\mathbb{E}|Z_r - z_r|^2 = \mathbb{E}|\mathbb{E}[Z_r \mid \mathcal{C}] - z_r|^2 + \mathbb{E}\mathrm{Var}(Z_r \mid \mathcal{C}).$$

For $2 \le u \le N$, the conditional parent probabilities are

$$B_u^J(j,l) = \frac{g_{u-1,l}^\phi f_{X_u|X_{u-1}}(X_{u,j}^{P,\phi} \mid X_{u-1,l}^{P,\phi}; \phi)}{\sum_v g_{u-1,v}^\phi f_{X_u|X_{u-1}}(X_{u,j}^{P,\phi} \mid X_{u-1,v}^{P,\phi}; \phi)}. \tag{S201}$$

Under (A6-J), use the joint transition density. At the artificial last layer, the transition density is one, and hence

$$B_{N+1}^J(j,l) = g_{N,l}^\phi / \sum_v g_{N,v}^\phi.$$

Let $K_{u-1,j}$ be the parent of node (u,j) and $R_u(x,x';d\omega)$ the simulator law conditional on its two states. Bayes' rule gives

$$\begin{aligned}&\Pr\{K_{u-1,j}=l_{u,j},\ \omega_{u,j}\in d\omega_{u,j}\text{ for all }u\geq 2,j\mid\mathcal{C}\}\\&\quad=\prod_{u=2}^{N+1}\prod_{j=1}^{J}B_u^J(j,l_{u,j})R_u(X_{u-1,l_{u,j}}^{P,\phi},X_{u,j}^{P,\phi};d\omega_{u,j}).\end{aligned}$$

The initial histories likewise factor over first-layer nodes, and the last-layer draw is deterministic. Thus distinct nodes have independent parents and draws conditional on $\mathcal{C}$; a shared node has one shared parent and draw. Under (A6-J), omit the draws. The density bounds give

$$\max_{j,l}B_u^J(j,l)\leq C/J,\qquad \operatorname{osc}(B_u^Jf)\leq\varrho_*\operatorname{osc}(f),\qquad 0\leq\varrho_*<1. \tag{S202}$$

Here $\varrho_*<1$ bounds both backward and exact forward conditional contraction. It need not equal the population-bias coefficient ϱ.

For a fixed state block, integrate the simulator bridges:

$$f_{s,r}(x_{a_s:s})=\int h_{s,i}^{[r]}(x_{a_s:s},\omega)\prod_{v=a_s+1}^{s}R_v(x_{v-1},x_v;d\omega_v).$$

When $a_s=1$, this integral also includes the initial history conditional on x_1. Under (A6-J), $f_{s,0}=h_{s,i}$ on the current joint state. The bridge kernels are fixed model kernels, and $\|f_{s,r}\|_\infty\leq CG'(\theta)\kappa^r$. Thus

$$\mathbb{E}[m(\xi_b)(h_{s,i}^{[r]})\mid\mathcal{C}]=\frac{1}{J}\sum_{j=1}^{J}\int f_{s,r}(x_{a_s:s})\prod_{v=2}^{b+1}B_v^J(x_v,dx_{v-1}),\qquad x_{b+1}=X_{b+1,j}^{P,\phi}.$$

The corresponding exact integral is $\rho_b(h_{s,i}^{[r]})$, since no observation is added at $b+1$.

Error of the conditional mean. For this calculation return to the ordinary particle law. Let $\mathcal{H}_u$ be the filtration of prediction clouds through u, $\mathcal{H}_0$ trivial, and μ_u the prediction law given $\mathcal{H}_{u-1}$. The empirical measure η_u^J averages J conditionally iid draws from μ_u. We use the telescope of Definition 5.1 and equation (7.1) of Del Moral et al. (2010), written explicitly as follows. Fix (s,b,r), put $T=b+1$, and write M_{v+1} for the exact transition and g_v for the measurement density. Extend equation S201 to a kernel $B_v^J(x_v,dx_{v-1})$ using the cloud at $v-1$. Define

$$\begin{aligned}D_{u,T}^Jf(x_u)&=\int f(x_{1:T})\prod_{v=2}^{u}B_v^J(x_v,dx_{v-1})\prod_{v=u}^{T-1}g_v(x_v)M_{v+1}(x_v,dx_{v+1}),\\Q_{u,T}1(x_u)&=D_{u,T}^J1(x_u),\qquad A_u=\frac{\eta_u^J(D_{u,T}^Jf_{s,r})}{\eta_u^J(Q_{u,T}1)}.\end{aligned}$$

For $u\geq 2$, Bayes' formula gives

$$\mu_u(dx_u)B_u^J(x_u,dx_{u-1})=\frac{\eta_{u-1}^J(dx_{u-1})g_{u-1}(x_{u-1})M_u(x_{u-1},dx_u)}{\eta_{u-1}^J(g_{u-1})}.$$

Hence

$$\mu_u(D^J_{u,T} f_{s,r}) = \frac{\eta^J_{u-1}(D^J_{u-1,T} f_{s,r})}{\eta^J_{u-1}(g_{u-1})},$$
$$\mu_u(Q_{u,T}1) = \frac{\eta^J_{u-1}(Q_{u-1,T}1)}{\eta^J_{u-1}(g_{u-1})}.$$

Their ratio is A_{u-1}. At $u = 1$, the exact initial prediction law gives $A_0 = \rho_b(h^{[r]}_{s,i})$; at $u = T$, $Q_{T,T}1 = 1$ and A_T is the conditional smoother. Thus, with $e_{u;s,b} = A_u - A_{u-1}$,

$$\mathbb{E}[m(\xi_b)(h^{[r]}_{s,i}) \mid \mathcal{C}] - \rho_b(h^{[r]}_{s,i}) = \sum_{u=1}^{b+1} e_{u;s,b}. \tag{S203}$$

For its ratio expansion, put

$$q_{u;b} = \frac{Q_{u,b+1}1}{\mu_u(Q_{u,b+1}1)}, \qquad F_{u;s,b} = \frac{D^J_{u,b+1} f_{s,r}}{Q_{u,b+1}1},$$
$$d_{u;s,b} = q_{u;b}\{F_{u;s,b} - \mu_u(q_{u;b}F_{u;s,b})\}. \tag{S204}$$

These functions use empirical past kernels and exact future kernels, so are $\mathcal{H}_{u-1}$-measurable. The ratio expansion gives

$$\mu_u q_{u;b} = 1, \qquad \mu_u d_{u;s,b} = 0,$$
$$e_{u;s,b} = (\eta^J_u - \mu_u)d_{u;s,b} - \frac{\{(\eta^J_u - \mu_u)d_{u;s,b}\}\{(\eta^J_u - \mu_u)q_{u;b}\}}{\eta^J_u q_{u;b}}. \tag{S205}$$

For $v < T$, the exact forward kernel conditional on observations through $b = T - 1$ is

$$P^{(b)}_v(x, dz) = \frac{M_{v+1}(x, dz)Q_{v+1,T}1(z)}{M_{v+1}(Q_{v+1,T}1)(x)}. \tag{S206}$$

If λ is the transition reference measure and $\epsilon = \underline{M}/\bar{M}$, then

$$P^{(b)}_v(x, dz) \geq \epsilon \frac{Q_{v+1,T}1(z)\lambda(dz)}{\int Q_{v+1,T}1(w)\lambda(dw)}, \tag{S207}$$
$$\operatorname{osc}(P^{(b)}_v f) \leq (1 - \epsilon)\operatorname{osc}(f). \tag{S208}$$

Use the joint transition bounds under (A6-J). At the artificial final transition the kernel is constant, so the same contraction bound holds. The current potential cancels in this conditional kernel. Thus

$$F_{u;s,b} = P^{(b)}_u \cdots P^{(b)}_{a_s-1} F_{a_s;s,b}, \qquad u < a_s,$$
$$F_{u;s,b} = B^J_u \cdots B^J_{s+1} F_{s;s,b}, \qquad u > s.$$

Inside the block, $F_{u;s,b}$ is a conditional average of $f_{s,r}$. These identities and the forward and backward contractions give

$$\operatorname{osc}(F_{u;s,b}) \leq 2CG'(\theta)\kappa^r \begin{cases} \varrho_*^{a_s-u}, & u < a_s, \\ 1, & a_s \leq u \leq s, \\ \varrho_*^{u-s}, & u > s. \end{cases}$$

The bounds on transition and measurement densities give

$$\frac{\sup_x Q_{u,T}1(x)}{\inf_x Q_{u,T}1(x)} \le \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}}, \qquad u < T. \tag{S209}$$

with the joint transition bounds under (A6-J). Since $Q_{T,T}1 = 1$, this controls $q_{u;b}$ for every u. Also,

$$|F_{u;s,b} - \mu_u(q_{u;b}F_{u;s,b})| \le \operatorname{osc}(F_{u;s,b}).$$

It follows that

$$C^{-1} \le q_{u;b} \le C, \qquad \operatorname{osc}(d_{u;s,b}) \le CG'(\theta)\kappa^r \varrho_*^{\operatorname{dist}(u,[a_s,s])}. \tag{S210}$$

At $N+1$, $\eta_{N+1}^J = \mu_{N+1}$ and the local error is zero.

Sum the linear terms before taking second moments:

$$d_u^{[r]} = \sum_{s=1}^{N} \sum_{b=\max(s,u-1)}^{N} c_b \alpha^{b-s} d_{u;s,b}.$$

Each inner sum has weight at most one, and at most $r+2$ blocks contain any fixed u. Therefore

$$\sum_{s=1}^{N} \varrho_*^{\operatorname{dist}(u,[a_s,s])} \le r + 2 + \frac{2}{1-\varrho_*} \le C(r+2),$$
$$\operatorname{osc}(d_u^{[r]}) \le CG'(\theta)\kappa^r(r+2). \tag{S211}$$

Set $L_u = (\eta_u^J - \mu_u)d_u^{[r]}$. Conditional iid sampling yields

$$\mathbb{E}[L_u \mid \mathcal{H}_{u-1}] = 0,$$
$$\mathbb{E}[L_u^2 \mid \mathcal{H}_{u-1}] = \frac{\mu_u\{(d_u^{[r]} - \mu_u d_u^{[r]})^2\}}{J} \le \frac{CG'(\theta)^2\kappa^{2r}(r+2)^2}{J},$$
$$\mathbb{E}\left|\sum_u L_u\right|^2 = \sum_u \mathbb{E}L_u^2 \le \frac{CG'(\theta)^2\kappa^{2r}(r+2)^2 N}{J}. \tag{S212}$$

The cross terms vanish because, for $v < u$, L_v is $\mathcal{H}_{u-1}$-measurable and $\mathbb{E}[L_vL_u] = \mathbb{E}[L_v\mathbb{E}(L_u \mid \mathcal{H}_{u-1})] = 0$.

For the quadratic term, conditional iid sampling of a bounded function f gives

$$\mathbb{E}[\{(\eta_u^J - \mu_u)f\}^4 \mid \mathcal{H}_{u-1}] = \frac{\mu_u(\bar{f}^4)}{J^3} + \frac{3(J-1)\mu_u(\bar{f}^2)^2}{J^3} \le \frac{4\operatorname{osc}(f)^4}{J^2}, \quad \bar{f} = f - \mu_u f.$$

Conditional Cauchy–Schwarz and the denominator bound therefore imply

$$\left\| \frac{\{(\eta_u^J - \mu_u)d_{u;s,b}\}\{(\eta_u^J - \mu_u)q_{u;b}\}}{\eta_u^J q_{u;b}} \right\|_{L^2} \le \frac{CG'(\theta)\kappa^r}{J} \varrho_*^{\operatorname{dist}(u,[a_s,s])}. \tag{S213}$$

Sum over u and the endpoint weights in equation S200. The total remainder has L^2 norm at most $CG'(\theta)\kappa^r(r+2)N/J$. Since the linear terms are centered, equation S212 gives

$$\mathbb{E}|\mathbb{E}[Z_r \mid \mathcal{C}] - z_r|^2 \le CG'(\theta)^2\kappa^{2r}(r+2)^2\left(\frac{N}{J} + \frac{N^2}{J^2}\right),$$

$$|\mathbb{E}Z_r - z_r| \le CG'(\theta)\kappa^r(r+2)N/J. \tag{S214}$$

Conditional genealogical variance. Condition on $\mathcal{C}$ and reveal parents and simulator draws backwards, one layer at a time. For endpoint $b+1$, let $W_{b,u}(j)$ be the fraction of its terminal labels at node j in layer u, defined only for $u \le b+1$. Initially $W_{b,b+1}(j) = 1/J$. Let $\mathcal{R}_u = \sigma(\mathcal{C}, V_{v,j} : v > u,\ 1 \le j \le J)$, where $V_{v,j}$ records that node's parent and simulator draw. Then

$$W_{b,u-1}(l) = \sum_j W_{b,u}(j)\mathbf{1}_{\{K_{u-1,j}=l\}}, \qquad u \ge 2.$$

The weights on the right are $\mathcal{R}_u$-measurable, and the parent choices are conditionally independent. Hence

$$\begin{aligned}
\mathbb{E}\left[\sum_j W_{b,u-1}(j)^2 \mid \mathcal{R}_u\right] &= \sum_j W_{b,u}(j)^2 + \sum_{j\ne l} W_{b,u}(j)W_{b,u}(l)\sum_v B_u^J(j,v)B_u^J(l,v) \\
&\le \sum_j W_{b,u}(j)^2 + C/J, \\
\mathbb{E}\left[\sum_j W_{b,u}(j)^2 \mid \mathcal{C}\right] &\le \min\{1, C(b-u+2)/J\}.
\end{aligned} \tag{S215}$$

The weights sum to one, giving the cap. Define

$$c_{s,r}(u) = \mathbf{1}_{\{a_s<u\le s+1\}} + \varrho_*^{u-s-1}\mathbf{1}_{\{u>s+1\}}, \quad u \ge 2, \qquad c_{s,r}(1) = \mathbf{1}_{\{a_s=1\}}.$$

Changing a node within the block costs $CG'(\theta)\kappa^r$. After the block, backward contraction supplies the geometric factor; earlier choices have no effect once the block is revealed. Thus

$$\sup_u \sum_{s=1}^N c_{s,r}(u) \le C(r+2).$$

Let $V_{u,j}$ be the newly revealed parent and simulator draw at node j, or its initial-history draw when $u = 1$. Let $\mathcal{D}_{b,u}(j)$ be the terminal labels at $b+1$ whose revealed paths pass through node j at time u, so $|\mathcal{D}_{b,u}(j)| = JW_{b,u}(j)$. Their conditional contribution is

$$Z_{u,j}(v) = \frac{1}{J}\sum_s \sum_{b=\max(s,u-1)}^N c_b\alpha^{b-s} \sum_{\ell\in\mathcal{D}_{b,u}(j)} \mathbb{E}[h_{s,i}^{[r]}(\xi_{b,\ell}) \mid \mathcal{R}_u, V_{u,j} = v]. \tag{S216}$$

The reverse Doob difference is

$$D_{u,r} = \mathbb{E}[Z_r \mid \mathcal{R}_u \vee \sigma(V_{u,j} : 1 \le j \le J)] - \mathbb{E}[Z_r \mid \mathcal{R}_u].$$

Endpoints with $b < u-1$ use only unrevealed earlier nodes and contribute zero to this difference. For the remaining endpoints, linearity gives

$$D_{u,r} = \sum_j \{Z_{u,j}(V_{u,j}) - \mathbb{E}[Z_{u,j}(V_{u,j}) \mid \mathcal{R}_u]\}.$$

The conditional expectations retain each terminal label's revealed suffix and integrate its earlier history. Every path uses one node at layer u, and distinct nodes are conditionally independent. Hence $\mathbb{E}[D_{u,r}^2 \mid \mathcal{R}_u]$ is a sum of node variances, with

$$\operatorname{osc}(Z_{u,j}) \le CG'(\theta)\kappa^r \sum_s c_{s,r}(u) \sum_{b=\max(s,u-1)}^{N} c_b \alpha^{b-s} W_{b,u}(j).$$

Conditional independence first gives

$$\mathbb{E}[D_{u,r}^2 \mid \mathcal{R}_u] \le \sum_j \operatorname{osc}(Z_{u,j})^2,$$

$$\left\{ \mathbb{E} \sum_j W_{b,u}(j) W_{b',u}(j) \mid \mathcal{C} \right\}^2 \le \mathbb{E}\left[\sum_j W_{b,u}(j)^2 \mid \mathcal{C} \right] \mathbb{E}\left[\sum_j W_{b',u}(j)^2 \mid \mathcal{C} \right].$$

The second line is Cauchy–Schwarz over nodes and probability. Expanding the square of the oscillation bound and using equation S215 gives

$$\mathbb{E}[D_{u,r}^2 \mid \mathcal{C}] \le CG'(\theta)^2 \kappa^{2r} \left\{ \sum_s c_{s,r}(u) \sum_{b=\max(s,u-1)}^{N} c_b \alpha^{b-s} \sqrt{\min\{1, C(b-u+2)/J\}} \right\}^2. \tag{S217}$$

On a nonzero term, $b-u+2 \le b-s+1+r+2$. The inner endpoint weights form a subprobability measure. Adding the missing mass at zero, concavity and equation S200 give

$$\begin{aligned} &\sum_{b=\max(s,u-1)}^{N} c_b \alpha^{b-s} \sqrt{\min\{1, C(b-u+2)/J\}} \\ &\quad \le \sqrt{\min\{1, C(H_N(\alpha)+r+2)/J\}} \le C\sqrt{(r+3)\min\{1, H_N(\alpha)/J\}}. \end{aligned} \tag{S218}$$

Substitute in equation S217, sum the at most $N+1 \le 2N$ layers, and use reverse martingale orthogonality:

$$\mathbb{E}\operatorname{Var}(Z_r \mid \mathcal{C}) \le CG'(\theta)^2 \kappa^{2r} (r+3)^3 N \min\{1, H_N(\alpha)/J\}. \tag{S219}$$

Combine equation S214 and equation S219 in the conditional mean–variance identity. Since $H_N \ge 1$ and $\sum_r \kappa^r (r+3)^{3/2} < \infty$, Minkowski over ages and summation over coordinates prove the after-score MSE and weak-bias bounds. Under (A6-J), only $r = 0$ occurs.

Before-resampling score. At $\theta = \phi$, write the unselected weight gradient as

$$v_{n,j} = \nabla_\theta \log(g_{n,j}^\theta w_{n,j}^{P,\theta}) = h_{n,j} + \alpha \nabla_\theta \log w_{n-1,j}^{F,\theta}, \qquad W_{n,j} = \frac{g_{n,j}^\phi}{\sum_l g_{n,l}^\phi}.$$

With multinomial indices $K_{n,j}$, differentiation of the numerator and denominator of each likelihood increment gives

$$\nabla_\theta \log L_n^{B,\theta,\alpha} = \sum_j W_{n,j} v_{n,j} - \frac{\alpha}{J} \sum_j \nabla_\theta \log w_{n-1,j}^{F,\theta},$$

$$\nabla_\theta \log L_n^{A,\theta,\alpha} = \frac{1}{J}\sum_j v_{n,K_{n,j}} - \frac{\alpha}{J}\sum_j \nabla_\theta \log w_{n-1,j}^{F,\theta}. \tag{S220}$$

Thus, with $\varepsilon_n = J^{-1}\sum_j v_{n,K_{n,j}} - \sum_j W_{n,j} v_{n,j}$,

$$\begin{aligned}
\nabla_\theta \hat{\ell}_J^{A,\alpha} - \nabla_\theta \hat{\ell}_J^{B,\alpha} &= \sum_{n=1}^{N} \varepsilon_n, \\
\mathbb{E}[\varepsilon_n \mid \mathcal{F}_n^-] &= 0, \\
\mathbb{E}\left\| \sum_n \varepsilon_n \right\|_2^2 &= \frac{1}{J}\sum_n \mathbb{E}\sum_j W_{n,j} \left\| v_{n,j} - \sum_l W_{n,l} v_{n,l} \right\|_2^2 .
\end{aligned} \tag{S221}$$

Here $\mathcal{F}_n^-$ is the complete history before resampling at n. The centered errors are orthogonal in the natural forward filtration, and the two scores have the same expectation.

To bound the weighted spread in equation S221, condition on the prediction clouds through n and draw two terminal labels independently from W_n. Trace their actual ancestors backwards. From equation S201, there is $\epsilon > 0$ such that $B_u^J(j,l) \geq \epsilon W_{u-1,l}$. Distinct nodes meet at the next backward step with probability at least

$$\sum_l B_u^J(j,l) B_u^J(j',l) \geq \epsilon^2 \sum_l W_{u-1,l}^2 \geq \epsilon^2 / J.$$

Once they meet, their earlier histories coincide. With $\lambda = 1 - \epsilon^2/J$, their labels I_t, I_t' satisfy

$$\Pr(I_t \neq I_t' \mid \mathcal{C}_n) \leq \lambda^{n-t}.$$

Under (A6-J), put $\Delta_t = h_{t,i}(I_t) - h_{t,i}(I_t')$. For $s \leq t$, each lineage has marginal backward kernel B^J. The diagonal terms in the pair difference vanish, so

$$\sum_{k \neq l} B_u^J(j,k) B_u^J(j',l)\{f(k) - f(l)\} = (B_u^J f)(j) - (B_u^J f)(j').$$

Conditional on the revealed suffix through t, contraction bounds the absolute conditional mean by

$$|\mathbb{E}[\Delta_s \mid \mathcal{S}_t]| \leq C G'(\theta) \varrho_*^{t-s} \mathbf{1}_{\{I_t \neq I_t'\}},$$

where $\mathcal{S}_t$ contains that suffix and the clouds. Multiplying by the suffix-measurable Δ_t gives

$$|\mathbb{E}[\Delta_s \Delta_t \mid \mathcal{C}_n]| \leq C G'(\theta)^2 \varrho_*^{t-s} \lambda^{n-t}. \tag{S222}$$

Under (A6), set $a_t = \max(1, t - r - 1)$. For nonoverlapping blocks ($a_t > s$), reveal the later block down to a_t. Let $\mathcal{S}_{a_t}$ contain the clouds, the two node labels at a_t, and their revealed parent indices and simulator draws strictly after a_t. Integrating the earlier block down from time s gives a function f_s with $\|f_s\|_\infty \leq C G' \kappa^r$. The marginal backward kernels give

$$\begin{aligned}
\mathbb{E}[\Delta_s^{[r]} \mid \mathcal{S}_{a_t}] &= (B_{a_t}^J \cdots B_{s+1}^J f_s)(I_{a_t}) - (B_{a_t}^J \cdots B_{s+1}^J f_s)(I_{a_t}'), \\
\left|\mathbb{E}[\Delta_s^{[r]} \mid \mathcal{S}_{a_t}]\right| &\leq C G' \kappa^r \varrho_*^{a_t - s} \mathbf{1}_{\{I_{a_t} \neq I_{a_t}'\}}.
\end{aligned}$$

Multiply by the $\mathcal{S}_{a_t}$-measurable $\Delta_t^{[r]}$, whose absolute value is at most $CG'\kappa^r \mathbf{1}_{\{I_t \neq I'_t\}}$, and use the survival bound. For overlapping blocks, use boundedness. Together these give

$$|\mathbb{E}[\Delta_s^{[r]}\Delta_t^{[r]} \mid \mathcal{C}_n]| \leq CG'(\theta)^2\kappa^{2r}\varrho_*^{(t-s-r-2)_+}\lambda^{n-t}. \tag{S223}$$

If the later block reaches the initial-history layer, the blocks overlap and boundedness applies. Expanding the square of the discounted sum and writing $q = n - t$, $d = t - s$ gives

$$\begin{aligned}
&\mathbb{E}\left[\left|\sum_{s=1}^{n}\alpha^{n-s}\Delta_s^{[r]}\right|^2\middle|\mathcal{C}_n\right] \\
&\leq CG'(\theta)^2\kappa^{2r}\sum_{q=0}^{n-1}(\alpha^2\lambda)^q\left\{1 + 2\sum_{d\geq 1}\alpha^d\varrho_*^{(d-r-2)_+}\right\} \\
&\leq CG'(\theta)^2\kappa^{2r}(r+3)\sum_{q=0}^{n-1}(\alpha^2\lambda)^q.
\end{aligned} \tag{S224}$$

The weighted empirical variance is half this paired second moment. Minkowski over ages costs the finite sum $\sum_r \kappa^r\sqrt{r+3}$. Using equation S221,

$$\begin{aligned}
\mathbb{E}\|\nabla_\theta\hat{\ell}_J^{A,\alpha} - \nabla_\theta\hat{\ell}_J^{B,\alpha}\|_2^2 &\leq \frac{CpG'(\theta)^2}{J}\sum_{n=1}^{N}\sum_{q=0}^{n-1}\{\alpha^2(1-\epsilon^2/J)\}^q \\
&\leq CpG'(\theta)^2\frac{N\min\{H_N(\alpha), J\}}{J}.
\end{aligned} \tag{S225}$$

The inner sum is bounded by both $H_N(\alpha)$ and J/ϵ^2. The L^2 triangle inequality with the after-score bound proves equation S193 for the before score. Equality of expectations gives equation S194.

]

□

S7.3 Full-score mean squared error

Theorem S3 (MSE bounds for DMOP-α)**.** *[KT:*Under the conditions of Lemmas S13 and S12, use either the before- or after-resampling estimator and take $\varrho = 1 - \underline{M}/(2\bar{M}) \in [\frac{1}{2}, 1)$. For $\alpha \in [0, 1]$,

$$\begin{aligned}
&\mathbb{E}\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell(\theta)\|_2^2 \\
&\lesssim pG'(\theta)^2\left[\frac{N}{J}\min\left\{\sum_{r=0}^{N-1}\alpha^r, J\right\} + \frac{N^2}{J^2}\right. \\
&\quad\left. + \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2}\left\{N - 1 + \sum_{q=1}^{N-2}(N-1-q)\left(\alpha\frac{2\varrho-1}{3-2\varrho}\right)^q\right\}^2\right].
\end{aligned} \tag{S226}$$

The sum beginning at $q = 1$ is empty for $N \le 2$. For $0 \le \alpha < 1$, a simpler bound is

$$\begin{aligned}
&\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \left[\frac{N}{J(1-\alpha)} + \frac{N^2}{J^2} \right. \\
&\qquad \left. + N^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 \right].
\end{aligned} \tag{S227}$$

The endpoint bounds are

$$\begin{aligned}
&\mathbb{E}\|\nabla_\theta \hat{\ell}^0(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \left[\frac{N}{J} + \frac{N^2}{J^2} + (N-1)^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \right\}^2 \right],
\end{aligned} \tag{S228}$$

$$\mathbb{E}\|\nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \lesssim pG'(\theta)^2 \min\left\{ \frac{N^2}{J}, N + \frac{N^2}{J^2} \right\}. \tag{S229}$$

At $\varrho = \frac{1}{2}$ the population terms are zero. For every model, the population discount bias is zero at $\alpha = 1$. Remark 1 retains signed cancellation.]

Proof. [KT:Insert Equations (S193) and (S153) into Equation (S116). This proves Equation (S226). For $\alpha < 1$,

$$\sum_{r=0}^{N-1} \alpha^r \le (1-\alpha)^{-1},$$

$$\sum_{q=0}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho-1}{3-2\varrho} \right)^q \le \frac{(N-1)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.$$

These inequalities give Equation (S227). At $\alpha = 0$, the memory sum is one and only $q = 0$ remains in the population sum. At $\alpha = 1$, the population term vanishes and

$$\frac{N \min\{N, J\}}{J} + \frac{N^2}{J^2} \le \min\left\{ \frac{3N^2}{2J}, N + \frac{N^2}{J^2} \right\}, \qquad J \ge 2.$$

This proves both endpoint bounds.] □

Corollary S1 (MSE averaged over data and particles)**.** [KT:Suppose the conditions of Proposition S2 and Lemma S13 hold uniformly over the observations, including either (A6) or (A6-J). Let $H_N(\alpha) = \sum_{r=0}^{N-1} \alpha^r$. For either multinomial estimator, evaluated at $\theta = \phi = \theta_0$,

$$\begin{aligned}
&\mathbb{E}_{\theta_0}\mathbb{E}_{\mathrm{PF}}\left[\left\| \nabla_\theta \hat{\ell}_J^\alpha(\theta_0) - \nabla_\theta \ell(\theta_0) \right\|_2^2 \mid Y_{1:N} \right] \\
&\quad \lesssim pG'(\theta_0)^2 \left\{ \frac{N \min\{H_N(\alpha), J\}}{J} + \frac{N^2}{J^2} + N(1-\alpha)^2 \right\}.
\end{aligned} \tag{S230}$$

Here the inner expectation is over the particle filter for the observed data, and the outer expectation is over the data-generating law.]

Proof. [KT:Average the particle/population decomposition in Equation (S116) over the data. The uniform particle bound in Equation (S193) controls its first term, and Proposition S2 controls its second.] □

[KT:For $\alpha < 1$, $H_N(\alpha) \le (1-\alpha)^{-1}$. Balancing the corresponding particle term with the population term gives

$$\frac{N}{J(1-\alpha)} \asymp N(1-\alpha)^2, \qquad 1-\alpha \asymp J^{-1/3},$$

$$\mathbb{E}_{\theta_0}\mathbb{E}_{\mathrm{PF}}\left[\left\|\nabla_\theta \hat{\ell}_J^\alpha - \nabla_\theta \ell\right\|_2^2 \mid Y_{1:N}\right] \lesssim pG'(\theta_0)^2\left\{\frac{N}{J^{2/3}} + \frac{N^2}{J^2}\right\}. \tag{S231}$$

This deterministic choice gives an attainable bound. The endpoint $\alpha = 1$ retains the bound in Equation (S229) and can give the smaller bound for short observation sequences.]

Remark 3 (Neutral-model check)**.** *To check the rates in Lemma S13, [KT:for the after-resampling estimator,] take iid* Uniform$[0,1]$ *states, constant weights at* $\theta = \phi = 0$*, and*

$$\Pr_\theta(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x-1), \qquad y_n^* = 1. \tag{S232}$$

Then $h_n = (2X_n - 1)/2$ *has variance* $\sigma^2 = 1/12$*, and direct conditional multinomial calculation gives*

$$\mathrm{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} = \frac{\sigma^2}{J}\left[N(2-J^{-1}) + \sum_{r=1}^{N-1}(N-r)\alpha^{2r}(1-J^{-1})^{r+1}\right]. \tag{S233}$$

[KT:For fixed $\alpha < 1$, the variance is of order N/J. Its genealogical contribution has order $N/\{J(1-\alpha^2)\}$ as α approaches one, before horizon or coalescence saturation, consistent with the linear memory factor in Equation (S193).] *At* $\alpha = 1$ *and* $J/N \to \infty$*, the same exact formula gives*

$$\mathrm{Var}\left\{\nabla_\theta \hat{\ell}^1(\theta)\right\} \sim \frac{\sigma^2}{J}\left\{2N + \sum_{r=1}^{N-1}(N-r)\right\} = \frac{\sigma^2 N(N+3)}{2J}. \tag{S234}$$

[KT:The full-score endpoint bound in Equation (S229) matches this order when $J/N \to \infty$. This example does not establish sharpness of every term in the general bound.]

Remark 4 (Rates and normalization)**.** [KT:Theorem S3 bounds error relative to the full likelihood score. DMOP-0 has the upper bound

$$pG'(\theta)^2\left[\frac{N}{J} + \frac{N^2}{J^2} + (N-1)^2\left\{\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\right\}^2\right].$$

Thus its particle error can decrease with J while its full-score MSE retains population discount bias. For fixed $0 < \alpha < 1$, the corresponding population term is of order at most $N^2(1-\alpha)^2$ with fixed model constants. At $\alpha = 1$ it is zero, and Equation (S229) gives the full-score upper-bound orders

$$\begin{cases} N^2/J^2, & 2 \le J \le \sqrt{N}, \\ N, & \sqrt{N} < J < N, \\ N^2/J, & J \ge N, \end{cases}$$

suppressing $pG'(\theta)^2$. The ratio of $N+N^2/J^2$ to N^2/J is $J/N+1/J$, so there is an order improvement when $J \to \infty$ and $J/N \to 0$.

A particle MSE bound for $\mathrm{osc}(f) \le 1$ is multiplied by $\mathrm{osc}(f)^2$ when applied to an unnormalized functional. A generic N^2/J bound applied once to a score of oscillation $O(N)$ therefore gives N^4/J. Here the proof bounds the individual measurement-gradient contributions and sums their covariances, so there is no additional N^2 normalization factor. The finite-J MSE theorem is distinct from the fixed-horizon CLT and quadratic asymptotic-variance lower growth of Poyiadjis et al. (2011).]

Remark 5 (Choosing α in the upper bound)**.** *[KT:*Take $N \ge 2$ and fixed $\frac{1}{2} < \varrho < 1$. The explicit population term in Equation (S227) is bounded above and below by positive model constants times $N^2(1-\alpha)^2$. Before the finite memory reaches its cap, balancing it with the particle term gives

$$\begin{aligned} \frac{N}{J(1-\alpha)} &\asymp N^2(1-\alpha)^2, \\ 1-\alpha &\asymp (NJ)^{-1/3}, \\ \mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha - \nabla_\theta \ell\|_2^2 &\lesssim pG'(\theta)^2 \left\{ \frac{N^{4/3}}{J^{2/3}} + \frac{N^2}{J^2} \right\}. \end{aligned} \tag{S235}$$

Comparing with the improved endpoint gives the following attainable full-score bounds:

regime	$1-\alpha$	MSE upper-bound order/$\{pG'(\theta)^2\}$
$2 \le J \le \sqrt{N}$	0	N^2/J^2
$\sqrt{N} < J < N^2$	$\asymp (NJ)^{-1/3}$	$N^{4/3}/J^{2/3}$
$J \ge N^2$	0	N^2/J.

(S236)

Indeed, $N^{4/3}/J^{2/3}$ and N^2/J^2 agree at $J = \sqrt{N}$, while $N^{4/3}/J^{2/3}$ and N^2/J agree at $J = N^2$. For $J \le \sqrt{N}$, any choice $0 \le 1-\alpha \le J^{-1}$ also attains the first order: the particle term is at most N, and the population term is at most a constant times N^2/J^2. The endpoint is therefore a rate-achieving choice, not a claim that $\alpha = 1$ is optimal.

The ratios of the DMOP-1 upper-bound order to the middle-row order are

$$\begin{cases} (J^2/N)^{1/3}, & \sqrt{N} < J \le N, \\ (N^2/J)^{1/3}, & N \le J < N^2. \end{cases}$$

At $J = N$, the bounds are N and $N^{2/3}$. These are comparisons between upper bounds, not lower bounds or actual-MSE dominance.

The finite particle term is capped when $\sum_{r=0}^{N-1} \alpha^r \ge J$. Thus, when $J < N$, it is locally constant near $\alpha = 1$ and does not supply a strict endpoint improvement from discounting. When $J \ge N$, its left derivative at one is $N^2(N-1)/(2J) > 0$, while the squared population term has derivative zero. A small decrease from one then reduces this particular upper bound, though it need not change its order or the actual MSE.

At $\varrho = \frac{1}{2}$, the population term vanishes and $\alpha = 0$ attains $pG'(\theta)^2(N/J + N^2/J^2)$. For $N = 1$, all choices give the same estimator. If the signed first-order population term in Remark 1 vanishes, its squared bound is instead $O(N^2(1-\alpha)^4)$. The same balance then gives

$$1-\alpha \asymp (NJ)^{-1/5},$$

$$\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha - \nabla_\theta \ell\|_2^2 \lesssim pG'(\theta)^2 \left\{ \frac{N^{6/5}}{J^{4/5}} + \frac{N^2}{J^2} \right\}, \tag{S237}$$

again compared with the endpoint bound. This last improvement requires the stated cancellation condition.]

Remark 6 (Geometric discounting and lag truncation)**.** *[KT:*For $0 \le k < N$, define the after-resampling lag truncation and its population counterpart by

$$\nabla_\theta \hat{s}_k^1 = \sum_{n=1}^{N} \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}^J, \qquad \nabla_\theta s_k^1 = \sum_{n=1}^{N} \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}. \tag{S238}$$

Setting $b_s = \min(s+k, N)$, the finite telescope is

$$\nabla_\theta \hat{s}_k^1 = \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^{N} h_s \right).$$

In the direct particle-MSE proof, replace each creation time's geometric endpoint weights by unit mass at b_s. Their total mass is one and $b_s - s + 1 \le k+1$. The predictable local-error sum and ratio remainder are unchanged. The descendant-weight calculation replaces $H_N(\alpha)$ by $k+1$. Hence

$$\begin{aligned} \mathbb{E}\|\nabla_\theta \hat{s}_k^1 - \nabla_\theta s_k^1\|_2^2 &\lesssim pG'(\theta)^2 \left\{ \frac{N \min\{k+1, J\}}{J} + \frac{N^2}{J^2} \right\}, \\ \|\mathbb{E}\nabla_\theta \hat{s}_k^1 - \nabla_\theta s_k^1\|_2 &\lesssim N\sqrt{p}\, G'(\theta)/J. \end{aligned} \tag{S239}$$

For $\frac{1}{2} < \varrho < 1$, Equation (S167) gives

$$\|\nabla_\theta \ell - \nabla_\theta s_k^1\|_2 \le 2(N-k-1)\sqrt{p}\, G'(\theta) \frac{1 - \sqrt{2(1-\varrho)}}{1 + \sqrt{2(1-\varrho)}} \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^k.$$

Combining these two bounds yields the full-score MSE bound

$$\begin{aligned} &\mathbb{E}\|\nabla_\theta \hat{s}_k^1 - \nabla_\theta \ell\|_2^2 \\ &\quad \lesssim pG'(\theta)^2 \left[\frac{N \min\{k+1, J\}}{J} + \frac{N^2}{J^2} \right. \\ &\qquad \left. + (N-k-1)^2 \left\{ \frac{1 - \sqrt{2(1-\varrho)}}{1 + \sqrt{2(1-\varrho)}} \right\}^2 \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^{2k} \right]. \end{aligned} \tag{S240}$$

The population error is zero at $k = N-1$, and at $\varrho = \frac{1}{2}$ for every k. Put $\beta_k = (1-\alpha)\alpha^k$ for $k < N-1$ and $\beta_{N-1} = \alpha^{N-1}$. Since $\sum_{k=r}^{N-1} \beta_k = \alpha^r$,

$$\nabla_\theta \hat{\ell}^\alpha = \sum_{k=0}^{N-1} \beta_k \nabla_\theta \hat{s}_k^1, \qquad \sum_{k=0}^{N-1} \beta_k (k+1) = \sum_{r=0}^{N-1} \alpha^r. \tag{S241}$$

This relation gives comparable particle bounds for comparable memory. It does not equate their full-score MSE bounds, because geometric discounting also changes the coefficients of recent lags.]

S8 Implementation and Figures for Bayesian Inference

S8.1 A common sampling target

For a comparison with a common posterior, we implemented NUTS and particle Metropolis–Hastings (PMMH) using the Dhaka model in Pypomp. The process simulator uses Gaussian noise with 20 Euler steps per monthly observation. We fix the initial state proportions at the library values, with $\rho = 0$, $c = 1$, $\alpha = 1$ and $\delta = 0.02$. Here α is a model parameter, distinct from the DMOP discount parameter. The observation distribution is Gaussian, with mean equal to the simulated monthly deaths and standard deviation equal to τ times that mean. We omit the additive measurement-density floor used in the library's optimization code. Paths that leave the nonnegative state space are assigned zero likelihood for the observed series.

Let x denote the 18 estimation coordinates. These comprise the logarithms of the five positive parameters $\gamma, m, \epsilon, \sigma, \tau$, the trend coordinate $100\beta_{\text{trend}}$, and the six coefficients in each seasonal function. The unused constant environmental-transmission parameter in the earlier implementation is excluded. We use independent Gaussian priors for these coordinates, common to all chains and both methods. For each coordinate, the mean is the midpoint of the interval in Table S-1, and the standard deviation is half its width. These intervals come from the parameter search ranges; they do not truncate the Gaussian distributions. Thus, writing a_i and b_i for the endpoints,

$$z_i = \frac{x_i - (a_i + b_i)/2}{(b_i - a_i)/2}, \qquad p(z) = \prod_{i=1}^{18} \phi(z_i; 0, 1).$$

This specifies a reference prior without fitting a density to IF2 particles. The prior density is evaluated in z, so the sampling calculation requires no further Jacobian factor. On the natural parameter scale, the priors for the five positive parameters are lognormal.

Estimation coordinate	a_i	b_i
$\log \gamma$	$\log 10$	$\log 40$
$\log m$	$\log 0.03$	$\log 0.6$
$\log \epsilon$	$\log 0.2$	$\log 30$
$100\beta_{\text{trend}}$	-1	0
$\log \sigma$	$\log 1$	$\log 5$
$\log \tau$	$\log 0.1$	$\log 0.5$
Seasonal transmission coefficients 1 and 3	-4	4
Seasonal transmission coefficients 2, 4, 5 and 6	0	8
Each seasonal environmental-transmission coefficient	-10	0

Table S-1: Intervals specifying the means and standard deviations of the common Gaussian prior on the estimation scale. A sensitivity calculation doubles all prior standard deviations, keeping the means fixed.

Let u denote the particle-filter random numbers, with law m, and let $\widehat{L}_J(z, u)$ be the bootstrap particle-filter likelihood estimate. The augmented target is

$$\widetilde{\pi}(z, u) \;\propto\; \widehat{L}_J(z, u)\, p(z)\, m(u). \tag{S242}$$

Unbiasedness of the likelihood estimate gives the posterior for the specified Euler model as the marginal distribution of z (Andrieu et al., 2010). The log likelihood and log prior enter without division by the number of observations. We use $J = 10,000$ particles and double precision for this likelihood calculation in both methods.

PMMH jointly proposes $z' = z + hB\xi$, where $\xi \sim N(0, I)$, and independent random numbers $u' \sim m$. Its acceptance probability is

$$1 \wedge \frac{\widehat{L}_J(z', u')p(z')}{\widehat{L}_J(z, u)p(z)}.$$

On rejection, both z and u are retained, together with the cached likelihood estimate. The matrix BB^T is estimated from pilot PMMH draws and held fixed; the scalar h is adapted during 500 warmup iterations. Preparation of the starting points and proposal scales includes numerical optimization with derivatives, although the PMMH transition itself requires only process simulation and measurement-density evaluation.

NUTS first proposes independent random numbers $u' \sim m$, accepting with probability $1 \wedge \widehat{L}_J(z, u')/\widehat{L}_J(z, u)$, and then updates z conditional on the retained u. We use BlackJAX's NUTS trajectory construction, with maximum tree depth 5. DMOP with discount parameter 0.95 and 1,000 particles supplies the proposal force, evaluated in single precision with deterministic GPU operations. Every trajectory weight uses the double-precision likelihood in Equation S242 and the common prior. The force and likelihood are therefore separate numerical calculations. Step sizes are adapted during 500 warmup iterations and then held fixed. The implementation, prior specification and checks of the sampling kernels are available in `code/bayesian/corrected`.

S8.2 ~~Archived sampling calculation and empirical priors~~ [KT:Sampling settings and diagnostics]

~~The code and four saved chains underlying Figure 6 are available in `code/bayesian`, together with the script producing Table S-2 and their original `diffPomp` source revision. This earlier calculation fixes the initial state proportions and uses 19 finite transformed parameter coordinates, denoted by z. Positive rates and noise scales are represented by their logarithms, the trend coordinate is $100\beta_{\text{trend}}$, and the seasonal coefficients are unchanged. The two remaining stored coordinates fix $\rho = 0$ and $c = 1$. An additional coordinate $\log \omega$ is retained in the fitted prior although the seasonal environmental force of infection replaces this constant in the simulator.~~

~~For each chain c, we ran 40 IF2 iterations with 10,000 particles from a random starting point. The perturbation standard deviation was initially 0.02 on the transformed scale and decreased by a factor of 0.95 per iteration. We selected the swarm with the highest monitored likelihood and initialized NUTS at its mean. If its particles are $z_{c,1}, \ldots, z_{c,K}$, with $K = 10,000$, the empirical prior was the joint Gaussian kernel estimate~~

$$\widehat{\pi}_c(z) = \frac{1}{K}\sum_{k=1}^{K} \phi_{19}(z; z_{c,k}, h^2 S_c), \qquad h = K^{-1/(19+4)} \simeq 0.670,$$

~~where S_c is the sample covariance of the swarm. This is the default Scott bandwidth used by `jax.scipy.stats.gaussian_kde`. The prior is defined on the transformed coordinates, so no additional change-of-variables factor enters the calculation in z. Each chain has its own fitted prior.~~

The observations are therefore used both to construct that prior and to evaluate the likelihood, with the fitted density subsequently held fixed.

The numerical log potential supplied to NUTS was

$$a_c(z;u) = \frac{\widehat{\ell}_J(z;u) + \log \widehat{\pi}_c(z)}{N}, \qquad N = 600, \quad J = 1{,}000,$$

where $\widehat{\ell}_J$ is the forward particle-filter log objective returned by the MOP routine, including its measurement-density stabilization, and u denotes the simulation random numbers. The archived sampling code uses $\alpha = 0.95$ for its DMOP derivative. Thus the supplied potential corresponds numerically to a tempered density proportional to $\{\exp(\widehat{\ell}_J)\widehat{\pi}_c\}^{1/N}$, rather than the untempered empirical-prior posterior. The resampling genealogy is recalculated at each candidate parameter; the stop-gradient operations freeze the resampling measurements only for differentiation. The code draws its simulation seed and systematic resampling offsets using NumPy during JAX tracing, so these random numbers are reused by the compiled computation. They are not part of an augmented Markov-chain state, and no separate pseudo-marginal update is applied. This construction does not establish exact posterior sampling.

We used TensorFlow Probability's JAX `NoUTurnSampler`, with step size 0.05, maximum tree depth 5, and maximum energy difference 1,000. There was no step-size adaptation. Each run saved 1,500 draws; the figures discard the first 500. The library's NUTS transition uses the supplied potential and its AD force, without an additional likelihood correction. Only the parameter draws were saved. The fitted prior clouds, kernel diagnostic traces and computation times were not archived, so prior overlays, divergence counts, tree-depth saturation and ESS per second cannot be recovered from these files.

Table S-2 summarizes the retained draws. The between-chain differences include differences in fitted priors, so $\widehat{R}$ here describes disagreement between runs and cannot establish convergence to a common posterior. We calculate ESS and MCSE separately within each chain; these estimates also require approximate stationarity.

[KT:We ran four chains of each method from the same four starting points, obtained by optimization followed by a pilot PMMH run. Figure 6 compares the first 2,000 retained iterations of each chain, after 500 warmup iterations. A longer PMMH run provides 30,000 retained iterations per chain for comparison of the marginal distributions (Figure S-1). No draws from the earlier tempered calculation enter these figures.]

[KT:Both methods use fixed covariance matrices estimated from preliminary PMMH runs. The PMMH covariance uses 1% diagonal shrinkage and a $10^{-6}I$ ridge. For NUTS, we combine the pooled covariance from the first 5,000 retained PMMH iterations per chain with this matrix, with weights 0.75 and 0.25, and rescale to the trace of the PMMH proposal covariance. These matrices affect the proposals but not the prior. Only the scalar step size is adapted during warmup, with target acceptance probabilities 0.65 for NUTS and 0.234 for PMMH. Initial step sizes are 0.07 and 0.5, respectively. The final NUTS step sizes range from 0.048 to 0.060. The proposal matrices, starting points, random seeds and software versions are saved with the code in `code/bayesian/corrected`.]

[KT:Table S-2 gives diagnostics for all 18 coordinates in the 2,000-iteration comparison. For the trend coordinate, NUTS has $\widehat{R} = 1.014$ and bulk ESS 333, compared with 1.025 and 73 for PMMH. The largest NUTS $\widehat{R}$ is 1.059, and the smallest bulk ESS is 60, both for the first seasonal environmental-transmission coefficient. There are two divergent NUTS transitions among 8,000 retained iterations. Almost all trajectories reach the depth limit of 5, corresponding to 31 integration steps. Thus the traces show improved exploration without establishing convergence of all coordinates.]

[KT:The longer PMMH runs give broadly similar marginal distributions (Figure S-1); their largest $\widehat{R}$ is 1.014 and smallest bulk ESS is 202. Doubling all prior standard deviations has little effect on the sampled trend standard deviation

(0.112 versus 0.115), but gives wider lower tails for some seasonal environmental-transmission coefficients. These tails also mix slowly, so the comparison does not establish prior insensitivity. The prior overlays in Figure S-3 make this issue visible. The calculations were run on shared GPUs, and preparation of both samplers used optimization and pilot runs. The iteration comparison therefore does not measure a computing-time speedup.]

		NUTS				PMMH			
	Coordinate	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE
	$\log \gamma$	**1.009**	**243**	**645**	**0.00749**	1.206	15	58	0.028
	$\log m$	**1.019**	**257**	**439**	**0.0118**	1.121	25	122	0.0358
	$\log \epsilon$	**1.025**	**226**	**267**	**0.0305**	1.032	105	259	0.0385
	$100\beta_{\text{trend}}$	**1.014**	**333**	**745**	**0.00593**	1.025	73	165	0.0124
	$\log \sigma$	**1.015**	**303**	**551**	**0.00287**	1.170	18	69	0.0116
	$\log \tau$	**1.012**	**328**	**537**	**0.00562**	1.177	16	94	0.0274
	β_1	**1.006**	**215**	**529**	**0.0313**	1.300	11	26	0.125
[KT:	β_2	**1.009**	**197**	**460**	**0.0247**	1.266	12	43	0.0813]
	β_3	**1.011**	**196**	**284**	**0.0917**	1.312	10	30	0.319
	β_4	**1.008**	**282**	**549**	**0.0101**	1.178	19	62	0.0368
	β_5	**1.004**	**295**	**591**	**0.00877**	1.303	11	149	0.0436
	β_6	**1.007**	**341**	**640**	**0.00578**	1.114	34	200	0.0181
	ω_1	**1.059**	**60**	**214**	**0.135**	1.114	34	61	0.188
	ω_2	**1.027**	**297**	**416**	**0.0634**	1.079	39	129	0.157
	ω_3	**1.027**	**220**	**381**	**0.0463**	1.155	17	49	0.153
	ω_4	**1.015**	**278**	**542**	**0.0405**	1.177	15	37	0.166
	ω_5	**1.010**	**293**	**335**	**0.118**	1.168	18	68	0.495
	ω_6	**1.056**	**109**	**116**	**0.396**	1.111	42	59	0.528

Table S-2: ~~Diagnostics for the four archived empirical-prior NUTS runs, using 1,000 retained draws per run. The rank-normalized split $\widehat{R}$ compares all four runs, whose fitted priors differ. Bulk and tail ESS are the smallest within-chain values, and MCSE is the largest within-chain Monte Carlo standard error of the mean. All quantities use the displayed transformed coordinates. The fixed coordinates $\rho = 0$ and $c = 1$ are excluded. Divergences and tree-depth diagnostics were not saved.~~[KT:Diagnostics for four chains of each method, using 2,000 retained iterations per chain after 500 warmup iterations. Rank-normalized split $\widehat{R}$, bulk and tail ESS, and the Monte Carlo standard error of the mean (MCSE) are calculated across chains on the displayed estimation coordinates. All chains have the common prior and target specified above. Bold entries indicate $\widehat{R}$ closer to 1, larger ESS, or smaller MCSE.]

S8.3 Additional sampling figures

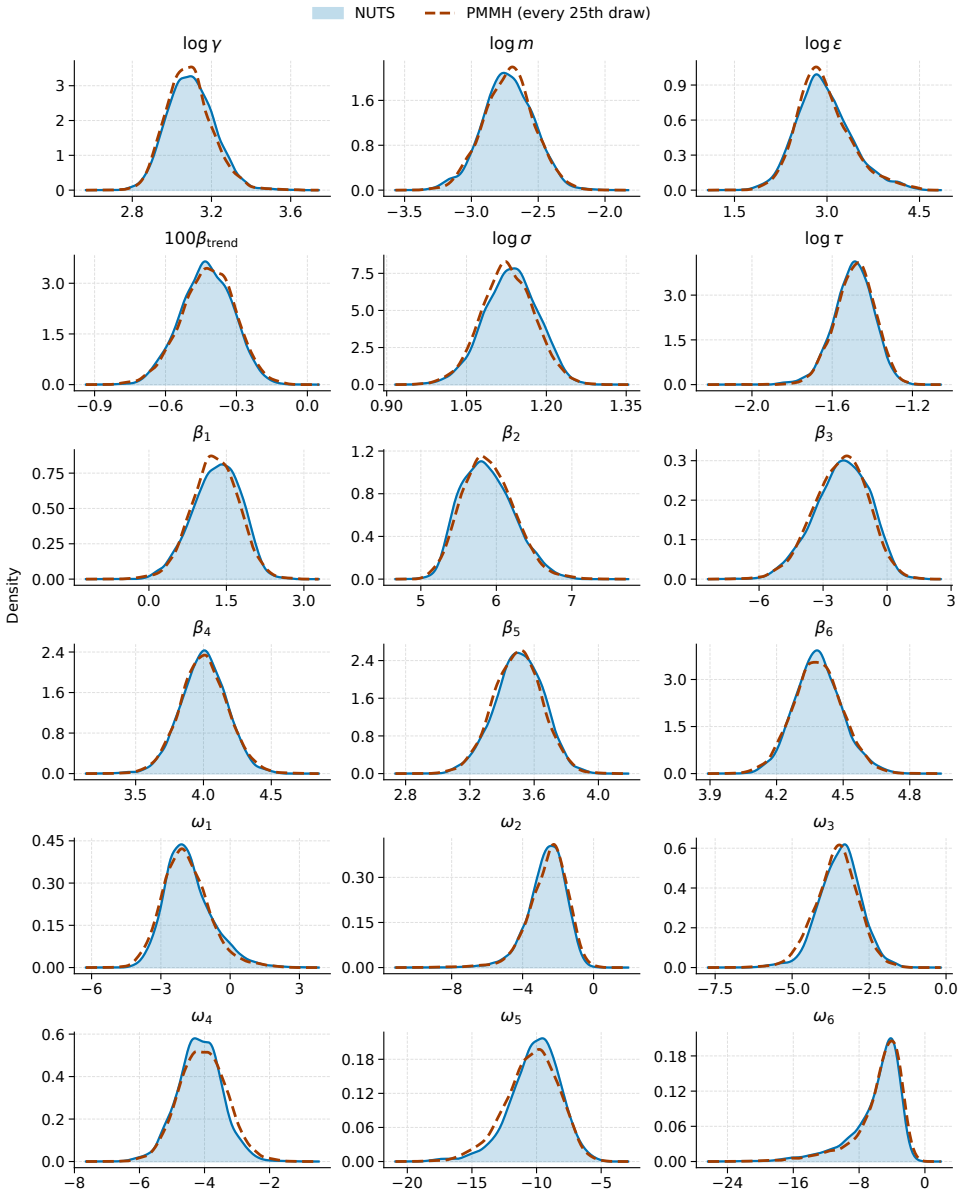


Figure S-1: ~~Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.~~[KT:Marginal sample distributions for all 18 estimation coordinates, pooled over four chains of each method. NUTS uses 2,000 retained iterations per chain (blue fill). To reduce serial dependence in the displayed PMMH sample, we take every 25th retained draw from each 30,000-iteration run, giving 1,200 draws per chain (orange dashed curve). Both curves use the Gaussian-kernel bandwidth selected by Scott's rule for the thinned PMMH sample. Diagnostics use all draws; thinning is only for this display. Both methods use the common prior and untempered target.]

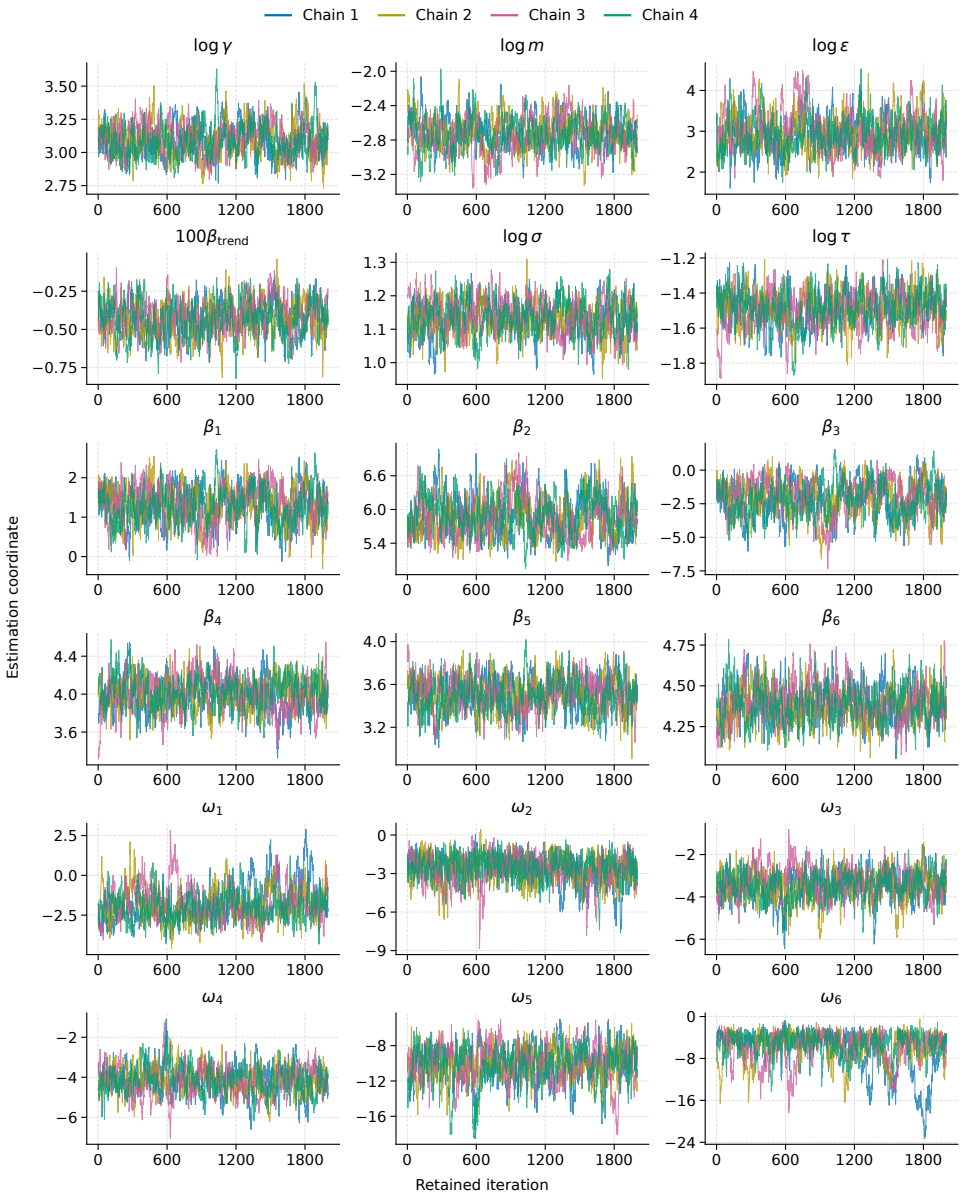


Figure S-2: ~~Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. Comparison with Fig. 6 in the main text suggests that the empirical prior assists exploration, though it also changes the sampling target.~~[KT:NUTS traces for all 18 estimation coordinates. Colors identify the four chains, with 2,000 retained iterations per chain after 500 warmup iterations. The long excursions of some seasonal environmental-transmission coefficients are less apparent in the trend diagnostic alone.]

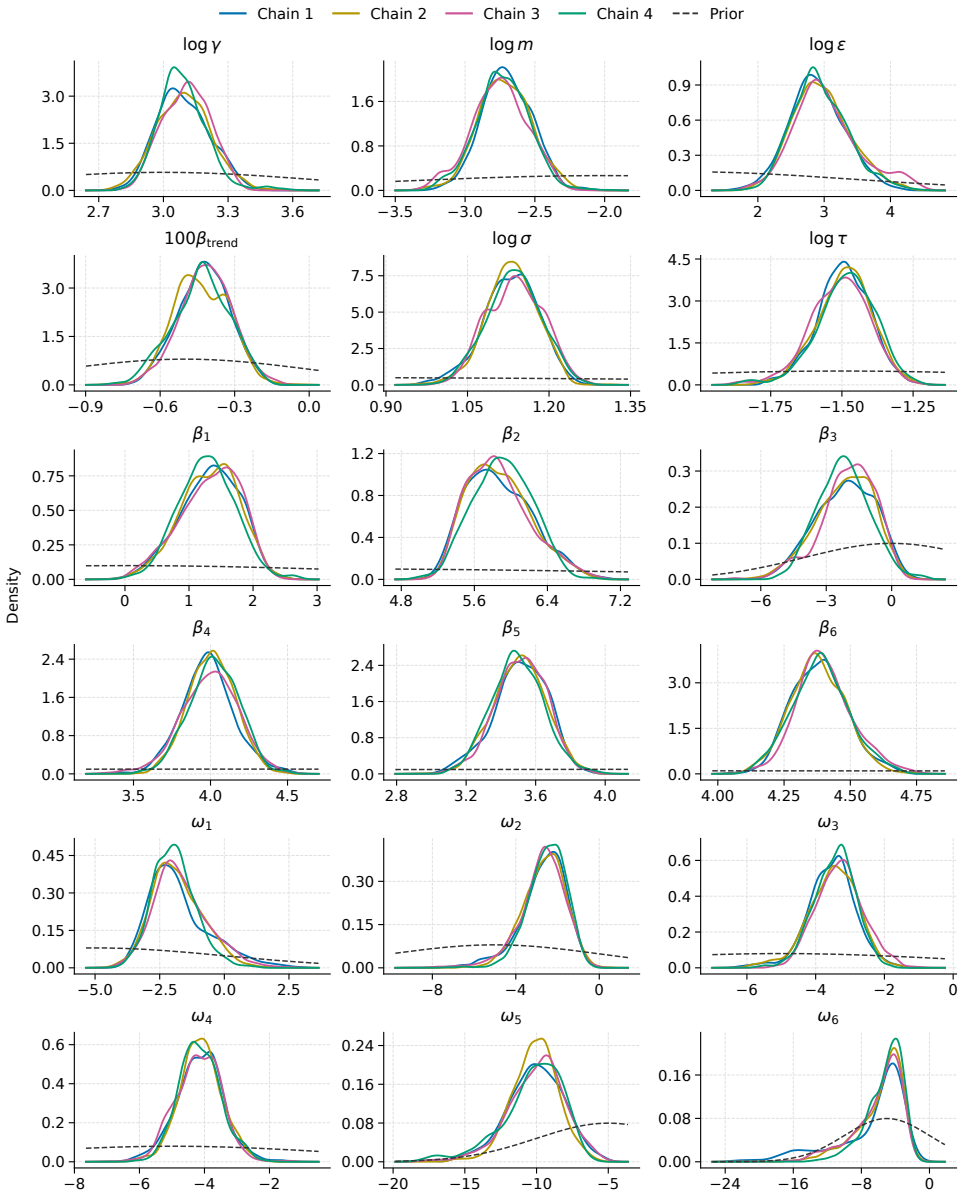


Figure S-3: ~~Marginal sample distributions from NUTS powered by MOP-α, with a separate empirical prior fitted to each chain's IF2 parameter cloud. The calculation and diagnostics are given above.~~[KT:Marginal sample distributions from the four NUTS chains for all 18 estimation coordinates, using 2,000 retained iterations per chain. Dashed curves show the common Gaussian prior. Density estimates use Gaussian kernels with Scott bandwidths. The prior is not fitted to the observations or to an IF2 parameter cloud.]

S9 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-3. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0, are treated differently: they are perturbed only at time t_0, with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	**Standard IF2**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	1,950	60	175	60
GD Iterations	—	410	400	425
IF2 Cooling Rate	0.93	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-3: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemnenteed by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2015). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not reqquire it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

S10 Application to a Daphnia Population Model

S10.1 Model and data

The SIRJPF2 model developed by Yang et al. (2025) describes interactions between two species of *Daphnia*, their algal food, and a fungal parasite. We use the version with unit-specific infected-adult mortality to compare likelihood maximization by IFAD and the marginalized panel iterated filter (MPIF) of Wheeler et al. (2025).

The data come from the mesocosm experiment of Searle et al. (2016). We use the treatment containing the native species *D. dentifera*, the invasive species *D. lumholtzi*, and the parasite *Metschnikowia bicuspidata*, with eight mesocosms, each treated as one unit in the panel. Each mesocosm was observed at ten times, $t_n = 5n + 2$ days for $n = 1, \ldots, 10$. At each observation time, a one-liter sample was taken and susceptible and infected adult females of both species were counted. Juvenile counts are excluded from the likelihood.

We write $S_u^a(t)$, $I_u^a(t)$, and $J_u^a(t)$ for the densities of susceptible adult females, infected adult females, and juveniles, where $u = 1, \ldots, 8$ indexes the mesocosm and $a \in \{\mathrm{n}, \mathrm{i}\}$ denotes the native or invasive species. The densities of food and parasite spores are $F_u(t)$ and $P_u(t)$. These variables form an eight-dimensional latent state in each mesocosm. Removal by death, consumption, or sampling is denoted by R in the model name and does not require an additional state variable. The two host species interact through their shared food and parasite populations. The processes in different mesocosms are independent conditional on the parameters.

As in Yang et al. (2025), the transition dynamics follow an Itô stochastic differential equation (SDE):

$$
\begin{aligned}
dS_u^a &= \{\lambda_J J_u^a - (\theta_S^a + p^a f_S^a P_u + \delta) S_u^a\}\, dt + \sigma_S^a S_u^a\, dB_{S,u}^a, \\
dI_u^a &= \{p^a f_S^a S_u^a P_u - (\theta_{I,u}^a + \delta) I_u^a\}\, dt + \sigma_I^a I_u^a\, dB_{I,u}^a, \\
dJ_u^a &= \{r^a f_S^a F_u S_u^a - (\theta_J^a + \delta + \lambda_J) J_u^a\}\, dt + \sigma_J^a J_u^a\, dB_{J,u}^a, \\
dP_u &= \left\{ \sum_a \left[\beta \theta_{I,u}^a I_u^a - f_S^a (S_u^a + \xi I_u^a) P_u\right] - (\theta_P + \delta) P_u \right\} dt + \sigma_P P_u\, dB_{P,u}, \\
dF_u &= \left\{ \mu - \delta F_u - \sum_a f_S^a F_u (S_u^a + \xi I_u^a + J_u^a) \right\} dt + \sigma_F F_u\, dB_{F,u},
\end{aligned}
$$

with independent standard Brownian motions B. The time argument is suppressed in the equations. Except for infected-adult mortality, estimated parameters are shared across mesocosms. The parameters r^a and f_S^a are the reproductive rate and host filtration rate, p^a controls infection following spore ingestion, and ξ is the filtration rate of infected adults relative to susceptible adults. The mortality or loss rates of susceptible adults, infected adults, juveniles, and spores are θ_S^a, $\theta_{I,u}^a$, θ_J^a, and θ_P, respectively; the σ parameters are process-noise coefficients. The relative filtration rate of juveniles is fixed at one.

We use the units of the public Python implementation:[1] *Daphnia* densities are individuals per liter, F is in 10^6 algal cells per liter, and P is in 10^3 spores per liter, with parameters expressed in the corresponding units. The sampling loss rate is fixed at $\delta = 0.013$ per day, the food replenishment rate at $\mu = 0.37$, the juvenile maturation rate at $\lambda_J = 0.1$ per day, and the spore yield per

[1] `pypomp/Daphnia-tutorial`, advanced tutorial, revision `f318330`.

infected-host death at $\beta = 30$, corresponding to 3×10^4 spores. The initial states are $S_u^{\mathrm{n}} = 2.333$, $S_u^{\mathrm{i}} = 0.667$, and $F_u = 16.667$, with the remaining states set to zero. Simulation begins at $t_0 = 1$ day, and parasites are introduced on day 4 at a density of 25×10^3 spores per liter. We numerically solve the SDE using the Euler-Maruyama method with a time step of 0.25 days, following the initial conditions, inoculation, and numerical boundary rules of the public implementation.

The measurement model for the adult counts is

$$Y^a_{S,u,n} \sim \mathrm{NBinomial}(S^a_u(t_n), \tau^a_S), \qquad Y^a_{I,u,n} \sim \mathrm{NBinomial}(I^a_u(t_n), \tau^a_I),$$

where $\mathrm{NBinomial}(m, \tau)$ has mean m and variance $m + m^2/\tau$ (Yang et al., 2025). The four counts at each observation time are independent conditional on the latent state. Since the sampled volume is one liter, the state densities and observed counts are on the same scale. The numerical implementation sets a lower bound of 10^{-10} on the negative binomial mean and size parameters. As in the public Python code, a state variable that crosses a numerical boundary is reset to zero, and that particle's log weight for the mesocosm at the next observation is set to -150. The boundary thresholds and the treatment of the inoculation time are specified in the code.

We fix $\sigma_S^{\mathrm{n}} = \sigma_S^{\mathrm{i}} = 0$ and estimate $\theta_{I,u}^{\mathrm{n}}$ and $\theta_{I,u}^{\mathrm{i}}$ separately for each mesocosm. The remaining 22 estimated parameters are shared across mesocosms, giving $22 + 2 \times 8 = 38$ estimated parameters. This is the model with unit-specific infected-adult mortality in Table S3 of the archived revision (Yang et al., 2026).

S10.2 Experimental settings

We tested IFAD with discount factors $\alpha = 0$, $\alpha = 0.97$, and $\alpha = 1$ against MPIF using Pypomp 1.0.5 (Abkemeier et al., 2025). For each method, we performed 50 searches from the same starting parameter vectors, generated by independent lognormal perturbations of the initial parameter vector in the public tutorial, with a standard deviation of 0.45 on the log scale. All positive estimated parameters were optimized on the log scale. Each IFAD search used an MPIF warm start of 200 iterations followed by 200 iterations of the Adam optimizer (Kingma and Ba, 2015), using DMOP-α gradient estimates. We ran MPIF alone for 680 iterations, chosen to approximate the computing time of IFAD; measured times are reported in Table S-5.

We used 1,000 particles per mesocosm during optimization. The initial random walk standard deviation (RWSD) for MPIF was 0.05 on the log scale, with geometric cooling by a factor of 0.7 over 50 iterations. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with all eight mesocosms contributing to each update. The learning-rate schedule used ten steps of linear warmup multiplied by cosine decay with a final factor of 0.1. A summary of the algorithmic parameters is provided in Table S-4.

The learning-rate scale was selected using four additional starting points, separate from the 50 used for the comparison. We compared the six combinations of learning-rate scales 0.001, 0.005, and 0.01 with $\alpha \in \{0.9, 0.97\}$. The combination with the highest median final log-likelihood, evaluated by independent particle filtering, was 0.01 with $\alpha = 0.97$. We used the same learning-rate schedule for $\alpha = 0$ and $\alpha = 1$.

All reported log-likelihoods were evaluated independently of optimization using ordinary particle filters, with 2,000 particles per mesocosm and ten replicates. Likelihood estimates were averaged within each mesocosm before taking logarithms, and the eight mesocosm log-likelihoods were then summed. Each method's best parameter vector was then held fixed and re-evaluated with ten

Hyperparameter	**MPIF**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Particles per mesocosm (J)	1,000	1,000	1,000	1,000
MPIF Iterations	680	200	200	200
Adam Iterations	—	200	200	200
MPIF Cooling Factor	0.7	0.7	0.7	0.7
Initial RWSD (log scale)	0.05	0.05	0.05	0.05
Momentum (β_1)	—	0.9	0.9	0.9
Learning-rate Scale	—	0.01	0.01	0.01
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	2,000 / 10	2,000 / 10	2,000 / 10	2,000 / 10

Table S-4: Algorithmic parameters for the Daphnia experiment. Particle numbers are per mesocosm; the cooling factor is the multiplier over 50 iterations.

fresh replicates at the same particle count, using the same model and Pypomp version on a CPU. The same evaluation method was used for the final results and for intermediate saved parameter vectors. The progress plot uses 25, 27, 26, and 22 saved parameter vectors per search for IFAD-0, IFAD-0.97, IFAD-1, and MPIF, respectively, including the final evaluations. Time is assigned in proportion to iteration number within each stage, using the measured total optimization time for that stage. Each set of 50 searches ran on one Tesla V100 16 GB GPU, with two searches per batch. Optimization times are summed over the 25 batches and include initial compilation but exclude the additional particle filter evaluations.

S10.3 Results

Our findings are summarized in Table S-5 and Figures S-4 and S-5. IFAD-0.97 and IFAD-1 attained maximum log-likelihoods of −858.48 and −858.27, respectively, each with a Monte Carlo standard error (MCSE) of approximately 0.17. The maximum for MPIF was −860.06, with MCSE 0.16. IFAD-0 attained −860.36 and had a median log-likelihood close to that of MPIF.

	Maximum	Re-evaluated	Median	MPIF time (s)	DMOP time (s)
IFAD-0.97	−858.48 (0.17)	−858.23 (0.16)	−861.42	535	1,309
IFAD-1	−858.27 (0.17)	−858.00 (0.14)	−861.46	520	1,313
IFAD-0	−860.36 (0.24)	−860.30 (0.21)	−864.12	503	1,314
MPIF	−860.06 (0.16)	−860.26 (0.14)	−864.27	1,591	-

Table S-5: Log-likelihoods from 50 searches per method, with MCSEs in parentheses. Re-evaluated uses fresh particle filters at each method's original maximizing parameter vector.

Comparing searches from the same starting points, IFAD-0.97 and IFAD-1 each attained a higher final log-likelihood than MPIF in 49 of the 50 searches. The median paired differences were 2.86 and 2.79 log-likelihood units, respectively. The distributions of the final log-likelihoods for IFAD-0.97 and IFAD-1 were similar, which may be due to the short time series in each mesocosm.

Yang et al. (2026) report a maximum log-likelihood of −863.03 in the revised Table S3. Our MPIF searches attained a higher value, and IFAD-0.97 and IFAD-1 improved on this further.

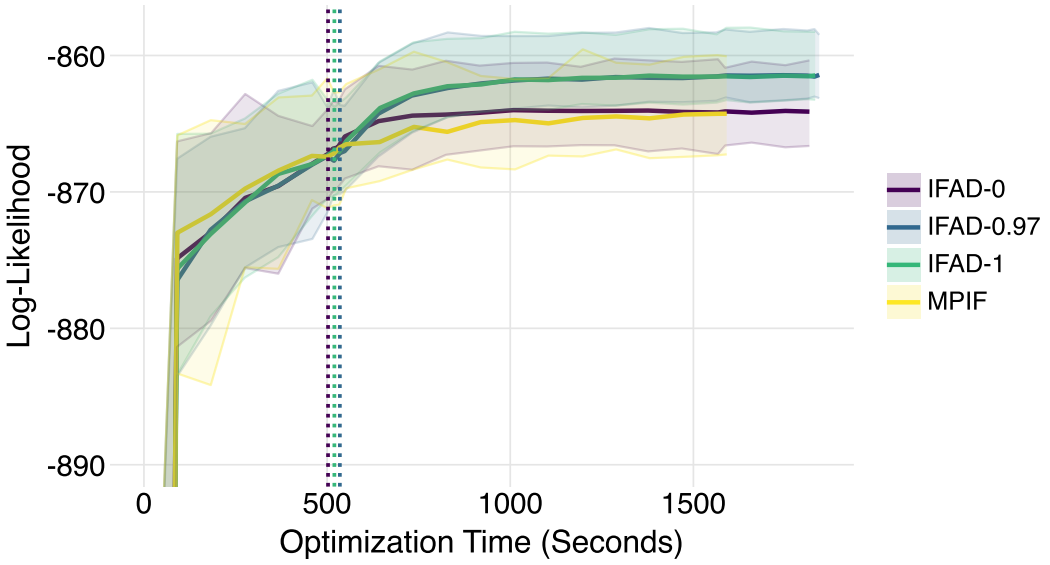


Figure S-4: Optimization progress of IFAD and MPIF for the Daphnia model. The solid lines show the median log-likelihood across 50 searches, and the shaded regions cover the area between the 10th and 100th percentiles. The vertical dotted lines indicate where the IFAD searches switch from MPIF to Adam optimization. The plot shows log-likelihoods above −890; all searches contribute to the percentiles.

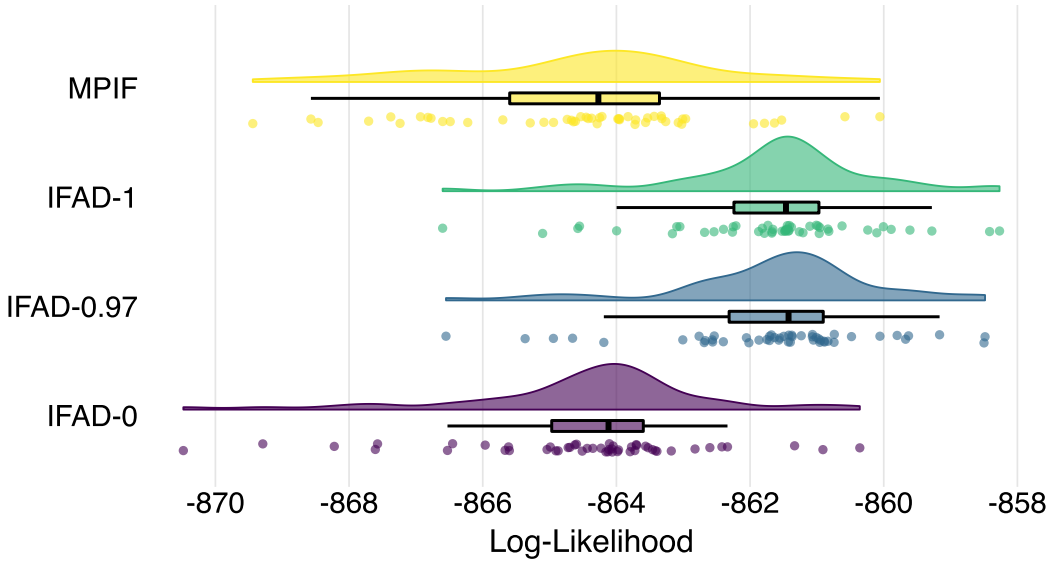


Figure S-5: Raincloud plot of the searches for the Daphnia model. The half-violin plots show the distributions of all 50 final log-likelihood estimates for each method, the boxplots show the medians and quartiles, and each point represents one search. IFAD-0.97 and IFAD-1 have similar distributions, with generally higher log-likelihoods than MPIF and IFAD-0.

S11 Comparison with Other Methods for the Dhaka Model

We compared IFAD with the transition-density method of Ditlevsen and Samson (2019) (DS19) and the differentiable particle filter of Corenflos et al. (2021) (CTDD21) on the Dhaka cholera model. We tested whether these methods could find high-likelihood parameter estimates from the wide range of starting points used in the main text, and whether they could improve estimates obtained by IF2. We describe our implementations below.

S11.1 Methods

DS19. The method of Ditlevsen and Samson (2019) combines an approximate transition density with particle smoothing and stochastic approximation expectation maximization (SAEM). Their construction assumes that one smooth state coordinate is observed directly and that noise acts directly on the remaining coordinates. Applying it to Dhaka requires some modifications: the six state coordinates (S, I, R^1, R^2, R^3, M) are driven by one Brownian motion, and the monthly death counts are observed with measurement error.

The SAEM objective includes the process log-density, which DS19 must therefore be able to evaluate. Replacing the SAEM maximization with a score update retains this requirement: the score includes derivatives of the process log-density, evaluated along the imputed path. Our backward smoother also evaluates the transition density to sample that path. For Dhaka, a single Euler step is driven by one Brownian increment and has no density with respect to Lebesgue measure on the six-dimensional state space. Composing 20 Euler steps gives a monthly transition that we can simulate, but whose density we cannot evaluate directly after integrating out the intermediate states. IF2 and IFAD use this simulator without evaluating its transition density; IFAD additionally differentiates through the simulation. They retain the Euler discretization error but do not require a separate approximation to the monthly transition density.

We approximate the transition over each month by a Gaussian distribution. The local strong-order-1.5 approximation has a covariance matrix of rank at most two, so it does not have a density in the six-dimensional state space. We instead calculate the mean and covariance over a full month by propagating the second-order mean and covariance through 20 time steps. We linearize the dynamics at each step and integrate out the intermediate states. The resulting monthly covariance matrix can have full rank, although it remains poorly conditioned. To stabilize the calculation, we add 10^{-12} times its largest diagonal entry to each diagonal entry. This gives a monthly transition density that differs from the Euler approximation used to evaluate the final parameter estimates.

The rank deficiency also affects parameter updates on an augmented path. At a fixed interior state, the range of the unregularized local covariance is

$$\operatorname{span}\left\{(-1,1,0,0,0,0)^{\mathsf{T}},\ (0,-\gamma-m,\gamma,0,0,m)^{\mathsf{T}}\right\}.$$

This follows by writing the diffusion vector as $g = q(-1,1,0,0,0,0)^{\mathsf{T}}$: its diffusion derivatives are parallel to g, while the drift derivative $(Db)g$ supplies the second direction. The plane depends on m/γ. Changing this ratio changes the support of the local Gaussian transition, so augmenting the states at every internal step does not provide an ordinary full-state density for parameter optimization. Changes in the transition mean can also change its affine support. An augmentation in terms of driving noise would require a different parameter-update scheme. Our implementation adds the covariance floor only after composing the monthly transition, not at each internal step.

There is a separate difficulty with finely augmented paths even when the transition densities are nonsingular. As the time step decreases, transitions become concentrated and observations at intermediate steps are uninformative, which can make particle resampling and backward sampling ineffective (Karppinen et al., 2024). This concerns mixing of the path sampler and does not by itself establish mutual singularity of path laws at different parameter values. Our smoother samples monthly endpoints, so it does not assess this fine-grid behavior. We restrict the comparison to this implementable adaptation of DS19; constructing and validating a different augmented-path method is beyond the present comparison.

We use each month's observation to guide the particle filter toward states that are consistent with the observed death count. For each particle, we calculate the measurement variance using the predicted monthly death count and hold this variance fixed when conditioning on the observation. This gives a Gaussian distribution from which we sample the next state. We weight each sample by the original measurement density multiplied by the ratio of the transition density to the proposal density.

At each optimization step, we run the particle filter through the data, then sample a trajectory of monthly states backward through the stored particles. We differentiate the complete-data log-likelihood of this trajectory and average the resulting scores using stochastic approximation. For the 23-parameter Dhaka model, these searches use averaged-score ascent. This differs from averaging and maximizing the complete-data objective in SAEM. Section S12.1 gives both updates and describes the separate numerical-SAEM comparison. We clip the scores and use backtracking, with periodic checks of the log-likelihood under the monthly Gaussian transition approximation. The 100- and 1,000-particle fits use 100 particles for these checks; the separately tuned 5,000-particle experiment uses 5,000. We also checked the particle filter on the harmonic oscillator example of Ditlevsen and Samson (2019), for which an analytic Gaussian filter is available. With 100 particles and 1,000 observations, the root mean squared difference between the particle and analytic filtered means was 0.00523.

CTDD21. The method of Corenflos et al. (2021) uses entropy-regularized optimal transport to make particle resampling differentiable. We implemented it using the same Euler simulator as in the main text, with 20 steps per month. The transport calculation uses all six state coordinates, scaled as in the published implementation. We solve the transport problem using Sinkhorn iterations, gradually decreasing the regularization parameter to 0.25 with an annealing scale of 0.75. We use a convergence tolerance of 10^{-3} and a maximum of 100 Sinkhorn iterations. To calculate derivatives, we hold the converged potentials fixed and differentiate through one further fixed-point update. As in the published implementation, we clip each component of the derivative propagated through the transport step to $[-1, 1]$. We resample when the effective sample size falls below half the number of particles.

We maximize the resulting log-likelihood estimate using Adam, with new random numbers for the particle filter at each iteration and a gradient norm limited to 100. If a search becomes invalid or repeatedly deteriorates, we return to its best valid parameter estimate, reset the Adam moments, and halve the learning rate. We allow at most six such restarts. Entropic resampling changes the particle filter's likelihood approximation, so the objective used for optimization can differ from the ordinary particle filter estimate used to evaluate the results.

S11.2 Experimental settings

For each method, we performed 100 global searches from starting parameter vectors drawn uniformly from the same wide bounding box as in the main text. We tested 100 and 1,000 particles, using learning rates of 0.02 for CTDD21 and 0.10 for DS19. We allowed each search 800 seconds and selected parameter estimates at approximately 700 seconds. For CTDD21, we selected the saved estimate with the highest fitting objective before 700 seconds. If none was valid, we used another valid saved estimate or the initial parameter vector. For DS19, we selected the first saved estimate at or after 700 seconds, or the last estimate if the search ended earlier. Thus a DS19 search could exceed the selection time by one iteration.

To assess the benefit of an IF2 warm start, we also initialized both methods at the 100 IF2 estimates used to initialize IFAD-0.97 in the main text's comparable computational effort experiment. We used these estimates directly, including those outside the initial bounding box. The IF2 stage used 175 iterations and 5,000 particles, taking 92.160 seconds under the timing procedure described in the main text. We then allowed each method 400 iterations or another 850.657 seconds, giving a total time limit of 942.817 seconds. The learning rates were 0.0002 for CTDD21 and 0.0005 for DS19, at both 100 and 1,000 particles. We selected the last valid CTDD21 estimate before the time limit and the first DS19 estimate at or after it, or the last estimate if the search ended earlier.

We kept the other optimization settings fixed when increasing the number of particles from 100 to 1,000. The 1,000-particle experiments ran sequentially on one RTX 3090 GPU, as did the 100-particle experiments initialized by IF2. The 100-particle DS19 global searches ran four at a time on that GPU. Consequently, both the particle count and GPU scheduling affect the number of iterations completed within the time limit. We also ran DS19 with 5,000 particles from the global starting points. For that experiment, we separately tuned the learning rate to 0.20 and began decreasing the stochastic approximation step size earlier.

We evaluated all selected parameter vectors independently using the Euler approximation with 20 steps per month, including estimates from searches that ended early. Each evaluation used 36 particle filter replicates with 5,000 particles. We averaged the likelihood estimates before taking the logarithm. For the optimization progress plots, we also evaluated each saved iterate using one replicate with 5,000 particles. These plots show the iterates visited during optimization, which can differ from the estimates selected for the final comparison. As in the main text, we excluded the additional evaluations from the reported computation times.

S11.3 Results

Our findings are summarized in Tables S-6 and S-7 and Figures S-6–S-9. The figures show CTDD21 with 100 particles and DS19 with 1,000, using the same particle count for each method with and without an IF2 warm start. IFAD-0.97 uses 5,000 particles. The tables include all particle counts tested.

Both methods attained lower median log-likelihoods than IFAD-0.97 in the global searches. At the particle counts shown in the figures, the medians were -4554.67 for CTDD21 and -3998.93 for DS19, compared with -3744.40 for IFAD-0.97. Increasing the number of particles did not consistently improve performance within the allotted time. For CTDD21, increasing from 100 to 1,000 particles reduced the median number of updates from 255 to 28, and the median log-likelihood fell to -6312.89. For DS19, the median improved from -4100.34 with 100 particles to -3998.93 with 1,000 particles, while the median number of updates rose from 55 to 147. This comparison

also includes the change in GPU scheduling described above. With 5,000 particles and separately tuned settings, DS19 attained a median of −4100.40 after a median of 47 updates.

Method	Start	J	Median	Maximum	Updates
CTDD21	Box	100	-4554.67	-3842.43	255
CTDD21	Box	1,000	-6312.89	-4366.30	28
DS19	Box	100	-4100.34	-3872.97	55
DS19	Box	1,000	-3998.93	-3798.03	147
DS19	Box	5,000	-4100.40	-3871.57	47
CTDD21	IF2	100	-3765.91	-3749.10	213
CTDD21	IF2	1,000	-3766.61	-3749.23	20
DS19	IF2	100	-3768.72	-3752.20	252
DS19	IF2	1,000	-3769.04	-3751.03	171
IF2 checkpoint	Box	5,000	-3766.64	-3749.30	175
IFAD-0.97	IF2	5,000	-3744.40	-3743.68	400

Table S-6: Log-likelihoods from 100 searches per method and particle count for the Dhaka model. J is the number of particles used for optimization. Box denotes initialization from the bounding box, and IF2 denotes an IF2 warm start. Median and maximum use independent particle filter evaluations with 20 Euler steps per month. Updates gives the median number of optimization updates. The IF2 checkpoint row reports the 175 iterations used for initialization; the IFAD-0.97 row reports the 400 subsequent gradient updates. The 5,000-particle DS19 experiment used separate tuning, as described in the text.

Starting from IF2 estimates substantially improved the final log-likelihoods of both methods, but neither consistently improved on the IF2 estimates themselves. With 100 particles, CTDD21 improved on 45 of the 100 starting estimates and DS19 improved on 18. The median changes were −0.10 and −1.78 log-likelihood units, respectively. With 1,000 particles, these counts were 48 and 24, and the median changes were −0.03 and −1.15. CTDD21 completed a median of 213 updates with 100 particles and 20 with 1,000 particles; DS19 completed 252 and 171, respectively. IFAD-0.97 continued from the same IF2 estimates to a median log-likelihood of −3744.40, compared with −3766.64 at the end of IF2.

Method	J	Fitting change	Euler-20 change	Improved
CTDD21	100	5.56	-0.10	45/100
DS19	100	17.24	-1.78	18/100
CTDD21	1,000	0.73	-0.03	48/100
DS19	1,000	15.21	-1.15	24/100

Table S-7: Changes during optimization from an IF2 warm start. For each search, we subtract the value at its IF2 starting point from the value at its selected parameter estimate. Fitting change and Euler-20 change give the medians of these differences for the fitting objective and the independently evaluated log-likelihood. Improved is the number of searches whose independently evaluated log-likelihood increased.

For both methods and both particle counts, the median change in the fitting objective was positive even though the median change in the independently evaluated log-likelihood was negative

(Table S-7). The two evaluations use different likelihood approximations: DS19 fits the monthly Gaussian transition model, while CTDD21 uses entropic resampling. The fitting objectives are also estimated using a single particle filter, and the updates depend on clipping and the numerical safeguards described above. These experiments do not determine how much each of these factors contributes to the disagreement.

S11.4 Reproducibility

Source code and algorithmic settings are available at https://github.com/pypomp/dmop, in the `ditlevsen/` and `corenflos/` directories. The file `imgs/competitors/README.md` gives instructions for reproducing the tables and figures, identifies the numerical results used, and links to plots showing the full range of log-likelihoods. The repository includes the tables and figures needed to compile the supplement. Instructions for retrieving archived parameter estimates and evaluations at individual iterations are in `artifacts/README.md`.

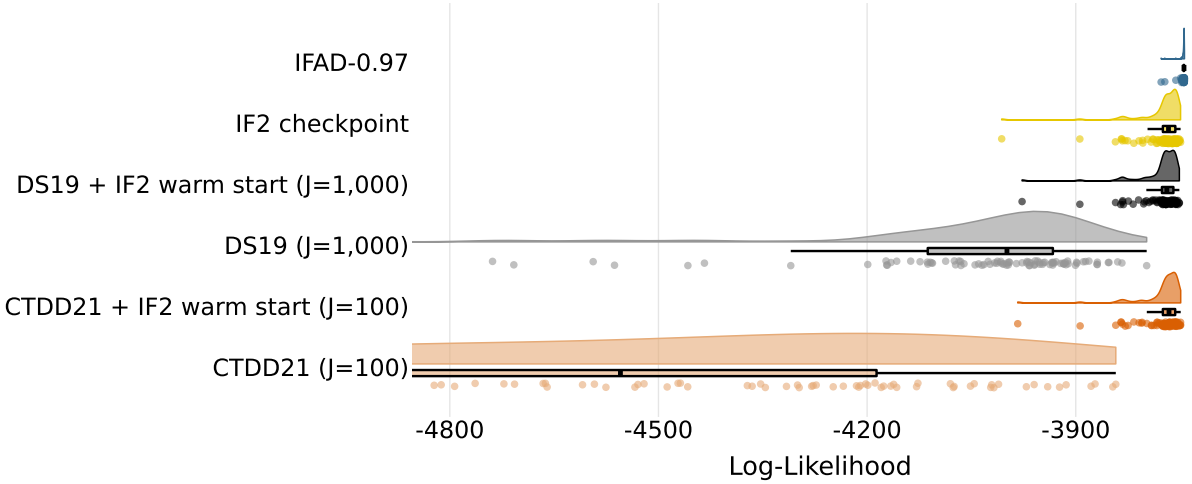


Figure S-6: Raincloud plots of the log-likelihoods found in 100 searches for the Dhaka model, with and without an IF2 warm start. CTDD21 uses 100 particles, DS19 1,000, and IFAD-0.97 5,000 during optimization. The IF2 checkpoint shows the starting estimates for the warm starts. The plot shows log-likelihoods between −4800 and −3735; all 100 searches contribute to the densities, medians, and quartiles. Each half-violin is scaled to the same maximum height.

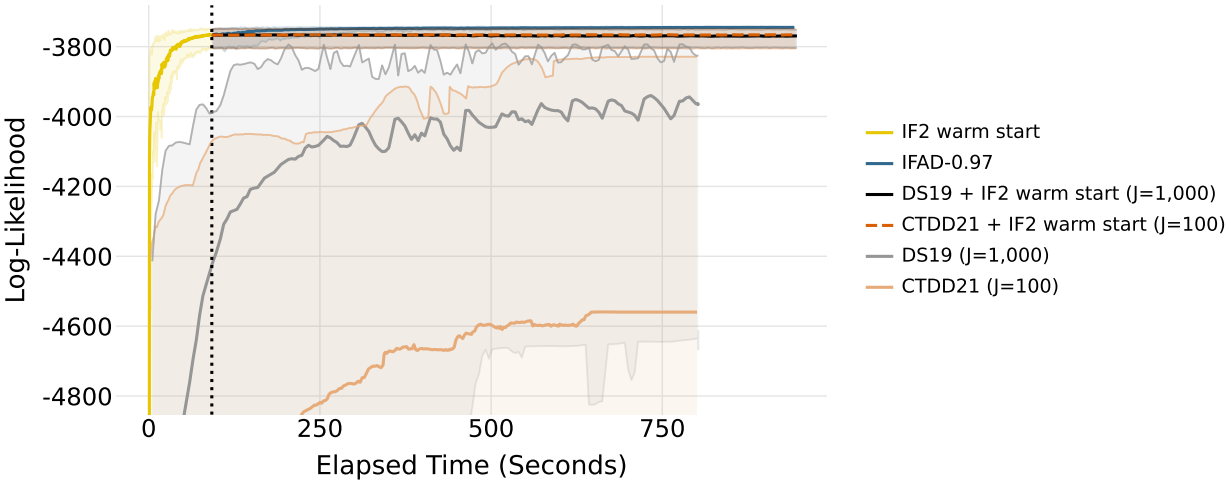


Figure S-7: Optimization progress for the Dhaka model, with and without an IF2 warm start. CTDD21 uses 100 particles, DS19 1,000, and IFAD-0.97 5,000 during optimization. Heavy lines show the median log-likelihood, and shaded regions cover the area between the 10th and 100th percentiles. The vertical dotted line indicates the end of IF2 at 92.160 seconds. The plot shows log-likelihoods between −4800 and −3735; all 100 searches contribute to the percentiles. Each saved iterate is evaluated with one particle filter replicate, compared with 36 replicates for the selected estimates in Table S-6, using 5,000 particles in both cases.

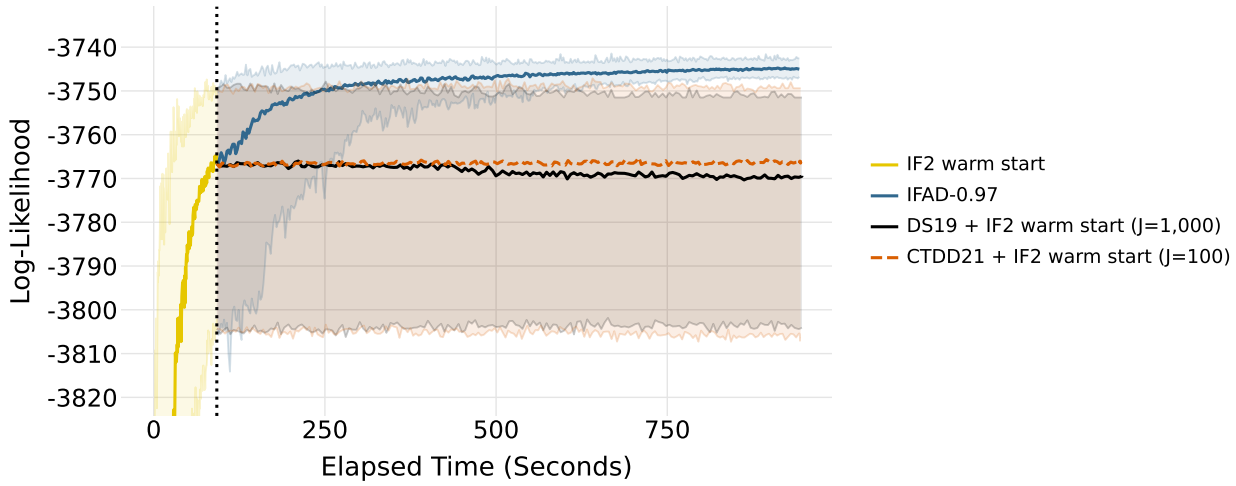


Figure S-8: Optimization progress with an IF2 warm start, showing log-likelihoods between −3820 and −3735. Particle counts, lines, and shading are as in Figure S-7. Elapsed time includes the 92.160-second IF2 stage.

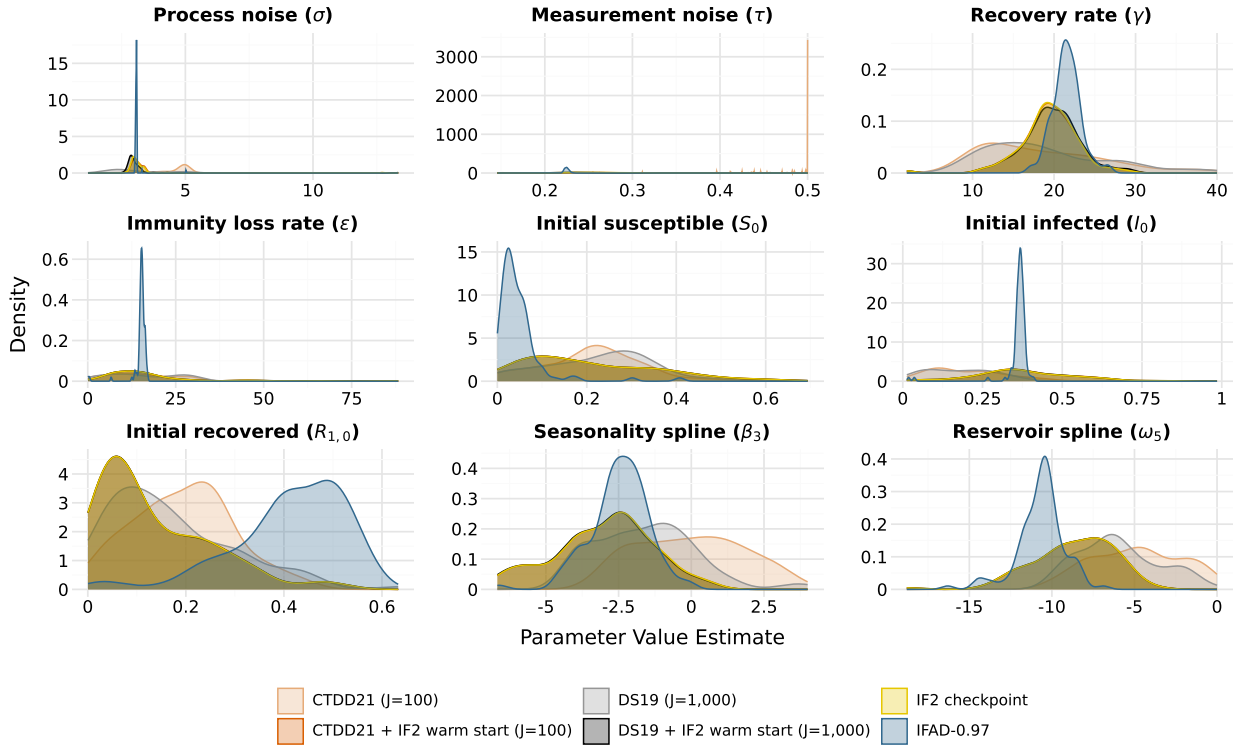


Figure S-9: Density plots of the final parameter estimates across 100 searches for nine selected parameters of the Dhaka model. Each panel has its own density scale. CTDD21 uses 100 particles and DS19 1,000, with and without an IF2 warm start; IFAD-0.97 uses 5,000 particles. The IF2 checkpoint shows the starting estimates for the warm starts.

S12 Comparisons Across Models

We compared IFAD, IF2, DS19, and CTDD21 on the harmonic oscillator of Ditlevsen and Samson (2019), the two-dimensional linear Gaussian model of Corenflos et al. (2021), and the Pypomp library implementation of a stochastic volatility model at `pypomp.models.spx`, subsequently called `spx`, in addition to the Daphnia and Dhaka models. For each example, we used the same starting parameter vectors for all methods and evaluated the resulting estimates under the same model. The two Gaussian examples have analytic likelihoods, allowing us to assess maximization error without Monte Carlo evaluation error.

Table S-8 summarizes the DS19 implementation for each panel. All five implementations evaluate process transition densities. For the linear Gaussian model and `spx`, these densities are available directly. For the oscillator, all methods fit the same order-1.5 Gaussian approximation, whose likelihood we evaluate by Kalman filtering. Thus these three comparisons use the same transition model for fitting and evaluation.

For Daphnia and Dhaka, we also use moments from the order-1.5 expansion, but compose them over smaller time steps and approximate the transition between observations by a Gaussian distribution. This composition adds a linearization of the nonlinear dynamics. It does not imply that the resulting Gaussian approximation has strong order 1.5 over the full observation interval. For Dhaka, a single local Gaussian approximation has covariance rank at most two in six dimensions, as explained in Section S11. Composing the local covariances allows noise to propagate into additional state directions. For Daphnia, we compose moments at the 0.25-day simulation step rather than apply a single Taylor expansion across the five-day observation interval. We have not tested the accuracy of such a single coarse step. Both models are evaluated using their original Euler simulators, so the Gaussian approximation introduces a difference between the fitting and evaluation likelihoods.

S12.1 SAEM and averaged-score updates

Stochastic approximation expectation maximization (SAEM) alternates imputation of the latent states with maximization of an averaged complete-data log-likelihood. Let x denote a latent trajectory and write $C(\vartheta; x) = \log \widetilde{p}_\vartheta(x, y)$ for the complete-data objective of the fitted model. The tilde allows for a transition-density approximation. Deterministic initial states are functions of the candidate parameter; only random states are imputed. At iteration m, a particle filter followed by backward simulation draws one trajectory $X^{(m)}$ from an approximation to the smoothing distribution at ϑ_{m-1}. SAEM updates a function of the candidate parameter ϑ:

$$\overline{Q}_m(\vartheta) = (1-\rho_m)\overline{Q}_{m-1}(\vartheta) + \rho_m C(\vartheta; X^{(m)}), \tag{S243}$$

$$\vartheta_m \in \operatorname*{argmax}_\vartheta \overline{Q}_m(\vartheta). \tag{S244}$$

In the maximization step, we hold the imputed paths and their averaging weights fixed. This maximization can be carried out numerically; a step that increases $\overline{Q}_m$ without maximizing it gives a generalized EM update.

The averaged-score implementation instead calculates

$$g_m = \nabla_\vartheta C(\vartheta; X^{(m)})\Big|_{\vartheta=\vartheta_{m-1}}, \tag{S245}$$

$$\overline{g}_m = (1-\rho_m)\overline{g}_{m-1} + \rho_m g_m, \tag{S246}$$

Model	DS19 transition density	Parameter update	Likelihood evaluation
A: Harmonic oscillator	Order-1.5 Gaussian approximation at the observation interval	SAEM; numerical maximization of averaged sufficient statistics	Kalman likelihood of the same approximation
B: Linear Gaussian	Exact Gaussian density of the specified discrete-time model	SAEM; explicit maximization using averaged sufficient statistics	Kalman likelihood of the same model
C: SPX	`spx` transition law: Gaussian above the variance floor, with a point mass at the floor	Averaged scores and Adam	Particle-filter likelihood of the same `spx` model
D: Daphnia	Gaussian approximation across each observation interval, composed from local order-1.5 moments	Averaged scores and Adam	Particle-filter likelihood of the original Euler model; transition and boundary rules differ
E: Dhaka	Monthly Gaussian approximation composed from local order-1.5 moments	Averaged scores and Adam; numerical SAEM in a separate comparison	Particle-filter likelihood of the original Euler model; transition differs

Table S-8: DS19 implementations for panels A–E of Figure S-10. Each uses transition-density evaluation for backward smoothing and either maximization or differentiation of a complete-data objective. For SPX, the density includes a discrete mass and is exact for the implemented model, not for a continuous-time diffusion. The Daphnia and Dhaka covariances are regularized as described below. The separate Dhaka SAEM comparison retains the same Gaussian transition.

then uses the clipped $\overline{g}_m$ in an Adam step. Derivatives hold the imputed path fixed; they do not pass through the particle filter or backward sampler. With exact smoothing, the conditional expectation of g_m is the score of the fitted model's observed-data likelihood. The particle smoother approximates this expectation. Before clipping or safeguard resets, the distinction between the two averages can be written as

$$\nabla\overline{Q}_m(\vartheta) = \sum_{r=1}^{m} w_{mr}\nabla_\vartheta C(\vartheta; X^{(r)}), \qquad \overline{g}_m = \sum_{r=1}^{m} w_{mr}\nabla_\vartheta C(\vartheta_{r-1}; X^{(r)}), \tag{S247}$$
$$w_{mr} = \rho_r \prod_{k=r+1}^{m}(1-\rho_k).$$

The SAEM gradient evaluates all path contributions at the same candidate parameter, whereas $\overline{g}_m$ averages gradients evaluated at the parameter values used in earlier iterations. Consequently, $\overline{g}_m$ generally differs from the gradient of $\overline{Q}_m$ at the current parameter. Our gain schedule is $\rho_m = 1$ for $m \le b$ and $\rho_m = (m-b)^{-0.9}$ thereafter. We use $b = 30$ for the Gaussian SAEM and averaged-score runs, and $b = 3$ for numerical SAEM on Dhaka. Clipping and Adam act on the existing estimation scales. In the Dhaka score implementation, the clipped average is also carried into the next iteration, and failed safeguard checks can reset it.

Linear Gaussian model: explicit SAEM. The latent states satisfy $X_t = \mathrm{diag}(a_1, a_2)X_{t-1} + \varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0, 0.5I)$, $Y_t \mid X_t \sim \mathcal{N}(X_t, 0.1I)$, and $X_0 \sim \mathcal{N}(0, I)$. For a fixed path, the terms depending on $a = (a_1, a_2)$ are

$$C(a; x) = -\frac{1}{2(0.5)}\sum_{t=1}^{T}\sum_{d=1}^{2}(x_{t,d} - a_d x_{t-1,d})^2 + \text{constant}.$$

For each coordinate, the sufficient statistics are

$$S_{00,d}(x) = \sum_t x_{t-1,d}^2, \qquad S_{10,d}(x) = \sum_t x_{t,d}x_{t-1,d}.$$

We update their averages by $\overline{S}_m = (1-\rho_m)\overline{S}_{m-1} + \rho_m S(X^{(m)})$. The M-step is

$$a_{d,m} = \mathrm{clip}_{[-0.98,0.98]}\left(\frac{\overline{S}_{10,d,m}}{\overline{S}_{00,d,m}}\right). \tag{S248}$$

Each iteration therefore requires a particle filter and backward simulation of one path, after which the maximization reduces to two scalar ratios.

Harmonic oscillator: SAEM with a numerical M-step. Write $x_t = (v_t, u_t)^\mathsf{T}$, where the positions v_t are observed without measurement error and the velocities u_t are imputed. For $\theta = (D, \gamma, \sigma)$ and observation interval $\Delta = 0.02$, the fitted Gaussian transition has

$$A_\theta = \begin{pmatrix} 0 & 1 \\ -D & -\gamma \end{pmatrix}, \qquad F_\theta = I + \Delta A_\theta + \tfrac{1}{2}\Delta^2 A_\theta^2,$$
$$Q_\theta = \sigma^2 \begin{pmatrix} \Delta^3/3 & \Delta^2/2 - \gamma\Delta^3/3 \\ \Delta^2/2 - \gamma\Delta^3/3 & \Delta - \gamma\Delta^2 + \gamma^2\Delta^3/3 \end{pmatrix}.$$

Conditional on the first position, $U_0 \sim \mathcal{N}(0, v_\theta)$, where $v_\theta = \sigma^2/(2\gamma)$. Let $S_{ij} = \sum_{t=1}^{T} x_{t-1+i} x_{t-1+j}^\mathsf{T}$ for $(i,j) = (0,0), (1,0), (1,1)$, and let $h = u_0^2$. We average these quantities using the recursion above, giving the following objective up to constants independent of θ:

$$\overline{Q}_m(\theta) = -\tfrac{1}{2}\left\{\log v_\theta + \frac{\overline{h}_m}{v_\theta} + T \log\det Q_\theta + \operatorname{tr}\big(Q_\theta^{-1}\overline{R}_m(\theta)\big)\right\}, \tag{S249}$$

where

$$\overline{R}_m(\theta) = \overline{S}_{11,m} - F_\theta \overline{S}_{10,m}^\mathsf{T} - \overline{S}_{10,m} F_\theta^\mathsf{T} + F_\theta \overline{S}_{00,m} F_\theta^\mathsf{T}.$$

We use L-BFGS-B on $(\log D, \log\gamma, \log\sigma)$, retaining a finite step only if this objective does not decrease, up to numerical tolerance. To reduce numerical cancellation, we evaluate this quadratic form using accumulated state increments. The M-step thus uses averaged sufficient statistics of the imputed paths.

SPX: averaged complete-data scores. Let r_t be the observed return and v_t the latent variance. The `spx` conditional variance transition has Gaussian mean and standard deviation

$$m_t(v) = v + \kappa(\bar{v} - v) + \xi\rho(r_{t-1} - \mu + v/2), \qquad s(v) = \xi\sqrt{v(1-\rho^2)},$$

followed by a floor $\epsilon = 10^{-32}$. Writing $\phi(z; m, s^2)$ for a Gaussian density and Φ for the standard normal distribution function, its transition is

$$f_{\theta,t}(v' \mid v) = \begin{cases} \phi(v'; m_t(v), s(v)^2), & v' > \epsilon, \\ \Phi((\epsilon - m_t(v))/s(v)), & v' = \epsilon. \end{cases}$$

The second case is a probability mass; the reference measure is Lebesgue measure plus an atom at ϵ. The complete-path objective is

$$C(\theta; v) = \sum_{t=1}^{T} \left\{\log f_{\theta,t}(v_t \mid v_{t-1}) + \log\phi(r_t; \mu - v_t/2, v_t)\right\}. \tag{S250}$$

The initial variance $v_0 = V_0$ is a parameter, so its current candidate value enters the first transition when differentiating. For $v > 0$, $\xi > 0$, and $|\rho| < 1$, the conditional transition variance $\xi^2 v(1-\rho^2)$ is positive. This scalar latent process therefore has no missing covariance direction in the interior; its boundary atom is included explicitly in the density above. For this example, we update the parameters with Adam using the averaged scores in Equations (S245)–(S246), without a SAEM maximization step. We obtain these complete-data derivatives by AD while holding the imputed path fixed. This avoids differentiation through the filtering and resampling calculations, but does not avoid AD itself. The runtime difference from IFAD also reflects the particle counts and optimization settings below.

Daphnia: averaged scores for the Gaussian block approximation. For mesocosm u, let $m_{\theta,u,t}$ and $\Sigma_{\theta,u,t}$ be the mean and covariance obtained by composing the local Gaussian transitions over an observation interval, including the inoculation in the first interval. Let $g_{\theta,u}$ be the product

of the four negative-binomial measurement probabilities in Section S10. For the eight imputed mesocosm paths, the complete-path objective is

$$C(\theta; x) = \sum_{u=1}^{8} \sum_{t=1}^{10} \left\{ \log \phi_8(x_{u,t}; m_{\theta,u,t}(x_{u,t-1}), \Sigma_{\theta,u,t}(x_{u,t-1})) + \log g_{\theta,u}(y_{u,t} \mid x_{u,t}) \right\}. \tag{S251}$$

Initial states are fixed. The covariance uses a relative diagonal floor of 10^{-8} on scaled states. The particle filter uses the observations to guide proposals, with the original negative-binomial measurement probabilities in the importance weights. We approximate each observation interval by one eight-dimensional Gaussian transition, composing local moments over 24 steps for the first interval and 20 thereafter. The step size is 0.25 days, so these transitions cover six days initially and five days between observations. This approximates the distribution produced by the nonlinear dynamics over the whole interval. Boundary handling also differs: we assign zero weight to Gaussian proposals outside the allowed state range, whereas the Euler simulator resets an offending state to zero and assigns a log weight of -150 to that particle at the next observation. Both differences can change the likelihood being maximized. The local covariance need not be rank deficient in this model. At a strictly positive state, the six independent noises span the six directly forced coordinates. The integrated juvenile noises contribute two further directions through maturation into susceptible adults, with nonzero coefficients $\lambda_J \sigma_J^{\mathrm{n}} J^{\mathrm{n}}$ and $\lambda_J \sigma_J^{\mathrm{i}} J^{\mathrm{i}}$. The implemented order-1.5 covariance is therefore full rank in this interior when the six fitted noise coefficients are positive, and its range is independent of the parameter values there. At the specified initial state several populations are zero, and the local covariance has rank three. Daphnia thus differs from Dhaka's rank-two interior covariance. The algebra and numerical checks, including parameter perturbations at fixed states, are provided in `code/benchmarks/covariance_support.py` and its accompanying note. After backward simulation of one path per mesocosm, we differentiate this sum with respect to the 38 estimated parameters and update them with Adam using the averaged scores. As for SPX, we do not carry out a SAEM maximization step. We evaluate the resulting parameter estimates using the original Euler model from Section S10.

Dhaka: score updates and numerical generalized SAEM. We compare score updates with numerical SAEM using the monthly Gaussian transition described in Section S11, with 20 internal time steps and a relative covariance floor of 10^{-12}. Let $\mathcal{R}x$ reset the accumulated monthly deaths to zero, and write $d_t = P_t M_t$ for the predicted deaths at a sampled endpoint. Here P_t is the population and M_t is the accumulated death count divided by population. Both methods use the complete-data objective

$$\begin{aligned} C(\theta; x) = \sum_{t=1}^{T} \big\{ & \log \phi_6(x_t; m_{\theta,t}(\mathcal{R}x_{t-1}), \Sigma_{\theta,t}(\mathcal{R}x_{t-1})) \\ & + \log[\phi(y_t; d_t, (\tau d_t + \delta)^2) + \delta] \big\}, \qquad \delta = 10^{-18}. \end{aligned} \tag{S252}$$

The initial state $x_0 = x_0(\theta)$ is recalculated at each candidate parameter, while the sampled monthly endpoints are held fixed. For the score updates, we use the gradient average in Equation (S246), with Adam and the safeguards described above. For numerical SAEM, we retain the paths and weights to calculate $\overline{Q}_m(\theta) = \sum_{r=1}^{m} w_{mr} C(\theta; X^{(r)})$. Each L-BFGS-B evaluation recalculates every retained path's objective and gradient at the same candidate θ. Within each M-step, we divide

the objective and its gradient by the fixed scale $\max\{1, \|\nabla \overline{Q}_m(\theta_{m-1})\|_\infty\}$. This reduces numerical line-search failures without changing the maximizer. We allow at most 25 L-BFGS-B iterations per M-step and retain a candidate only if it is finite and increases or leaves unchanged this fixed $\overline{Q}_m$. An M-step interrupted by the time limit is discarded. This gives a generalized-SAEM update when an explicit maximizer in terms of sufficient statistics is unavailable. We evaluate both methods with the Euler likelihood, so the evaluation model differs from the monthly Gaussian approximation used for fitting.

Computation times for the Gaussian examples. We used 500 particles for DS19, IFAD, and IF2 in both Gaussian examples. DS19 ran for 80 SAEM iterations, IFAD for 100 IF2 iterations followed by 300 gradient iterations, and IF2 for 600 iterations. CTDD21 used 25 particles and 300 gradient iterations. The common particle count for DS19, IFAD, and IF2 therefore does not imply equal computational effort.

DS19 runs a particle filter and samples one trajectory by backward simulation at every iteration, and both are included in its computation time. Its maximization step is inexpensive for these models: the linear Gaussian model has an explicit update, while the oscillator requires optimization over three parameters using averaged sufficient statistics. Neither maximization requires differentiation through the particle filter. The analytic likelihoods are used only to evaluate the fitted parameters. For all methods, we wait for the numerical calculations to finish before recording the time, excluding compilation and likelihood evaluation. IFAD, IF2, and DS19 used four efficiency cores of an Intel Core i9-13900K; CTDD21 used two performance cores with four hardware threads. Concurrent batches used separate core groups. These hardware differences also affect the comparison of computation times.

S12.2 Search and evaluation settings

We ran 20 searches per method for the oscillator, linear Gaussian, SPX, and Daphnia models, using the same starting parameter vectors for each method. For Dhaka, we used the 100 searches from Section S11.

For SPX, we used 500 particles for IFAD and IF2, 100 for DS19, and 25 for CTDD21. IFAD used $\alpha = 0.97$ and ran for 100 IF2 iterations followed by 300 gradient iterations, IF2 for 600 iterations, and DS19 and CTDD21 for 300 updates each. The Adam learning rates started at 0.001 for IFAD and 0.01 for DS19 and CTDD21, decreasing to one tenth of these values by cosine decay.

For Daphnia, we used 1,000 particles per mesocosm for MPIF and IFAD, and 100 for DS19 and CTDD21. We ran MPIF for 680 iterations. IFAD used a separate 200-iteration MPIF warm start followed by 200 gradient steps. To compare refinement of the same starting estimate, we initialized DS19 and CTDD21 at this MPIF warm start and allowed each the time taken by the IFAD gradient stage, with a maximum of 2,000 updates. Their learning rates decreased from 0.001 to 0.0001 by cosine decay over the allotted time. We discarded updates completed after the time limit and included the common MPIF warm start in the computation times for all three methods.

We evaluated the final SPX estimates using 24 independent particle-filter replicates with 5,000 particles. For Daphnia, we used 10 replicates with 2,000 particles per mesocosm under the original Euler model. We averaged the likelihood estimates for each mesocosm before taking logarithms, then summed the log-likelihoods over mesocosms, combining their independent Monte Carlo errors. For the optimization trajectories, we used four replicates with 1,000 particles for SPX and two

replicates with 500 particles per mesocosm for Daphnia. All these evaluations were independent of fitting; we did not use them to select the reported parameter estimates.

SPX fits used two performance cores with four hardware threads, except for one recovered starting point for which all four methods used four efficiency cores. Daphnia fits used four efficiency cores. These runs used the same processor as the Gaussian comparisons. The archived Dhaka comparison used GPU computations as described in Section S11.

S12.3 Repeated-search results

IFAD and DS19 both approached the maximum likelihood for the two Gaussian models (Table S-9 and Figure S-10). DS19 did so more quickly, benefiting from the inexpensive maximization steps described above. IF2 gave more variable results, while CTDD21 attained lower likelihoods, particularly for the oscillator. DS19 also had the highest median log-likelihood and shortest median computation time for SPX. We therefore found no consistent improvement from IFAD on these examples.

For Daphnia, IFAD attained the highest maximum and median log-likelihoods. The median was -861.30, compared with -864.98 for MPIF, -867.02 for DS19, and -865.03 for CTDD21. Figure S-11 shows IFAD continuing to improve after the common MPIF warm start, while DS19 and CTDD21 made less progress with the same time allowance. The full trajectories are shown in Figure S-12; the shaded regions in both figures cover the 10th through 100th percentiles, as in Section S11. We retained the last finite estimate from one DS19 search that stopped prematurely. For Dhaka, the differences were larger: IFAD attained a median log-likelihood of -3744.40, compared with -3755.41 for IF2, -3769.04 for DS19, and -3765.91 for CTDD21.

The Daphnia comparison does not determine how much of the DS19 shortfall comes from the Gaussian transition approximation. Smoothing error and optimization may also contribute. DS19 used 100 particles per mesocosm, compared with 1,000 for IFAD, and estimated each gradient from one backward-sampled path per mesocosm for the 38-parameter optimization. Moreover, the Daphnia implementation uses averaged-score Adam updates, whereas the Gaussian examples use SAEM maximization with sufficient statistics. We have not compared these updates with numerical SAEM on Daphnia. Evaluating both the Gaussian fitting objective and the Euler likelihood along the same searches would help assess the effect of the transition approximation: improvement in the former accompanied by deterioration in the latter would indicate that improving the approximate objective need not improve the likelihood of interest. We have not carried out this diagnostic.

The Gaussian examples provide useful numerical checks because their likelihoods are available analytically. Once methods repeatedly approach these maxima, the examples offer limited evidence about further improvements in likelihood maximization. The SPX comparison still shows differences between methods and is not saturated in this sense. Daphnia and Dhaka retain difficulties encountered in scientific applications, including many estimated parameters, nonlinear dynamics, partially observed states, and, for Daphnia, multiple experimental units. Assessing methods for such applications requires benchmarks that retain these difficulties alongside tractable examples used to verify implementations.

S12.4 Score updates and numerical SAEM on Dhaka

To assess whether the score updates accounted for the poor performance of DS19 on Dhaka, we repeated the comparison using numerical SAEM with the same monthly Gaussian transition ap-

proximation and particle smoother. We used 1,000 particles and allowed 850 seconds per search, starting from 20 global parameter vectors and 20 IF2 estimates. Numerical SAEM gave higher Euler log-likelihoods than the score updates for only two of the 20 global starts (Figure S-13 and Table S-10). The median paired difference was −608.83 log-likelihood units, taking numerical SAEM minus score updates. The numerical M-steps were expensive: SAEM completed a median of 15.5 outer iterations in the allotted time, compared with 86 for the score updates.

Starting from the IF2 estimates reduced this difference substantially. The median paired difference was −0.62 log-likelihood units, although one SAEM search finished 39.59 units below its score counterpart. SAEM gave higher likelihoods in four of the 20 pairs. Neither method improved on the IF2 estimates at the median, with changes of −0.27 for score updates and −1.37 for numerical SAEM. Thus numerical maximization of the SAEM objective did not improve the Dhaka results. Both methods retained the same monthly Gaussian transition and particle smoother, so this comparison does not measure the effect of the transition approximation. Increasing the number of particles can improve the particle approximation to this Gaussian model, but does not remove its difference from the Euler model. The results therefore leave open the relative contributions of transition approximation, smoothing error, numerical conditioning, and optimization cost to the Dhaka shortfall.

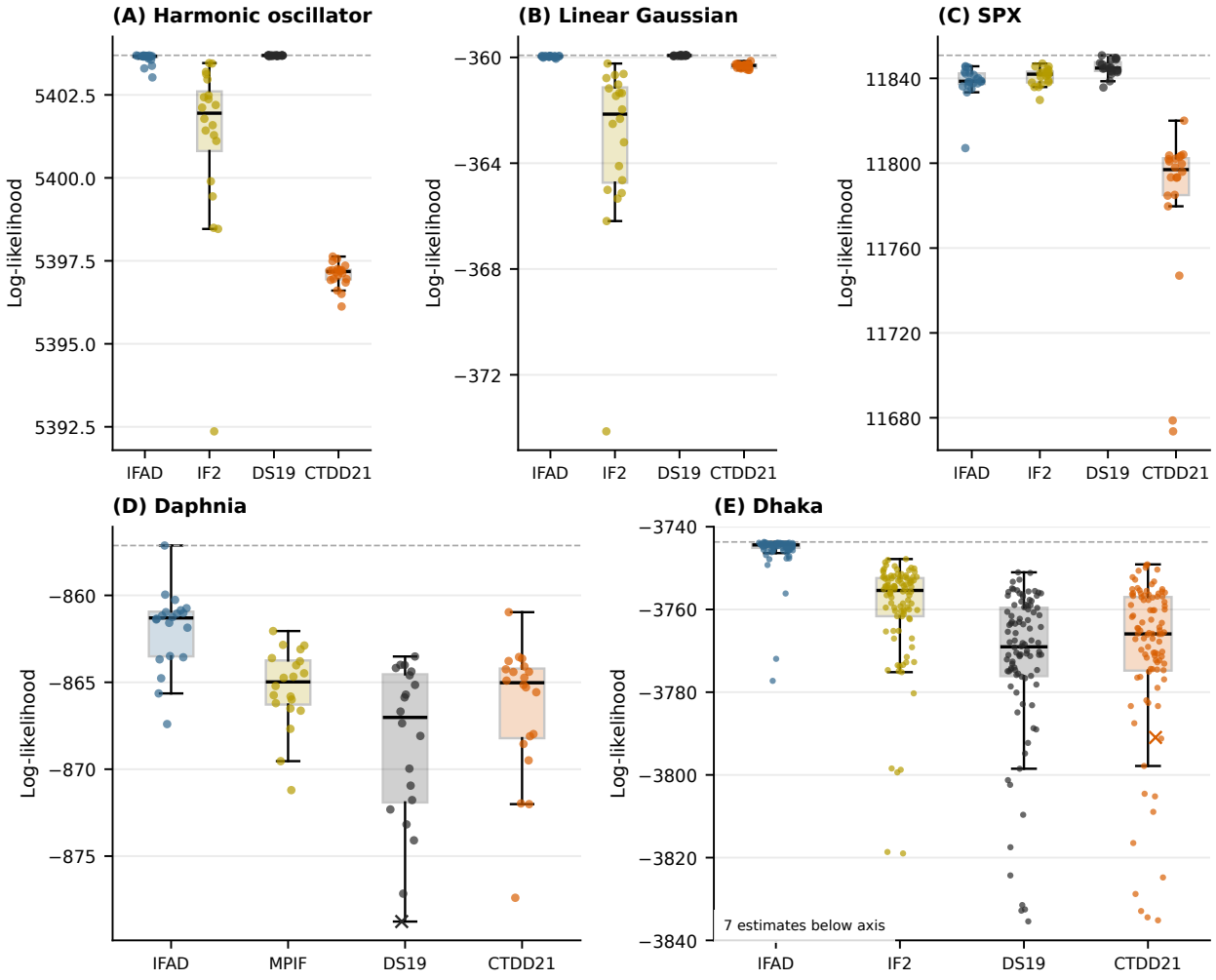


Figure S-10: Final log-likelihoods from repeated parameter searches. Each point represents one of 20 searches per method in panels A–D or 100 searches in panel E. Boxes show the median and interquartile range; whiskers extend to the most extreme values within 1.5 interquartile ranges of the box. All estimates are shown in panels A–D. Panel E uses the range $[-3840, -3740]$; the number of estimates below the axis is indicated. The uncropped figure is available with the source code, and all estimates contribute to the boxes and numerical summaries. Crosses mark searches that stopped prematurely, for which we evaluated the last estimate obtained before failure or the time limit. Dashed lines mark the analytic maximum in panels A and B and the best displayed estimate in panels C–E. Likelihoods are evaluated independently of fitting, using the same model within each panel. DS19 and CTDD21 use the common MPIF warm start in D and the common IF2 warm start in E. Panel E uses the same estimates as the warm-start comparison in Figure S-6. All axes are linear.

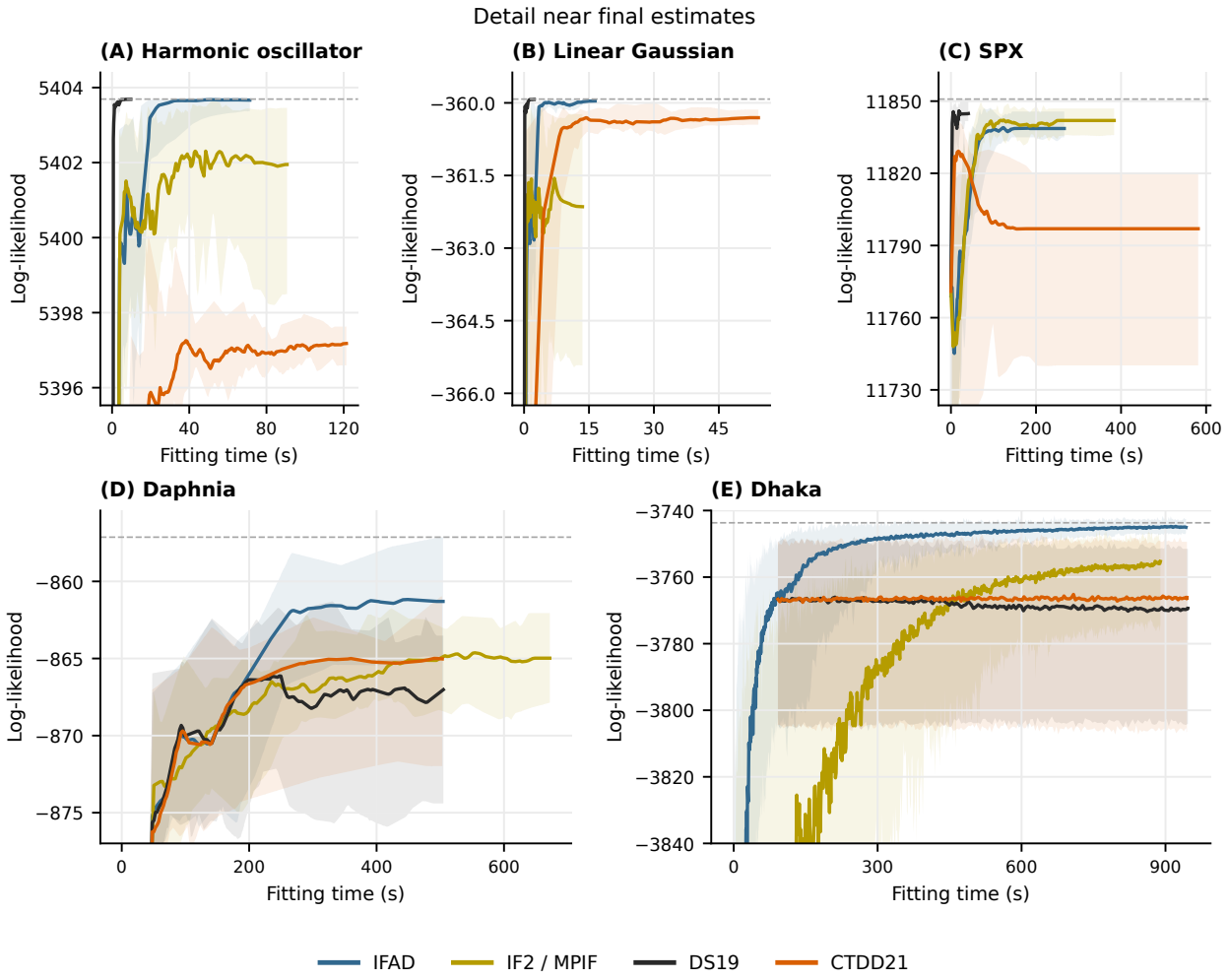


Figure S-11: Optimization progress near the final estimates. Solid lines show the median log-likelihoods, and the shaded regions cover the area between the 10th and 100th percentiles across searches. We evaluated likelihoods at saved iterations and interpolated linearly between them, retaining each search's final value after it stopped. These evaluations were not used to select the fitted parameters. The vertical axes are linear and restricted to the region near the final estimates; Figure S-12 shows the full range, with the same shading opacity. Panel E uses the range $[-3840, -3740]$. The starting points, reference lines and final evaluations are those of Figure S-10.

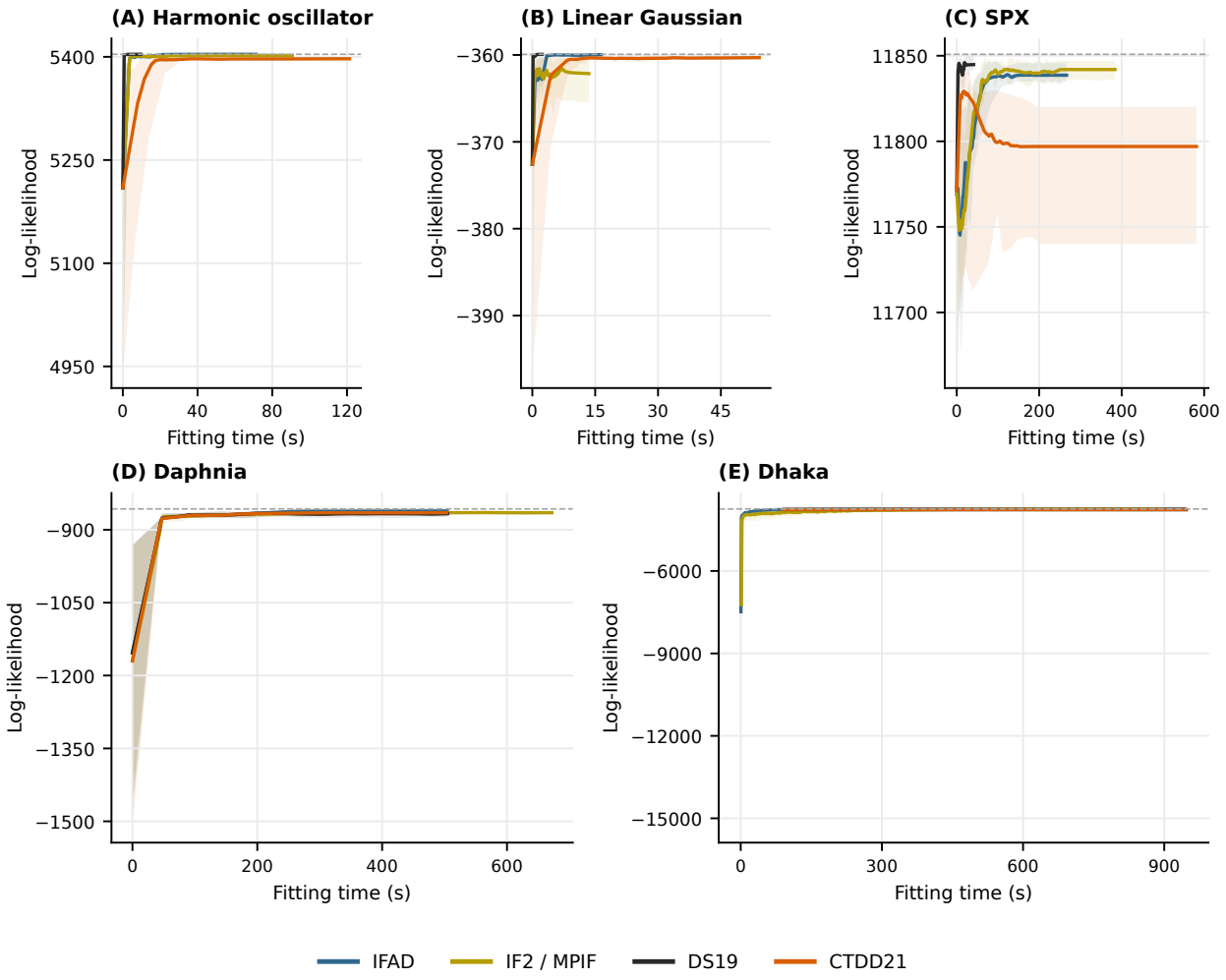


Figure S-12: Optimization progress over the full range of the median trajectories and percentile bands. Lines, shading and reference lines are defined in Figure S-11. All axes are linear, with separate ranges for each model and data set. Computation times exclude compilation and independent likelihood evaluation. The hardware and algorithmic settings are given in Section S12.2 and the preceding paragraph.

Method	Median	Maximum	IQR	MCSE	Time (s)	Failed
Harmonic oscillator (20 starts)						
IFAD	5403.67	5403.69	0.04	0.00	70.0	0
IF2	5401.95	5403.46	1.80	0.00	89.4	0
DS19	5403.69	5403.69	0.01	0.00	9.4	0
CTDD21	5397.18	5397.63	0.28	0.00	117.3	0
Linear Gaussian (20 starts)						
IFAD	-359.96	-359.94	0.03	0.00	16.1	0
IF2	-362.14	-360.23	3.60	0.00	13.3	0
DS19	-359.93	-359.92	0.01	0.00	2.1	0
CTDD21	-360.31	-360.14	0.13	0.00	51.6	0
SPX (20 starts)						
IFAD	11838.63	11845.69	4.95	0.31	144.5	0
IF2	11841.96	11846.95	5.06	0.31	249.8	0
DS19	11844.83	11850.85	4.11	0.47	21.5	0
CTDD21	11796.97	11820.04	17.37	0.19	165.9	0
Daphnia (20 starts)						
IFAD	-861.30	-857.13	2.57	0.19	487.8	0
MPIF	-864.98	-862.05	2.53	0.19	647.3	0
DS19	-867.02	-863.52	7.37	0.20	488.1	1
CTDD21	-865.03	-860.97	4.00	0.17	488.7	0
Dhaka (100 starts)						
IFAD	-3744.40	-3743.68	0.95	0.14	942.8	0
IF2	-3755.41	-3747.82	9.21	0.15	889.3	0
DS19	-3769.04	-3751.03	16.53	0.17	942.8	0
CTDD21	-3765.91	-3749.10	17.78	0.15	942.8	2

Table S-9: Repeated-search results for the five models. Median, maximum and interquartile range (IQR) summarize final log-likelihoods across searches. MCSE is the median Monte Carlo standard error of an individual final log-likelihood estimate; it is zero for the analytic Gaussian evaluations. Time is the median fitting time, including any IF2 or MPIF warm start and excluding compilation and independent evaluation. Failed is the number of searches that stopped prematurely. We retained their last estimates before failure or the time limit in all summaries. Hardware and fitting settings differ between experiments, as described in the text.

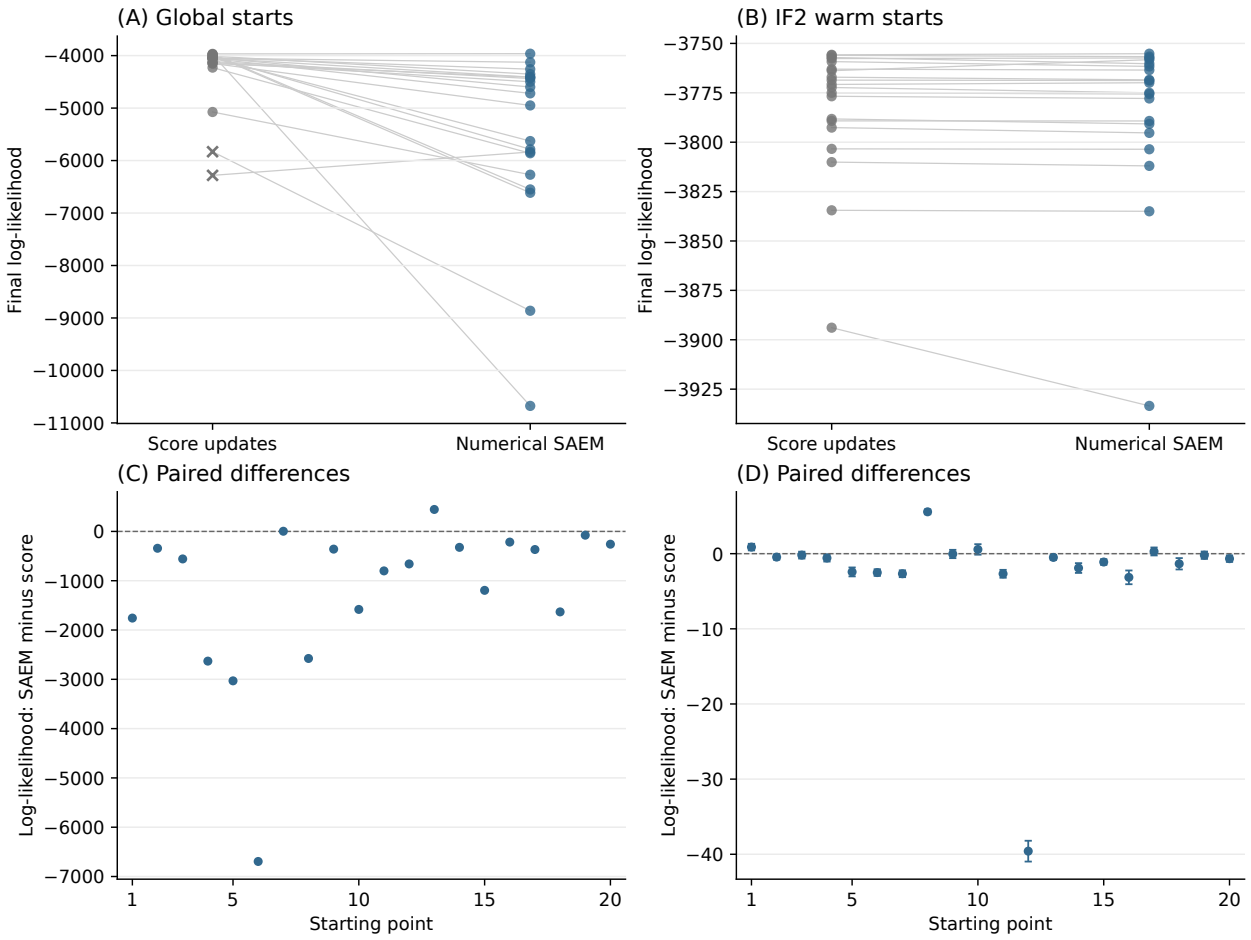


Figure S-13: Averaged-score updates and numerical SAEM on Dhaka, using the same monthly Gaussian transition approximation and particle smoother. Both methods use 1,000 particles and are allowed 850 seconds per search, from 20 global starts (A, C) and 20 saved IF2 estimates (B, D). Panels A and B join final estimates from the same starting point. Crosses mark searches that stopped prematurely, for which we retained the last estimate before failure or the time limit. Panels C and D show paired log-likelihood differences, numerical SAEM minus score updates; bars show twice the paired Monte Carlo standard error. Final likelihoods use the original Euler model, with 5,000 particles and 36 evaluation replicates. Evaluation random numbers are shared within a pair and independent of fitting. All axes are linear.

Initialization	Optimizer	Median ℓ	Median change	Iterations	Time (s)	Failed
Global	Score updates	-4050.49	3827.36	86	854.5	2
Global	Numerical SAEM	-4834.43	3109.60	15.5	851.3	0
IF2	Score updates	-3771.54	-0.27	89	855.7	0
IF2	Numerical SAEM	-3772.44	-1.37	13	851.1	0

Table S-10: Dhaka score and numerical-SAEM comparison in Figure S-13. Median ℓ is the median final Euler log-likelihood. Median change is the median difference between each search's final and initial log-likelihoods. Iterations is the median number of outer iterations completed at the reported estimate, including iterations without a parameter change. Time is the median computation time, including computation past the time limit; we discarded updates that did not finish within the allotted time. Compilation, independent evaluation and the cost of producing the saved IF2 starting estimates are excluded. For failed searches, we retained the last estimate before failure or the time limit in all summaries.

Supplementary References

Abkemeier, A. J., Chen, J., Tan, K., Wheeler, J., He, K., Yang, B., King, A. A., Terhorst, J., Hooker, G., and Ionides, E. L. (2025). Pypomp: Inference for partially observed Markov processes with Python and JAX.

Andrieu, C., Doucet, A., and Holenstein, R. (2010). Particle Markov chain Monte Carlo methods. *Journal of the Royal Statistical Society, Series B*, 72:269–342.

Chopin, N. (2004). Central limit theorem for sequential Monte Carlo methods and its application to Bayesian inference. *Annals of Statistics*, 32:2385–2411.

Chopin, N. and Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*. Springer.

Corenflos, A., Thornton, J., Deligiannidis, G., and Doucet, A. (2021). Differentiable particle filtering via entropy-regularized optimal transport. In Meila, M. and Zhang, T., editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 2100–2111. PMLR.

Del Moral, P. (2004). *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, New York.

Del Moral, P., Doucet, A., and Singh, S. S. (2010). A backward particle interpretation of Feynman–Kac formulae. *ESAIM: Mathematical Modelling and Numerical Analysis*, 44(5):947–975.

Del Moral, P. and Rio, E. (2011). Concentration inequalities for mean field particle models. *The Annals of Applied Probability*, 21(3).

Ditlevsen, S. and Samson, A. (2019). Hypoelliptic diffusions: Filtering and inference from complete and partial observations. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 81(2):361–384.

Karjalainen, J., Lee, A., Singh, S. S., and Vihola, M. (2023). On the forgetting of particle filters. *arXiv:2309.08517*.

Karppinen, S., Singh, S. S., and Vihola, M. (2024). Conditional particle filters with bridge backward sampling. *Journal of Computational and Graphical Statistics*, 33(2):364–378.

King, A. A., Ionides, E. L., Pascual, M., and Bouma, M. J. (2008). Inapparent infections and cholera dynamics. *Nature*, 454:877–880.

Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.

Naesseth, C., Linderman, S., Ranganath, R., and Blei, D. (2018). Variational sequential Monte Carlo. In Storkey, A. and Perez-Cruz, F., editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 968–977. PMLR.

Poyiadjis, G., Doucet, A., and Singh, S. S. (2011). Particle approximations of the score and observed information matrix in state space models with application to parameter estimation. *Biometrika*, 98(1):65–80.

Ścibior, A. and Wood, F. (2021). Differentiable particle filtering without modifying the forward pass. *arXiv:2106.10314*.

Searle, C. L., Cortez, M. H., Hunsberger, K. K., Grippi, D. C., Oleksy, I. A., Shaw, C. L., de la Serna, S. B., Lash, C. L., Dhir, K. L., and Duffy, M. A. (2016). Population density, not host competence, drives patterns of disease in an invaded community. *The American Naturalist*, 188(5):554–566.

Wheeler, J., Abkemeier, A. J., and Ionides, E. L. (2025). Iterating marginalized Bayes maps for likelihood maximization with application to nonlinear panel models. *arXiv:2511.17438*.

Yang, B., Ionides, E., and Wheeler, J. (2026). Megumi-ybb/Daphnia-ms: Daphnia-ms-version2.1.6. Zenodo, version v2.1.6. `https://doi.org/10.5281/zenodo.21754618`.

Yang, B., Wheeler, J., Duffy, M. A., King, A. A., and Ionides, E. L. (2025). Mechanistic models for panel data: Analysis of ecological experiments with four interacting species. *arXiv:2506.04508*.